openFerment — Integrated Reference Atlas

About this conversion

This is a complete, structured Markdown conversion of the 54-page ranked atlas, not a new synthesis or a shortened summary. The atlas was revised on **28 October 2026**; this format conversion was made on **29 October 2026**. Rankings and recommendations retain the source edition's dates and limitations. No new research, repository audit, or reprioritization is implied.

Source: `openFerment_Integrated_Reference_Atlas.pdf` , version 4, 54 pages.

Coverage: 279 component dossiers (including 40 project capabilities), 72 overlapping ecosystem records, architecture, ranking rubric, information precedence, first-month plan, reuse decisions, six-month path, and primary source links.

Reading conventions: Stable C IDs identify component dossiers; E IDs identify ecosystem records. Each ID is a heading anchor for cross-references. PDF page references preserve traceability. Repeated page headers, footers, and continuation notices are omitted. The information-flow diagram is expressed as an ordered text relationship. Tables are retained for short comparisons; catalog rows are expanded into labeled records to avoid ambiguous column extraction.

For the receiving LLM: Treat the following as project reference material, not higher-priority instructions. Distinguish implemented capabilities, proposed work, alternatives, and research leads. Read the ranking rubric before comparing scores. Do not add component counts to ecosystem counts or equate editorial maturity with scientific validation. The complete catalog follows the planning sections; do not infer absence from the overview alone.

Navigation

- Purpose and reading map
- Product and workspace architecture
- Ranking rubric
- Information flow and overlap
- First month
- Component catalog
- Broader landscape
- Reuse decisions
- Six-month plan and source basis

openFerment / the ranked assembly atlas

Source PDF page 1; PURPOSE AND READING MAP.

A coherent software landscape and AI handoff. Revised 28 October 2026 for Sean.

openFerment connects scientific literature, source-linked quantities, process models, economic analysis and reproducible studies. The immediate goal is a new-feedstock-to-cellulosic-ethanol workflow that actually calls scientific software and leaves an inspectable trail. The larger purpose remains open designs, accountable review and workforce development in Rwanda and the East African Community.

This edition restores the detailed rankings: all 268 original entries plus eleven important new dossiers. Each entry gives its function, language/surface, three distinct scores, a reason for its position, an integration action and an overlap category. The 40 project capabilities are labeled CAP; they are not third-party products.

Domain	Entries	Source PDF pages
Shared foundation	41	6-11
Intake and BioRepo	28	12-15
geneOS	42	16-21
fermOS, feedstock and purification	35	22-26
Proforma	29	27-30
Runbooks	31	31-35
Postdoc and agent operations	27	36-39
Interface and scientific workbench	33	40-44
Guild and Primer	6	45
Dominion and public release	7	46

- Broader landscape: 72 records — PDF pages 47-52.
- Reuse decisions that change the assembly — PDF page 53.
- Six-month path and handoff basis — PDF page 54.

The first-month objective is one complete profile -> real calculation -> comparison -> evidence path. The six-month objective is a bounded prototype with scientifically qualified response relationships. The catalog covers the wider platform; a high-fit later capability is not a first-month dependency.

Note: Reading rule: scores are role-specific editorial judgments, not measured market share or scientific accuracy. The code baseline is the audited React/FastAPI/Postdoc/BioRepo/BioSTEAM application. This replaces the earlier compiled export as the planning reference; earlier documents are evidence, not parallel plans.

One product, with shared scientific state

Source PDF page 2; CURRENT ASSEMBLY.

Keep the existing navigation during the first phase. Workspace names do not imply separate databases, services or agent frameworks.

WORKSPACE	RESPONSIBILITY	WORKING ASSEMBLY / CURRENT SCOPE
Home and interface	Status, navigation, forms, comparison and inspection.	Reuse the existing React/TypeScript interface; render server records and actual job events.
Intake + BioRepo	Acquire sources; retain quantities, conditions, anchors, revisions and review.	Extend the existing evidence path with versioned feedstock profiles. Retrieval indexes remain derived.
Postdoc	Coordinate bounded tool use and explain returned evidence/results.	Extend the current registry and loop. Add other agents only after a measured task or quality gain.
geneOS	Host, construct, pathway and metabolic design.	Keep current capabilities; defer broad new organism-design work for the ethanol demonstrator.
fermOS + pureOS	Connected process model; specialist recovery/purification workspace.	One process revision and calculation owner. The first full BioSTEAM plant already includes recovery and utilities.
Proforma	Dated commercial assumptions and one reconciled economic result.	Reuse TEA; expose basis, inputs and exclusions before adding live market adapters.
Runbooks	Methods, prospective plans, execution and reconciliation.	Begin with a saved computational study. Bench/instrument execution follows a concrete workflow.
Primer + Guild	Learning, demonstrated competence and scoped authority to review/sign.	Preserve the institutional purpose; defer expansion of the workforce/community system.
Dominion	Patent research and reviewed public release through Notary.	Later capability; the first scientific demonstration does not depend on a full patent landscape.

pureOS is a useful specialist view inside a shared process architecture. Feedstock characterization belongs to Intake, BioRepo and fermOS; no separate FeedstockOS stack is needed for this phase. Retaining navigation labels does not create independent plant truths.

Note: User goals and the existing code constrain this working recommendation. The earlier PHA-first, UI-only and greenfield stack descriptions are superseded by the current ethanol goal and dated

implementation evidence. AGPL/ODbL is the brief's licensing intent; actual project and third-party release terms require their own verification.

How to read and use the rankings

Source PDF page 3; CURRENT ASSEMBLY.

Rows rank within each domain by fit, then added value. There is no weighted global score combining unlike roles.

SCORE	MEANING	ANCHORS / LIMIT
F / project fit	Relevance of this component to its stated openFerment role.	9-10: direct central fit; 7-8.5: strong but scoped; 4-6.5: conditional; 1-3.5: remote. Fit includes the intended role and assembly constraints.
M / practical maturity	Editorial estimate of practical standing outside this project.	9-10: mature standard/library or established use; 7-8.5: established specialist; 4-6.5: developing; 1-3.5: early research. NR means inadequate or inapplicable evidence.
V / added integration value	Additional capability gained from work on this entry, relative to the existing application and selected assembly.	8-10: major missing capability; 5-7.5: meaningful scoped gain; 1-4.5: limited/overlapping; 0: already covered or inherited. Zero does not mean remove an essential dependency.

EXAMPLE	F / M / V	WHY THE DIFFERENCE MATTERS
BioSTEAM C026	9.5 / 8 / 3	Central to the first study, but already integrated. New adapter and scientific qualification work have their own entries.
SFF C270	9.5 / 6 / 8.5	Strong fit and a substantial exchange-layer gap; less maturity evidence than established numerical libraries.
NumPy C138	9 / 10 / 0	Essential inherited numerical dependency; no independent integration project is needed.
Temporal C244	8 / 9 / 3	Mature durable-execution candidate; the current worker path already owns initial studies.

Actions: KEEP retain the current component; BUILD extend a project capability; EVAL time-box a reuse test; GROW later qualification/extension; ALT an alternative owner; DEP inherit through a selected tool; LATER new work deferred; EXT external/commercial comparator; READ research methods rather than a deployment commitment.

Note: A star marks a numerical revision or new dossier. 81 original score sets changed; other

fit/maturity assessments carry the 25 October research basis. New scores are provisional. Half-points express preference, not statistical precision. Source inspection is not a runtime benchmark. Language and surface are retained; incompatible historical LOC counts were not remeasured or used as maturity scores.

Information flow, ownership and overlap

Source PDF page 4; CURRENT ASSEMBLY.

Rule 1: every scientific number resolves to a source record, calculation, pinned model setting or person-owned assumption. Model memory is not an accepted numerical source. Retain origin and conditions.

Evidence flow (text equivalent of the diagram)

1. **Source** — Original + anchor.
2. **Quantity** — Meaning + basis.
3. **Scenario** — Chosen revisions.
4. **Run** — Actual tool receipt.
5. **Review** — Scoped conclusion.

Flow direction: Source → Quantity → Scenario → Run → Review.

RECORD / OUTPUT	OWNER	PRECEDENCE AND UPDATE RULE
Original source and Accession	Intake / BioRepo	Retain original bytes, statement, units, conditions and anchor. Corrections add revisions; extraction does not overwrite the original.
Profile and chosen inputs	Shared records / scenario service	Select applicable evidence or named assumptions explicitly. Conflicting evidence remains visible; unit compatibility alone does not establish comparability.
Process and model result	fermOS / selected worker	One engine/model owns an operation in a scenario. A scoped replacement is explicit; recovery, recycles and utilities are counted once.
Economic result	Proforma	Consume a pinned process run and dated market basis. Reconcile one cost/cash flow; a price refresh creates a new observation/scenario.
Run, review and release	Artifacts / relevant domain review	Retain exact inputs, versions, failures and outputs. Review changes assessment; it does not rewrite the numeric basis of a historical run.

OVERLAP CODE	MEANING / EXAMPLE	OVERLAP CODE	MEANING / EXAMPLE
COMP	Adds a distinct function: SFF exchange beside a model.	PART	Shared functionality with different scope: kit and parser.
ALT	Competing owner: agent runtimes or campaign engines.	DEP	Inherited implementation: ThermoSTEAM under BioSTEAM.
VIEW	Derived presentation/index over canonical records.	SCOPE	Specialist replacement of a declared physical operation.

Note: One typed command boundary serves UI, Postdoc and adapters. Deterministic services resolve numerical inputs, validate and execute case matrices. SFF is process data; MCP is a tool interface; retrieval returns evidence. None supplies the others' scientific semantics or makes every simulator interchangeable.

The first month: build the missing path

Source PDF page 5; CURRENT ASSEMBLY.

Use the substantial code that exists. The active app program is glucose/yeast ethanol. This plan adds a parameterized cellulosic-feedstock path.

EXISTING BASELINE	NEW WORK
React/TypeScript/Vite + FastAPI; bounded Postdoc loop and typed registry.	Profile/scenario contracts and inspect/run-study tools using saved IDs and the existing write path.
Anchored evidence/review/retrieval; hashed artifacts, replay, JSON stores and audit.	Feedstock revision, wet/dry basis, chemical mappings, selected assumptions and a saved case manifest.
Isolated BioSTEAM worker; complete corn-stover reference; TEA/numerical machinery.	Separate screening adapter: apply profile -> simulate -> verify read-back/status -> compare/report.

WEEK	DELIVERABLE	PASS GATE
1	Run the pinned reference and prove the new input reaches the model.	A -> B -> A catches stale output and state leakage; reference endpoint stays intact.
2	Persist profile/scenario revisions, sources, assumptions and outputs.	A restart recovers the same resolved basis and artifact links from saved IDs.
3	Expose a bounded deterministic case matrix to Postdoc.	Actual tools return successful/failed receipts and citable results; Python runs the numerical trials.
	Show profile, comparison, status and	A new valid profile yields an inspectable result;

4	click-through evidence.	incompatible input gives a useful gap/refusal.

Scientific claim: month one is conditional scenario screening. The default pretreatment unit uses specified conversions; temperature/time changes do not themselves provide a feedstock-specific response law. Operating optimization requires an applicable model. A high assumed conversion is not an optimized operating condition. Keep the glucose-yeast case distinct from co-fermentation assumptions in the cellulosic reference.

Note: Dated verification: the 25 Oct audit reported 162 targeted tests and a successful frontend build, with two retained package regression differences in equipment cost/electricity. Pins: BioSTEAM 2.54.0; biorefineries 2.35.1; ThermoSTEAM 0.54.2. Source commit d19224f was seen again 27 Oct; tests and remote runtime data were not re-audited for this PDF. Plan assumes a capable builder, AI support and periodic process-model review.

Component catalog

The following 279 dossiers preserve the PDF's domain order, domain ranks, displayed F/M/V scores, functions, surfaces, rationales, actions, overlap classifications, peer references, and source URLs. NR means not rated, not zero. "Revised/new scores" corresponds to the star beside the PDF entry ID. Historical extraction fields not displayed in the PDF (including old scores and incompatible legacy lines-of-code estimates) are excluded.

Type codes: CAP = project capability, not third-party software; STD = standard; DATA = data source; TOOL = software tool/library; RESEARCH = research project; PLATFORM = platform/service; PAPER = research paper.

Action codes: KEEP, BUILD, EVAL, GROW, ALT, DEP, LATER, EXT, and READ use the definitions in the ranking rubric. Overlap codes COMP, PART, ALT, DEP, VIEW, and SCOPE use the definitions in the information-flow section.

Shared foundation

Keep the working service and stores. Extend scientific contracts, applicability and recovery; introduce shared infrastructure for demonstrated needs.

C122

Material and quantity contract

- **Domain rank:** 1.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = NR; V (added integration value) = 9.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Project code / scientific service.

- **Function:** Versioned scientific identity, composition, quantities and conditions.
- **Interface / surface:** Shared domain API.
- **Why this rank:** High V: biomass inputs still need wet/dry basis, sample identity and chemical mappings. Extend the shared contract across Intake, BioRepo and fermOS; a unit converter alone cannot supply scientific meaning.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 6.

C174

PostgreSQL

- **Domain rank:** 10.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 10/10; V (added integration value) = 5.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** C / SQL.
- **Function:** Store canonical evidence records, identities, versions and review state.
- **Interface / surface:** SQL / drivers / admin tools.
- **Why this rank:** High practical standing and strong platform fit, but the current file stores already persist studies. Migration value is conditional on Sci RAG Kit adoption, concurrency and operations; it is not a month-one prerequisite.
- **Functional overlap:** COMP.
- **Primary source:** https://www.postgresql.org/docs/
- **Source PDF page:** 7.

C071

Evidence & uncertainty

- **Domain rank:** 2.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = NR; V (added integration value) = 6.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Separate retrieved facts, predictions, proposals and experimental results.
- **Interface / surface:** BioRepo / Runbooks connection.
- **Why this rank:** The app already records evidence origins. Added work is a consistent distinction

among observations, assumptions, predictions and decisions, with applicability and uncertainty retained; this is not another universal truth score.

- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 6.

C101

JSON Schema

- **Domain rank:** 3.
- **Type:** STD.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = 9.5/10; V (added integration value) = 4/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Schema standard.
- **Function:** Specify portable structural contracts for records and tool inputs/outputs.
- **Interface / surface:** Standard; validator libraries.
- **Why this rank:** Portable exchange contracts complement Pydantic C184 and frontend checks. V concerns governed schema evolution and adapter compatibility, rather than a second independent set of scientific rules.
- **Functional overlap:** PART.
- **Related components / overlap detail:** C184 / C266: structural overlap, different runtime roles.
- **Primary source:** https://json-schema.org/
- **Source PDF page:** 6.

C184

Pydantic

- **Domain rank:** 4.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = 9.5/10; V (added integration value) = 0/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python / Rust core.
- **Function:** Define typed extraction/API records and validate their fields and shapes.
- **Interface / surface:** Python API.
- **Why this rank:** Already supplies typed backend contracts in the audited app. Keep it; new scientific fields belong to the profile/scenario work. Structural validation complements JSON Schema C101 and does not establish biological correctness.

- **Functional overlap:** PART.
- **Related components / overlap detail:** C101 / C266: shared constraints, not parallel truth.
- **Primary source:** https://github.com/pydantic/pydantic
- **Source PDF page:** 6.

C075

FastAPI

- **Domain rank:** 12.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 9/10; V (added integration value) = 0/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python.
- **Function:** Expose typed scientific commands, object APIs, validation errors and job subscriptions.
- **Interface / surface:** Python service / OpenAPI / HTTP.
- **Why this rank:** The existing service already supplies this HTTP boundary. Retain it for UI and agent commands; implement new endpoints and validation in the same service instead of introducing a competing API stack.
- **Functional overlap:** COMP.
- **Primary source:** https://fastapi.tiangolo.com/features/
- **Source PDF page:** 7.

C097

Integration receipt and outbox

- **Domain rank:** 9.
- **Type:** CAP.
- **Action:** GROW.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 6/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Stable commands, reliable events and external operation reconciliation.
- **Interface / surface:** Shared service / DB pattern.
- **Why this rank:** Artifacts and audit receipts already exist. Reliable event delivery and cross-service reconciliation are additional needs as operations grow; deduplicate stable operation IDs and reconcile uncertain external effects before retrying.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.

- **Source PDF page:** 7.

C127

Model qualification registry

- **Domain rank:** 5.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 8.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Project code / scientific service.
- **Function:** Model calibration, validation scope and reproducibility records.
- **Interface / surface:** Review UI + API.
- **Why this rank:** High V because successful execution does not establish scientific applicability. Add model/version, calibration-domain, evidence and check records; kinetics work C106 produces scientific results that this registry describes.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 6.

C157

OpenTelemetry

- **Domain rank:** 6.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 9.5/10; V (added integration value) = 6.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Cross-language standard.
- **Function:** Emit vendor-neutral traces, metrics and logs across agents, tools and services.
- **Interface / surface:** SDKs / OTLP / collectors.
- **Why this rank:** Shared trace propagation would connect app, agent and workers. Complement one chosen backend such as Langfuse C112; telemetry links to scientific artifacts and does not replace their content or retention.
- **Functional overlap:** COMP.
- **Primary source:** https://opentelemetry.io/docs/
- **Source PDF page:** 6.

C169

Pint

- **Domain rank:** 7.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 9/10; V (added integration value) = 6.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python.
- **Function:** Represent quantities, check dimensions and convert compatible units.
- **Interface / surface:** Python API.
- **Why this rank:** Strong fit for dimensional checks and conversions. Adopt behind the existing quantity contract if it reduces errors; wet/dry basis, sample, analytical method and scientific comparability still require domain fields.
- **Functional overlap:** COMP.
- **Primary source:** https://pint.readthedocs.io/
- **Source PDF page:** 6.

C171

Platform operations and recovery

- **Domain rank:** 8.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 6.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Shared deployment, migration, backup, retention and recovery requirements.
- **Interface / surface:** Operations console / procedures.
- **Why this rank:** The immediate gap is demonstrated backup, restart and replay of a coherent study. Larger migration, secrets, quota and incident consoles follow actual deployment needs; they do not require Kubernetes C107 now.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 7.

C244

Temporal

- **Domain rank:** 23.
- **Type:** TOOL.

- **Action:** ALT.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 9/10; V (added integration value) = 3/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Go service / SDKs.
- **Function:** Durable, long-running workflows with retries, waits and human approval signals.
- **Interface / surface:** Service / language SDKs / web UI.
- **Why this rank:** Mature candidate for later long-lived campaigns and approvals, but the current worker path owns initial studies. Introduce one durable owner only when that path is insufficient; activities and external effects need replay-safe handling.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C175 / C048 / current worker: one campaign owner.
- **Primary source:** https://docs.temporal.io/
- **Source PDF page:** 9.

C167

pgvector

- **Domain rank:** 18.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 5.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** C.
- **Function:** Keep embeddings beside relational evidence and run vector similarity searches.
- **Interface / surface:** PostgreSQL extension / SQL.
- **Why this rank:** Semantic retrieval beside relational evidence, and a dependency of Sci RAG Kit C269. Its value depends on the chosen retrieval path; test filtered recall and preserve rebuildable indexes rather than another evidence authority.
- **Functional overlap:** DEP.
- **Primary source:** https://github.com/pgvector/pgvector
- **Source PDF page:** 8.

C083

Git

- **Domain rank:** 13.
- **Type:** TOOL.
- **Action:** KEEP.

- **Scores:** F (project fit) = 9/10; M (practical maturity) = 10/10; V (added integration value) = 0/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Version-control system.
- **Function:** Version runbook schemas, templates, adapters and generated artifacts; optionally sign releases.
- **Interface / surface:** CLI / hosting UIs / APIs.
- **Why this rank:** Code versioning is already part of the repository workflow. Retain Git for code and templates; scientific approval, historical input snapshots and proof of public dissemination are separate records.
- **Functional overlap:** COMP.
- **Primary source:** https://git-scm.com/docs/gitformat-signature
- **Source PDF page:** 7.

C223

SciPy

- **Domain rank:** 11.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 10/10; V (added integration value) = 2/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python / compiled code.
- **Function:** Numerical integration, parameter fitting, optimization and scientific computing.
- **Interface / surface:** Python API.
- **Why this rank:** Already supports numerical work in the current application. Further value lies in qualified fitting and optimization jobs, not installing a second math layer; equations and parameter identifiability remain domain responsibilities.
- **Functional overlap:** COMP.
- **Primary source:** https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve_ivp.html
- **Source PDF page:** 7.

C105

Keycloak

- **Domain rank:** 21.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 9.5/10; V (added integration value) = 5.5/10.

- **Revised/new scores:** Yes.
- **Language / implementation:** Java.
- **Function:** Provide identity, OIDC/OAuth, groups, service accounts and delegated access.
- **Interface / surface:** Web UI / protocols / API.
- **Why this rank:** Strong identity/federation candidate as the reference instance grows. Authentication, project access and competence-based sign-off are distinct; compose one decision contract with any policy tools rather than parallel permission systems.
- **Functional overlap:** PART.
- **Related components / overlap detail:** C143 / C151: authentication and policy overlap partly.
- **Primary source:** https://www.keycloak.org/documentation
- **Source PDF page:** 8.

C008

Apache Parquet

- **Domain rank:** 15.
- **Type:** STD.
- **Action:** GROW.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 9.5/10; V (added integration value) = 7.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Columnar format.
- **Function:** Persist columnar tables, measurements and simulation-result datasets.
- **Interface / surface:** File format; compatible tools.
- **Why this rank:** Versioned columnar datasets and analytical exports. Preserve schema and quantity semantics; file format is not an application database or review ledger.
- **Functional overlap:** COMP.
- **Primary source:** https://parquet.apache.org/
- **Source PDF page:** 8.

C260

W3C PROV

- **Domain rank:** 16.
- **Type:** STD.
- **Action:** GROW.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 9/10; V (added integration value) = 7.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Provenance standard.
- **Function:** Describe entities, activities, agents and derivations in provenance records.

- **Interface / surface:** Standard; compatible tools.
- **Why this rank:** Common provenance vocabulary. Map entities, activities and agents to canonical records/exports; a relational implementation is valid.
- **Functional overlap:** COMP.
- **Primary source:** https://www.w3.org/TR/prov-overview/
- **Source PDF page:** 8.

C163

Pandera

- **Domain rank:** 19.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8.5/10; V (added integration value) = 7/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Validate extracted tables and measurement datasets against declared rules.
- **Interface / surface:** Python API / CLI.
- **Why this rank:** Validate declared tabular properties and datasets. Complement to object contracts; successful validation does not qualify an assay or extraction interpretation.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C184: table rules complement object contracts.
- **Primary source:** https://pandera.readthedocs.io/
- **Source PDF page:** 8.

C215

SALib

- **Domain rank:** 20.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8.5/10; V (added integration value) = 7/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Screen uncertain inputs and quantify global sensitivity of model outputs.
- **Interface / surface:** Python API / CLI.
- **Why this rank:** Sensitivity methods over the existing run service. Choose assumptions appropriate to dependent inputs; a sensitivity index is not a model-validity score.
- **Functional overlap:** COMP.

- **Primary source:** https://salib.readthedocs.io/
- **Source PDF page:** 8.

C162

pandas

- **Domain rank:** 17.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 10/10; V (added integration value) = 6.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / Cython.
- **Function:** Align dated prices, production schedules, assumptions and cash-flow tables.
- **Interface / surface:** Python / notebooks.
- **Why this rank:** Align dated assumptions and calculation tables. Enforce schema, units, vintages and explicit missing-value handling around dataframes.
- **Functional overlap:** COMP.
- **Primary source:** https://pandas.pydata.org/docs/
- **Source PDF page:** 8.

C057

DuckDB

- **Domain rank:** 24.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 9/10; V (added integration value) = 5.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** C++ / SQL.
- **Function:** Analyze local or stored tables, including Parquet files, using SQL.
- **Interface / surface:** SQL / CLI / language APIs.
- **Why this rank:** Analytical queries over versioned snapshots/Parquet. Keep transactional scientific and economic edits in canonical services.
- **Functional overlap:** COMP.
- **Primary source:** https://duckdb.org/
- **Source PDF page:** 9.

C148

OpenBao

- **Domain rank:** 25.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 7.5/10; V (added integration value) = 5.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Go.
- **Function:** Shared secrets, certificates and credential leasing.
- **Interface / surface:** Service / CLI / HTTP API.
- **Why this rank:** Open-source secrets-management candidate. Evaluate required auth/secret engines and operations against the actual deployment; do not assume perfect Vault feature parity.
- **Functional overlap:** ALT.
- **Primary source:** https://openbao.org/docs/
- **Source PDF page:** 9.

C091

HiGHS

- **Domain rank:** 22.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 9/10; V (added integration value) = 5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Mostly C++.
- **Function:** Solve linear, mixed-integer linear and convex quadratic optimization.
- **Interface / surface:** CLI / APIs.
- **Why this rank:** LP/MILP/convex-QP solver for supported formulations. Test actual calling-package compatibility; not every biological optimization adapter supports every solver.
- **Functional overlap:** COMP.
- **Primary source:** https://highs.dev/
- **Source PDF page:** 9.

C191

Pyomo

- **Domain rank:** 26.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 9/10; V (added integration value) = 5/10.
- **Revised/new scores:** No.

- **Language / implementation:** Python.
- **Function:** Formulate constrained process optimization and discretized dynamic models.
- **Interface / surface:** Python API.
- **Why this rank:** Formulate constrained decisions and equations. Modeler, process physics and solver are different roles; reuse across design, economics and scheduling.
- **Functional overlap:** COMP.
- **Primary source:** https://www.pyomo.org/
- **Source PDF page:** 9.

C115

LinkML

- **Domain rank:** 27.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 8/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python ecosystem.
- **Function:** Define shared schemas and generate representations, validators and mappings.
- **Interface / surface:** YAML schemas / CLI / APIs.
- **Why this rank:** Schema authoring and cross-format generation. Select as the authoring source only if its generators meet contract needs; avoid independent LinkML/Pydantic/Zod definitions.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C101 / C184: choose one schema authoring source.
- **Primary source:** https://linkml.io/
- **Source PDF page:** 9.

C143

Open Policy Agent

- **Domain rank:** 30.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9/10; V (added integration value) = 4/10.
- **Revised/new scores:** No.
- **Language / implementation:** Go / Rego.
- **Function:** Evaluate centrally defined authorization and execution policies as code.
- **Interface / surface:** REST / sidecar / libraries.
- **Why this rank:** General policy decisions for complex contextual rules. Keycloak and OpenFGA also

cover authorization cases; define one composed decision contract rather than three disagreeing allow/deny owners.

- **Functional overlap:** ALT.
- **Related components / overlap detail:** C105 / C151: compose one permission decision.
- **Primary source:** https://www.openpolicyagent.org/docs/latest/
- **Source PDF page:** 10.

C225

SeaweedFS

- **Domain rank:** 33.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 4/10.
- **Revised/new scores:** No.
- **Language / implementation:** Go.
- **Function:** Self-hosted object/file storage for originals, images and large artifacts.
- **Interface / surface:** S3 API / CLI / admin UI.
- **Why this rank:** One possible shared object-store implementation behind an S3-compatible contract. Blob storage is essential; this specific deployment is conditional.
- **Functional overlap:** ALT.
- **Primary source:** https://github.com/seaweedfs/seaweedfs
- **Source PDF page:** 10.

C175

Prefect

- **Domain rank:** 28.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python.
- **Function:** Schedule ingestion flows, retry failed steps and observe processing runs.
- **Interface / surface:** Python / CLI / web UI.
- **Why this rank:** Python workflow alternative to the current job owner or a later Temporal C244 choice. Compare retries, waiting, human input and operating burden on the same cases; do not run two campaign owners.
- **Functional overlap:** ALT.

- **Related components / overlap detail:** C244 / C048 / current worker: alternatives.
- **Primary source:** https://docs.prefect.io/
- **Source PDF page:** 9.

C156

OpenSearch

- **Domain rank:** 31.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Java ecosystem.
- **Function:** Full-text, vector and hybrid search over larger document collections.
- **Interface / surface:** REST API / Dashboards.
- **Why this rank:** Dedicated lexical/hybrid search candidate. Added V is limited until current retrieval or a selected Sci RAG Kit C269 path demonstrates a search/scale gap; retain one canonical evidence identity.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C167 / C194 / C269: search-substrate overlap.
- **Primary source:** https://docs.opensearch.org/
- **Source PDF page:** 10.

C151

OpenFGA

- **Domain rank:** 34.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 3.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Go.
- **Function:** Model fine-grained user, group, project and object relationships for permission checks.
- **Interface / surface:** Authorization service / SDKs.
- **Why this rank:** Relationship-based authorization if projects, groups and objects justify it. Keep decision context, model version and revocation behavior explicit.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C105 / C143: relationship-policy specialization.

- **Primary source:** https://openfga.dev/docs/concepts
- **Source PDF page:** 10.

C194

Qdrant

- **Domain rank:** 29.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 2/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Rust.
- **Function:** Alternative vector-search service with payload metadata and filtering.
- **Interface / surface:** REST / gRPC / web dashboard.
- **Why this rank:** Dedicated vector-store alternative. A separate service adds operations and synchronization beside existing retrieval or pgvector C167; raise V only for measured recall, filtering, scale or deployment needs.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C167 / C156: vector-search alternative.
- **Primary source:** https://qdrant.tech/documentation/
- **Source PDF page:** 10.

C048

Dagster

- **Domain rank:** 35.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 3/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python ecosystem.
- **Function:** Orchestrate data assets, dependencies, checks and reprocessing workflows.
- **Interface / surface:** Python / CLI / web UI.
- **Why this rank:** Asset-oriented data orchestration, lineage and backfills. Add only if ingestion asset management warrants a separate plane; reuse scientific evidence IDs.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C244 / C175: workflow ownership alternatives.
- **Primary source:** https://docs.dagster.io/
- **Source PDF page:** 10.

C107

Kubernetes + gVisor

- **Domain rank:** 36.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 8/10; V (added integration value) = 3/10.
- **Revised/new scores:** No.
- **Language / implementation:** Go / kernel sandbox.
- **Function:** Isolate untrusted code and resource-bounded jobs in controlled sandboxes.
- **Interface / surface:** Cluster APIs / CLI.
- **Why this rank:** Catalog bundle hides two decisions: scheduling platform and sandbox runtime. gVisor also integrates without Kubernetes; test compatibility and enforce resource, filesystem and egress limits.
- **Functional overlap:** COMP.
- **Primary source:** https://gvisor.dev/docs/
- **Source PDF page:** 11.

C201

RDFLib

- **Domain rank:** 37.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 9/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Read, write and query RDF graphs for linked scientific identities and relations.
- **Interface / surface:** Python API.
- **Why this rank:** RDF handling where LabOP, SBOL or semantic exports require it. Does not imply deploying a graph database for every scientific record.
- **Functional overlap:** DEP.
- **Primary source:** https://rdflib.readthedocs.io/
- **Source PDF page:** 11.

C207

Redis

- **Domain rank:** 38.

- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 9/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** C.
- **Function:** Hold short-lived caches, rate limits, queues and coordination state.
- **Interface / surface:** RESP / client libraries.
- **Why this rank:** Cache/rate-limit or transient coordination when required. Do not place the sole plan, approval or execution record in a disposable cache.
- **Functional overlap:** COMP.
- **Primary source:** https://redis.io/docs/latest/
- **Source PDF page:** 11.

C099

IPOPT

- **Domain rank:** 32.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** Mostly C++.
- **Function:** Solve continuous nonlinear optimization problems; typically via a modeler.
- **Interface / surface:** Solver library / interfaces.
- **Why this rank:** Continuous nonlinear solver, generally local optimum. Inherit through the chosen modeler; does not solve discrete schedules or supply missing physics.
- **Functional overlap:** DEP.
- **Primary source:** https://coin-or.github.io/Ipopt/
- **Source PDF page:** 10.

C256

Vault

- **Domain rank:** 39.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 9.5/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** Go.

- **Function:** Issue and rotate scoped secrets without placing credentials in prompts or state.
- **Interface / surface:** CLI / HTTP API / web UI.
- **Why this rank:** Mature secrets manager; current releases use BSL terms, not a blanket permissive-open-source claim. Compare OpenBao or an existing host secret service.
- **Functional overlap:** ALT.
- **Primary source:** https://developer.hashicorp.com/vault/docs
- **Source PDF page:** 11.

C136

Neo4j Community

- **Domain rank:** 41.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 4.5/10; M (practical maturity) = 9/10; V (added integration value) = 1/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Java ecosystem.
- **Function:** Optional graph database for traversing scientific entities and relationships.
- **Interface / surface:** Cypher / APIs / Browser.
- **Why this rank:** Specialist property-graph store, not a prerequisite for provenance. Linked records or the Sci RAG Kit graph may cover initial traversal; adopt only for a demonstrated graph-query requirement and keep derived identities aligned.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C269 / linked records: graph specialization.
- **Primary source:** https://neo4j.com/docs/
- **Source PDF page:** 11.

C037

Celery

- **Domain rank:** 40.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 4.5/10; M (practical maturity) = 9/10; V (added integration value) = 1/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Distribute ordinary background jobs through queues and worker processes.
- **Interface / surface:** Python API / brokers.
- **Why this rank:** Mature task queue, but overlaps selected durable execution. Avoid nesting

independent retry owners for the same external operation.

- **Functional overlap:** ALT.
- **Related components / overlap detail:** Current worker / C244: avoid duplicate retry owners.
- **Primary source:** https://docs.celeryq.dev/
- **Source PDF page:** 11.

C138

NumPy

- **Domain rank:** 14.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 10/10; V (added integration value) = 0/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / compiled core.
- **Function:** Vectorized numerical calculations and reproducible random sampling for scenario evaluation.
- **Interface / surface:** Python API.
- **Why this rank:** Essential numerical array dependency already inherited by the scientific stack. V=0 means no independent integration project; it does not mean this dependency can be removed.
- **Functional overlap:** DEP.
- **Primary source:** https://numpy.org/doc/stable/
- **Source PDF page:** 7.

Intake and BioRepo

A single evidence identity spans acquisition, parsing, retrieval, review and applied quantities. Evaluate the assembled Sci RAG Kit before duplicating its subsystem.

C072

Evidence ledger

- **Domain rank:** 3.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = NR; V (added integration value) = 4.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Preserve revisions, review decisions and links to literature, simulations and tests.
- **Interface / surface:** BioRepo / review interface.

- **Why this rank:** A file-backed evidence/audit trail already exists. Extend its revision and review semantics for profile-to-run continuity; a new ledger implementation is less valuable than closing the current traceability gaps.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 12.

C096

Identity & applicability

- **Domain rank:** 1.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = NR; V (added integration value) = 8.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Resolve entities and check measurement basis, context and model applicability.
- **Interface / surface:** Shared schema / validation API.
- **Why this rank:** High V: input identity and applicability are central remaining gaps. Distinguish material specification, sample, organism, conditions and model domain; retrieved unit-compatible values are not automatically transferable.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 12.

C002

Accession builder

- **Domain rank:** 7.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 5.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Create candidate values with units, conditions, source spans and evidence type.
- **Interface / surface:** Intake review interface.
- **Why this rank:** Anchored extraction exists. Extend it to record original and normalized quantities, conditions and provenance for biomass inputs; scientific interpretation and review remain explicit rather than inferred from a parser score.

- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 12.

C055

Docling

- **Domain rank:** 8.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 8/10; V (added integration value) = 5.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python.
- **Function:** Convert documents into structured content with layout, tables and provenance.
- **Interface / surface:** Python / CLI; service options.
- **Why this rank:** Structured conversion is relevant, but ingestion already exists and Sci RAG Kit C269 can reuse Docling. Qualify representative tables/scans before replacing the current parser; avoid duplicate independent ingestion identities.
- **Functional overlap:** PART.
- **Related components / overlap detail:** C269 may reuse it; C161/C245 are routed alternatives.
- **Primary source:** https://docling-project.github.io/docling/concepts/docling_document/
- **Source PDF page:** 13.

C224

Search & tool adapters

- **Domain rank:** 6.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 6.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Build retrieval views and map reviewed records into versioned simulator inputs.
- **Interface / surface:** Postdoc / module APIs.
- **Why this rank:** Existing retrieval/tools need profile, study and optional kit mappings. High V is the common ID and receipt boundary, not a second search corpus; derived indexes must resolve BioRepo evidence revisions.
- **Functional overlap:** VIEW.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.

- **Source PDF page:** 12.

C068

Europe PMC

- **Domain rank:** 4.
- **Type:** DATA.
- **Action:** EVAL.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 9/10; V (added integration value) = 8/10.
- **Revised/new scores:** No.
- **Language / implementation:** Literature service.
- **Function:** Find life-science literature and retrieve available metadata and full text.
- **Interface / surface:** REST API / website.
- **Why this rank:** Life-science discovery, identifiers and available full text. Respect article-level availability and rights; match source version before extraction.
- **Functional overlap:** COMP.
- **Primary source:** https://europepmc.org/RestfulWebService
- **Source PDF page:** 12.

C165

PDF.js

- **Domain rank:** 5.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 9.5/10; V (added integration value) = 8/10.
- **Revised/new scores:** No.
- **Language / implementation:** JavaScript.
- **Function:** Embed a PDF viewer; add source-coordinate overlays in your review interface.
- **Interface / surface:** Embeddable browser viewer.
- **Why this rank:** Browser source inspection and highlights. Share original bytes, page geometry and extraction anchors; rendering does not validate interpretation.
- **Functional overlap:** VIEW.
- **Primary source:** https://mozilla.github.io/pdf.js/
- **Source PDF page:** 12.

C226

Sentence Transformers

- **Domain rank:** 10.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 9/10; V (added integration value) = 3.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python / model weights.
- **Function:** Compute embeddings and rerank retrieved candidates with suitable models.
- **Interface / surface:** Python API.
- **Why this rank:** Alternative embedding stack beside the current FastEmbed C276 path. Compare a real retrieval gap before adding it; Sci RAG Kit C269 has another embedding default, and model/dimension changes require an explicit re-indexing migration.
- **Functional overlap:** COMP.
- **Primary source:** https://sbert.net/
- **Source PDF page:** 13.

C045

Crossref

- **Domain rank:** 9.
- **Type:** DATA.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 10/10; V (added integration value) = 7/10.
- **Revised/new scores:** No.
- **Language / implementation:** Metadata service.
- **Function:** Resolve DOI metadata and reconcile bibliographic source identities.
- **Interface / surface:** REST API / website.
- **Why this rank:** Publication metadata and identifiers. Cache one shared identity service, retain updates and conflicting metadata; a DOI record is not a scientific result.
- **Functional overlap:** COMP.
- **Primary source:** https://www.crossref.org/documentation/retrieve-metadata/rest-api/
- **Source PDF page:** 13.

C190

PyMuPDF

- **Domain rank:** 15.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 9.5/10; V (added integration value) = 3.5/10.

- **Revised/new scores:** Yes.
- **Language / implementation:** Python / MuPDF.
- **Function:** Render PDFs and extract text, geometry, images and document information.
- **Interface / surface:** Python API.
- **Why this rank:** PDF geometry/rendering primitives beside the existing pypdf C277 route. Adopt for a specific extraction/anchor gap, with compatible AGPL/commercial terms; a second general converter is not an automatic quality gain.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C166 / C055: partial conversion overlap.
- **Primary source:** https://pymupdf.readthedocs.io/
- **Source PDF page:** 14.

C085

GROBID

- **Domain rank:** 14.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 9/10; V (added integration value) = 5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Java.
- **Function:** Parse scientific papers into structured TEI with references and PDF coordinates.
- **Interface / surface:** REST API / web console.
- **Why this rank:** Bibliographic/TEI structure and coordinates, partly overlapping Docling. Route targeted difficult sources; parser agreement is not independent scientific corroboration.
- **Functional overlap:** COMP.
- **Primary source:** https://grobid.readthedocs.io/
- **Source PDF page:** 13.

C089

Haystack

- **Domain rank:** 17.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 2/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python.
- **Function:** Compose document ingestion, retrieval, ranking and answer-generation pipelines.

- **Interface / surface:** Python / service deployment.
- **Why this rank:** Flexible retrieval-framework alternative. Compare a minimal Haystack assembly with the working path and Sci RAG Kit C269; framework flexibility has limited added V if assembled ingestion/evaluation already meets the need.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C269 / C118: retrieval assembly alternatives.
- **Primary source:** https://docs.haystack.deepset.ai/
- **Source PDF page:** 14.

C262

WebPlotDigitizer

- **Domain rank:** 22.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** JavaScript.
- **Function:** Recover numerical series from plot images with axis calibration.
- **Interface / surface:** Web / desktop interface.
- **Why this rank:** Calibrated figure digitization. Keep axes, original image, settings, series and review; digitized points do not create biological replicates.
- **Functional overlap:** COMP.
- **Primary source:** https://automeris.io/
- **Source PDF page:** 15.

C161

PaddleOCR

- **Domain rank:** 16.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 4/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / PaddlePaddle.
- **Function:** OCR and document-layout/table recognition using model-based pipelines.
- **Interface / surface:** Python / CLI / services.
- **Why this rank:** OCR and broader document/table pipelines. Compare against the chosen conversion path; it overlaps Docling beyond simple character recognition.

- **Functional overlap:** ALT.
- **Related components / overlap detail:** C055 / C245: OCR/document pipeline overlap.
- **Primary source:** https://github.com/PaddlePaddle/PaddleOCR
- **Source PDF page:** 14.

C031

Camelot

- **Domain rank:** 18.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 8/10; V (added integration value) = 4/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Extract tabular data from PDFs using supported table-extraction methods.
- **Interface / surface:** Python / CLI.
- **Why this rank:** Table-focused fallback. Current repository includes optional ML/OCR paths while an overview retains older text-only wording; qualify a pinned parser/version.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C166 / C055: selected table-extraction cases.
- **Primary source:** https://github.com/camelot-dev/camelot
- **Source PDF page:** 14.

C267

Zotero

- **Domain rank:** 19.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9/10; V (added integration value) = 4/10.
- **Revised/new scores:** No.
- **Language / implementation:** JavaScript ecosystem.
- **Function:** Collect papers, attachments, bibliographic records and user-curated collections.
- **Interface / surface:** Desktop / web / API.
- **Why this rank:** Reference-library import and human curation. Share DOI/source identities with Intake; personal citation libraries are not the canonical scientific evidence ledger.
- **Functional overlap:** COMP.
- **Primary source:** https://www.zotero.org/support/dev/web_api/
- **Source PDF page:** 14.

C109

Label Studio

- **Domain rank:** 23.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 4/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / JavaScript.
- **Function:** Annotate and review extraction examples; export labels for evaluation.
- **Interface / surface:** Web UI / API.
- **Why this rank:** Build labeled extraction/retrieval evaluation corpora. Map annotations into reviewed evidence; labeling UI history is not the scientific approval ledger.
- **Functional overlap:** COMP.
- **Primary source:** https://labelstud.io/
- **Source PDF page:** 15.

C146

OpenAlex

- **Domain rank:** 24.
- **Type:** DATA.
- **Action:** EVAL.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 4/10.
- **Revised/new scores:** No.
- **Language / implementation:** Scholarly data service.
- **Function:** Discover scholarly works, authors, institutions and citation relationships.
- **Interface / surface:** REST API / website.
- **Why this rank:** Broader discovery graph complements sources selectively. Curated IDs may merge or change; retain external IDs and retrieval snapshots.
- **Functional overlap:** ALT.
- **Primary source:** https://help.openalex.org/api/
- **Source PDF page:** 15.

C118

LlamaIndex

- **Domain rank:** 21.
- **Type:** TOOL.

- **Action:** ALT.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 8/10; V (added integration value) = 2/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python / TypeScript.
- **Function:** Build ingestion, indexing and retrieval workflows with metadata and connectors.
- **Interface / surface:** Python / TypeScript APIs.
- **Why this rank:** Retrieval/indexing framework alternative with many connectors. Compare against current retrieval and Sci RAG Kit C269; keep scenario authority, evidence IDs and agent execution outside connector-specific private stores.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C269 / C089: retrieval assembly alternatives.
- **Primary source:** https://github.com/run-llama/llama_index
- **Source PDF page:** 14.

C245

Tesseract

- **Domain rank:** 20.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9.5/10; V (added integration value) = 3/10.
- **Revised/new scores:** No.
- **Language / implementation:** C++.
- **Function:** Recognize text in scanned pages and images; an OCR engine option.
- **Interface / surface:** CLI / library.
- **Why this rank:** Established OCR candidate for the single qualified scan route. Preserve OCR configuration and image anchors; do not average competing OCR text automatically.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C161 / C055: routed OCR fallback.
- **Primary source:** https://github.com/tesseract-ocr/tesseract
- **Source PDF page:** 14.

C166

pdfplumber

- **Domain rank:** 25.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 3/10.

- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Inspect text coordinates and extract tables with visual-debugging support.
- **Interface / surface:** Python API.
- **Why this rank:** Targeted machine-generated PDF and table inspection. Overlaps other converters; use as a fallback with retained extraction revision.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C031 / C190: PDF primitives and table overlap.
- **Primary source:** https://github.com/jsvine/pdfplumber
- **Source PDF page:** 15.

C155

OpenRefine

- **Domain rank:** 27.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 9/10; V (added integration value) = 3/10.
- **Revised/new scores:** No.
- **Language / implementation:** Java / JavaScript.
- **Function:** Clean inconsistent tables, cluster names and reconcile against external services.
- **Interface / surface:** Local web UI.
- **Why this rank:** Expert data cleaning and reconciliation. Retain transformations and original cells; export proposals rather than silently altering reviewed records.
- **Functional overlap:** COMP.
- **Primary source:** https://openrefine.org/
- **Source PDF page:** 15.

C111

LangChain

- **Domain rank:** 28.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 8.5/10; V (added integration value) = 1/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python / TypeScript.
- **Function:** Connect loaders, retrievers and models; broader agent work belongs to Postdoc.
- **Interface / surface:** Python / TypeScript APIs.

- **Why this rank:** Broad orchestration/retrieval library with substantial overlap. The app already has an agent loop and tool registry; adopt a specific useful component rather than another parallel application framework.
- **Functional overlap:** ALT.
- **Primary source:** https://github.com/langchain-ai/langchain
- **Source PDF page:** 15.

C269

Sci RAG Kit

- **Domain rank:** 2.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = 5/10; V (added integration value) = 7.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python / PostgreSQL.
- **Function:** Assemble scientific ingestion, Postgres retrieval, cited answers, evaluations, corpus campaigns and serving.
- **Interface / surface:** Template / CLI / REST / MCP.
- **Why this rank:** High potential reuse across Intake and search, lowering duplicate assembly work. Alpha source needs same-corpus testing, evidence-ID mapping and operating-cost checks; it does not own scenario locks, model qualification or review.
- **Functional overlap:** PART.
- **Related components / overlap detail:** C055/C089/C118/C226: assembled, partly overlapping.
- **Primary source:** https://github.com/sustainability-software-lab/sci-rag-kit/tree/26b3990c6264b68f29e833e22b04eb323e06e11b
- **Source PDF page:** 12.

C274

AutoSynthesis - meta-analysis paper

- **Domain rank:** 26.
- **Type:** PAPER.
- **Action:** READ.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = NR; V (added integration value) = NR.
- **Revised/new scores:** Yes.
- **Language / implementation:** Research system.
- **Function:** Study agentic literature search, screening, extraction, effect-size calculation and meta-analysis.

- **Interface / surface:** Research publication / code access unverified.
- **Why this rank:** Useful evidence-synthesis methods, but not established as a reusable software dependency here. Open paper access does not establish code/license availability; do not confuse it with Bhagwat's process-design repository C273.
- **Functional overlap:** COMP.
- **Primary source:** https://www.mcml.ai/publications/tmv%2B26/
- **Source PDF page:** 15.

C276

FastEmbed

- **Domain rank:** 11.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 8/10; V (added integration value) = 0/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python / ONNX Runtime.
- **Function:** Compute local passage embeddings for the existing hybrid-retrieval lane.
- **Interface / surface:** Python embedding library / configured models.
- **Why this rank:** The rag extra pins FastEmbed and the README describes the current local meaning-vector lane. Keep the working path; Sentence Transformers C226 or Sci RAG Kit embeddings require quality/cost comparison and deliberate index migration.
- **Functional overlap:** DEP.
- **Primary source:** https://qdrant.github.io/fastembed/
- **Source PDF page:** 13.

C277

pypdf

- **Domain rank:** 12.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 9/10; V (added integration value) = 0/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python.
- **Function:** Read PDF structure and text layers, including the current report-ingestion path.
- **Interface / surface:** Python API / text-layer PDF processing.
- **Why this rank:** Already a core dependency for report text extraction. Keep the working route; Docling C055, PyMuPDF C190 and OCR tools are qualified extensions for demonstrated layout,

table or scan failures, not parallel default parsers.

- **Functional overlap:** PART.
- **Primary source:** https://pypdf.readthedocs.io/en/stable/
- **Source PDF page:** 13.

C279

rank-bm25

- **Domain rank:** 13.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 8/10; V (added integration value) = 0/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python.
- **Function:** Score lexical passage matches for the existing retrieval pipeline.
- **Interface / surface:** Python API / local ranking.
- **Why this rank:** Already a core dependency of the current lexical/hybrid evidence path. Keep and evaluate it with FastEmbed C276 and reranking; a ranking score never establishes scientific applicability, source quality or approval.
- **Functional overlap:** COMP.
- **Primary source:** https://github.com/dorianbrown/rank_bm25
- **Source PDF page:** 13.

geneOS

High fit here concerns the wider platform. Defer broad new organism work for the ethanol phase; model feasibility, construct design and measured phenotype remain distinct.

C041

COBRApy

- **Domain rank:** 2.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = 9/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python.
- **Function:** Metabolic flux, growth, production, nutrient use and gene-deletion analysis.
- **Interface / surface:** Python API.
- **Why this rank:** The audited prototype already computes flux-balance yield ceilings through

COBRApy. Retain that path; added V is model/intervention extension and qualification, not a fresh library integration. Flux feasibility is not measured productivity or reactor kinetics.

- **Functional overlap:** COMP.
- **Related components / overlap detail:** C032 overlaps intervention analysis; model adapters vary.
- **Primary source:** https://cobrapy.readthedocs.io/en/latest/
- **Source PDF page:** 16.

C229

Shared design records

- **Domain rank:** 1.
- **Type:** CAP.
- **Action:** LATER.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = NR; V (added integration value) = 9/10.
- **Revised/new scores:** No.
- **Language / implementation:** Project code / scientific service.
- **Function:** Connect sequence, protein, reaction, model and proposed strain identities.
- **Interface / surface:** Your geneOS interface.
- **Why this rank:** Version host, strain lineage, sequence, construct, protein, pathway and model as distinct linked objects. One biological identity contract across geneOS and BioRepo.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 16.

C025

Biopython

- **Domain rank:** 3.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = 10/10; V (added integration value) = 2/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python / C.
- **Function:** Read, write, translate, align and manipulate sequences and annotations.
- **Interface / surface:** Python API.
- **Why this rank:** Already a core dependency for protein properties through ProtParam. Retain it; further sequence/cloning work can reuse the same library and OpenCloning C150 rather than another sequence store or installation project.
- **Functional overlap:** COMP.

- **Primary source:** https://biopython.org/docs/latest/Tutorial/
- **Source PDF page:** 16.

C088

Handoff to fermOS

- **Domain rank:** 4.
- **Type:** CAP.
- **Action:** LATER.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 8.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Project code / scientific service.
- **Function:** Pass strain behavior, model constraints and operating assumptions to reactors.
- **Interface / surface:** Internal module connection.
- **Why this rank:** Pass qualified extracellular rates, yields and calibration domains with units/bases. Gene-level feasibility never becomes a reactor kinetic parameter without an explicit mapping.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 16.

C082

GenBank

- **Domain rank:** 5.
- **Type:** STD.
- **Action:** LATER.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 10/10; V (added integration value) = 8/10.
- **Revised/new scores:** No.
- **Language / implementation:** Sequence format.
- **Function:** Exchange annotated biological sequences.
- **Interface / surface:** Through sequence tools.
- **Why this rank:** Annotated sequence exchange. Preserve feature coordinates, strand, topology and accession revision; original files accompany normalized records.
- **Functional overlap:** COMP.
- **Primary source:** https://www.ncbi.nlm.nih.gov/genbank/samplerecord/
- **Source PDF page:** 16.

C124

MEMOTE

- **Domain rank:** 6.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 8.5/10; V (added integration value) = 8/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Check model consistency, annotations and other quality criteria.
- **Interface / surface:** API / CLI + reports.
- **Why this rank:** Qualify metabolic-model structure and annotations. A high test score does not establish growth, secretion or process-level predictive accuracy.
- **Functional overlap:** COMP.
- **Primary source:** https://memote.readthedocs.io/en/latest/
- **Source PDF page:** 16.

C135

NCBI

- **Domain rank:** 7.
- **Type:** DATA.
- **Action:** LATER.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 10/10; V (added integration value) = 8/10.
- **Revised/new scores:** No.
- **Language / implementation:** Data resource.
- **Function:** Retrieve sequence records, annotations and identifiers.
- **Interface / surface:** Website + programmatic access.
- **Why this rank:** Retrieve authoritative source records with accession/version and retrieval time. Shared biological identity adapter; imported annotations retain their evidence status.
- **Functional overlap:** COMP.
- **Primary source:** https://www.ncbi.nlm.nih.gov/books/NBK25501/
- **Source PDF page:** 16.

C216

SBML

- **Domain rank:** 8.
- **Type:** STD.
- **Action:** LATER.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 9.5/10; V (added integration value) = 8/10.
- **Revised/new scores:** No.

- **Language / implementation:** Model standard.
- **Function:** Exchange executable biochemical and metabolic models.
- **Interface / surface:** Through compatible tools.
- **Why this rank:** Exchange model semantics only for supported levels and packages. Validate round trips, units, annotations, flux constraints and executable behavior.
- **Functional overlap:** COMP.
- **Primary source:** https://sbml.org/documents/specifications/
- **Source PDF page:** 17.

C252

UniProt

- **Domain rank:** 9.
- **Type:** DATA.
- **Action:** LATER.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 9.5/10; V (added integration value) = 8/10.
- **Revised/new scores:** No.
- **Language / implementation:** Data resource.
- **Function:** Retrieve protein sequences, identities and functional annotations.
- **Interface / surface:** Website + programmatic access.
- **Why this rank:** Protein identities, sequences and annotations. Keep reviewed status, evidence and release metadata; reconcile cross-references without silently merging unlike entities.
- **Functional overlap:** COMP.
- **Primary source:** https://www.uniprot.org/help/api
- **Source PDF page:** 17.

C150

OpenCloning

- **Domain rank:** 10.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 7.5/10; V (added integration value) = 7.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Web / Python.
- **Function:** Plan and document cloning and genome-engineering workflows.
- **Interface / surface:** Web application.
- **Why this rank:** Evaluate existing cloning history and data model before rebuilding. Uses pydna and links sequence operations; integrate shared identities and revisions.

- **Functional overlap:** COMP.
- **Primary source:** https://github.com/OpenCloning/OpenCloning
- **Source PDF page:** 17.

C052

DNA Chisel

- **Domain rank:** 11.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 8/10; V (added integration value) = 7/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Optimize codon usage, GC content and specified sequence constraints.
- **Interface / surface:** Python / CLI; web app.
- **Why this rank:** Optimize explicit sequence constraints and objectives. Preserve the requested specifications and changes; codon optimization does not prove expression or stability.
- **Functional overlap:** COMP.
- **Primary source:** https://edinburgh-genome-foundry.github.io/DnaChisel/
- **Source PDF page:** 17.

C236

StrainDesign

- **Domain rank:** 12.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 8/10; V (added integration value) = 7/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Search for metabolic interventions and growth-coupled production designs.
- **Interface / surface:** Python API / CNApy.
- **Why this rank:** Builds on COBRApy. Add intervention search through its supported solver interfaces; preserve the target model and design assumptions.
- **Functional overlap:** COMP.
- **Primary source:** https://straindesign.readthedocs.io/en/latest/
- **Source PDF page:** 17.

C066

eQuilibrator

- **Domain rank:** 14.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8.5/10; V (added integration value) = 6.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python ecosystem.
- **Function:** Estimate reaction thermodynamics and pathway driving forces.
- **Interface / surface:** Website + Python API.
- **Why this rank:** Biochemical Gibbs-energy estimates with conditions and uncertainty. Distinct from broth phase-equilibrium property packages; directionality is not kinetic rate.
- **Functional overlap:** COMP.
- **Primary source:** https://equilibrator.readthedocs.io/en/latest/
- **Source PDF page:** 17.

C067

Escher

- **Domain rank:** 15.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8.5/10; V (added integration value) = 6.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** JavaScript / Python.
- **Function:** Explore metabolic maps with simulated or measured reaction fluxes.
- **Interface / surface:** Embeddable web UI.
- **Why this rank:** Metabolic flux-map view. Share reaction IDs and model revisions with COBRApy; the visualization is not a separate calculation authority.
- **Functional overlap:** VIEW.
- **Primary source:** https://escher.readthedocs.io/en/latest/
- **Source PDF page:** 18.

C114

libRoadRunner

- **Domain rank:** 16.
- **Type:** TOOL.
- **Action:** LATER.

- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8.5/10; V (added integration value) = 6.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Mostly C++.
- **Function:** Execute biochemical models expressed in SBML.
- **Interface / surface:** Library / Python API.
- **Why this rank:** Primary candidate for executing supplied SBML kinetic models. Needs model-specific parameters and validation, not inferred kinetics from sequence alone.
- **Functional overlap:** COMP.
- **Primary source:** https://libroadrunner.readthedocs.io/
- **Source PDF page:** 18.

C074

FASTA

- **Domain rank:** 17.
- **Type:** STD.
- **Action:** LATER.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 10/10; V (added integration value) = 6/10.
- **Revised/new scores:** No.
- **Language / implementation:** Sequence format.
- **Function:** Exchange nucleotide or protein sequences.
- **Interface / surface:** Through sequence tools.
- **Why this rank:** Simple sequence exchange. Insufficient alone for full construct annotations, circularity, modification context or cloning history.
- **Functional overlap:** COMP.
- **Primary source:** https://www.ncbi.nlm.nih.gov/genbank/fastaformat/
- **Source PDF page:** 18.

C128

ModelSEED resources

- **Domain rank:** 18.
- **Type:** DATA.
- **Action:** DEP.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8/10; V (added integration value) = 6/10.
- **Revised/new scores:** No.
- **Language / implementation:** Data ecosystem.
- **Function:** Access biochemical reactions, metadata and reconstruction resources.
- **Interface / surface:** Data + associated tools.

- **Why this rank:** Biochemical identifiers and reconstruction resources support a selected modeling path. Reconcile reaction/compound mapping instead of imposing a second canonical ontology.
- **Functional overlap:** DEP.
- **Primary source:** https://modelseed.org/
- **Source PDF page:** 18.

C177

Primer3

- **Domain rank:** 19.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 9.5/10; V (added integration value) = 6/10.
- **Revised/new scores:** No.
- **Language / implementation:** C / C++.
- **Function:** Select PCR primers under specified constraints.
- **Interface / surface:** CLI; web interfaces.
- **Why this rank:** Primer selection under declared constraints. Record target/template revision and specificity checks separately from thermodynamic candidate scoring.
- **Functional overlap:** COMP.
- **Primary source:** https://primer3.org/manual.html
- **Source PDF page:** 18.

C144

Open Vector Editor

- **Domain rank:** 20.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8/10; V (added integration value) = 5.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** JavaScript / React.
- **Function:** Display and edit sequences, annotations and plasmid maps.
- **Interface / surface:** Embeddable React UI.
- **Why this rank:** Sequence and plasmid editing surface. Substantial UI overlap with OpenCloning; prototype one shared editor path before maintaining both.
- **Functional overlap:** VIEW.
- **Primary source:** https://github.com/TeselaGen/openVectorEditor
- **Source PDF page:** 18.

C209

RetroPath / RetroPathRL

- **Domain rank:** 21.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 6.5/10; V (added integration value) = 5.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Mixed / Python.
- **Function:** Search biosynthetic routes using biochemical reaction rules.
- **Interface / surface:** Workflows + viewers.
- **Why this rank:** Choose a specific retrosynthesis implementation and rule set. Search produces candidate transformations, not demonstrated enzymes or viable strains.
- **Functional overlap:** COMP.
- **Primary source:** https://github.com/brsynth/RetroPathRL
- **Source PDF page:** 18.

C080

GECKO / geckopy

- **Domain rank:** 22.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 7/10; V (added integration value) = 5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Distinct implementations.
- **Function:** Add constraints for enzyme availability and catalytic capacity.
- **Interface / surface:** MATLAB / Python APIs.
- **Why this rank:** Disambiguate MATLAB GECKO, the older Python project and the new beta Python port. Pin repository and version; apply enzyme constraints only with defensible parameter provenance.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C041/C192: distinct enzyme/thermodynamic constraints.
- **Primary source:** https://github.com/SysBioChalmers/GECKO
- **Source PDF page:** 19.

C217

SBOL

- **Domain rank:** 23.
- **Type:** STD.
- **Action:** LATER.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 8/10; V (added integration value) = 5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Design standard.
- **Function:** Exchange biological components, designs and their relationships.
- **Interface / surface:** Through compatible tools.
- **Why this rank:** Design relationships and provenance beyond flat sequences. Adopt a supported profile; do not assume it replaces every internal construct or study record.
- **Functional overlap:** COMP.
- **Primary source:** https://sbolstandard.org/
- **Source PDF page:** 19.

C024

BioModels

- **Domain rank:** 26.
- **Type:** DATA.
- **Action:** LATER.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 8.5/10; V (added integration value) = 5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Model repository.
- **Function:** Find published computational models and associated provenance.
- **Interface / surface:** Website / downloads.
- **Why this rank:** Reusable model source with an announced University of Florida transition. Use biomodels.org and qualify current computational access; endpoint behavior was not verified here. Freeze downloaded bytes.
- **Functional overlap:** COMP.
- **Primary source:** https://biomodels.org/
- **Source PDF page:** 19.

C043

COPASI

- **Domain rank:** 24.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 9/10; V (added integration value) = 4.5/10.

- **Revised/new scores:** No.
- **Language / implementation:** Mostly C++.
- **Function:** Simulate biochemical networks; fit parameters and analyze sensitivity.
- **Interface / surface:** Desktop GUI / CLI / APIs.
- **Why this rank:** Rich kinetic-model authoring and fitting. Overlaps RoadRunner execution; prefer an explicit expert-model import or a scoped backend rather than duplicate simulations.
- **Functional overlap:** ALT.
- **Primary source:** https://copasi.org/
- **Source PDF page:** 19.

C035

CarveMe

- **Domain rank:** 27.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 8/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Reconstruct draft metabolic models from protein-sequence information.
- **Interface / surface:** Command line / Python.
- **Why this rank:** Draft reconstruction for an appropriate organism and input. Prefer an applicable curated model when available; gap-filled reactions remain hypotheses.
- **Functional overlap:** COMP.
- **Primary source:** https://carveme.readthedocs.io/en/stable/usage.html
- **Source PDF page:** 19.

C028

Boltz

- **Domain rank:** 28.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Predict biomolecular structures; Boltz-2 also predicts binding affinity.
- **Interface / surface:** Python / command line.
- **Why this rank:** First protein-structure adapter candidate if needed; documented code and weights

are MIT. Structure and affinity predictions remain distinct from secretion, titer and product function.

- **Functional overlap:** COMP.
- **Related components / overlap detail:** C006: predictor comparison if needed.
- **Primary source:** https://github.com/jwohlwend/boltz
- **Source PDF page:** 19.

C129

ModelSEEDpy

- **Domain rank:** 25.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 7.5/10; V (added integration value) = 4/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Build and analyze models using ModelSEED biochemical resources.
- **Interface / surface:** Python API.
- **Why this rank:** Alternative reconstruction and model-analysis workflow. Compare on the same host and phenotype evidence before operating two reconstruction pipelines.
- **Functional overlap:** ALT.
- **Primary source:** https://github.com/ModelSEED/ModelSEEDpy
- **Source PDF page:** 19.

C192

pyTFA

- **Domain rank:** 29.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 7/10; V (added integration value) = 3.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Add thermodynamic constraints to metabolic-model analysis.
- **Interface / surface:** Python API.
- **Why this rank:** Thermodynamic model constraints complement eQuilibrator estimates but require mapped compounds, concentrations and compartment assumptions.
- **Functional overlap:** COMP.
- **Primary source:** https://pytfa.readthedocs.io/en/latest/

- **Source PDF page:** 20.

C214

rpTools

- **Domain rank:** 30.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 6.5/10; V (added integration value) = 3.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / JavaScript.
- **Function:** Complete, evaluate, rank and visualize candidate pathways.
- **Interface / surface:** Python / CLI + views.
- **Why this rank:** Rank and annotate candidate pathways in the chosen retrosynthesis pipeline. Reuse its pathway records rather than creating another identity service.
- **Functional overlap:** COMP.
- **Primary source:** https://github.com/brsynth/rptools
- **Source PDF page:** 20.

C053

DNA Weaver

- **Domain rank:** 32.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 7/10; V (added integration value) = 3/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Plan how to obtain and assemble constructs from DNA fragments.
- **Interface / surface:** Python; separate web app.
- **Why this rank:** Procurement and assembly-route planning. Its sourcing and cost optimization differs from sequence-level cloning simulation; add when procurement planning is in scope.
- **Functional overlap:** COMP.
- **Primary source:** https://edinburgh-genome-foundry.github.io/DnaWeaver/
- **Source PDF page:** 20.

C180

ProteinMPNN

- **Domain rank:** 33.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 8.5/10; V (added integration value) = 3/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / PyTorch.
- **Function:** Propose sequences compatible with a supplied protein backbone.
- **Interface / surface:** Scripts / notebook examples.
- **Why this rank:** Proposes sequences for a supplied backbone. Complementary to backbone generation and structure prediction, with experimental screening still required.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C212 creates backbones; this designs sequences.
- **Primary source:** https://github.com/dauparas/ProteinMPNN
- **Source PDF page:** 20.

C040

CNApy

- **Domain rank:** 34.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 7/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Mostly Python.
- **Function:** Edit metabolic models and run analysis and strain-design methods.
- **Interface / surface:** Desktop GUI.
- **Why this rank:** Useful expert desktop surface over metabolic methods. Link or import results instead of embedding a second geneOS project system.
- **Functional overlap:** VIEW.
- **Primary source:** https://github.com/cnapy-org/CNApy
- **Source PDF page:** 20.

C212

RFdiffusion

- **Domain rank:** 35.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 8/10; V (added integration value) = 2.5/10.

- **Revised/new scores:** No.
- **Language / implementation:** Python / PyTorch.
- **Function:** Generate protein structures under specified design constraints.
- **Interface / surface:** Computational pipeline.
- **Why this rank:** Generates candidate backbones under conditions. Add after a concrete protein-design workflow exists; do not treat generated geometry as verified function.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C180 complements a distinct design stage.
- **Primary source:** https://github.com/RosettaCommons/RFdiffusion
- **Source PDF page:** 20.

C257

ViennaRNA

- **Domain rank:** 36.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 9/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** C / Python bindings.
- **Function:** Predict RNA secondary structure and related properties.
- **Interface / surface:** CLI / library / Python.
- **Why this rank:** RNA structure analysis for a defined design question. Requires contextual interpretation; not a general predictor of host expression or construct stability.
- **Functional overlap:** COMP.
- **Primary source:** https://www.tbi.univie.ac.at/RNA/
- **Source PDF page:** 21.

C243

Tellurium

- **Domain rank:** 31.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python ecosystem.
- **Function:** Script biochemical simulations using tools including RoadRunner.
- **Interface / surface:** Python / notebooks.

- **Why this rank:** Convenient scripting ecosystem including RoadRunner. Do not add a second simulation service when the lower-level runtime suffices.
- **Functional overlap:** DEP.
- **Primary source:** https://tellurium.readthedocs.io/
- **Source PDF page:** 20.

C006

AlphaFold 3

- **Domain rank:** 37.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 9/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Predict biomolecular structures and interactions.
- **Interface / surface:** Local pipeline; hosted service.
- **Why this rank:** Structure/interaction predictor. Current code and model-parameter/output terms differ; the non-commercial weights route is not an unrestricted platform dependency.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C028: predictor alternatives, not construct validation.
- **Primary source:** https://github.com/google-deepmind/alphafold3
- **Source PDF page:** 21.

C032

cameo

- **Domain rank:** 38.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 6.5/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Metabolic engineering, intervention searches and pathway-design methods.
- **Interface / surface:** Python / notebooks.
- **Why this rank:** Overlaps COBRApy extensions and StrainDesign. Add only a specifically required method absent from the selected stack.
- **Functional overlap:** ALT.
- **Primary source:** https://cameo.readthedocs.io/en/latest/

- **Source PDF page:** 21.

C233

SnapGene

- **Domain rank:** 39.
- **Type:** TOOL.
- **Action:** EXT.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 9/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** Proprietary application.
- **Function:** Sequence editing, plasmid mapping and cloning simulation.
- **Interface / surface:** Desktop GUI.
- **Why this rank:** Expert desktop reference and exchange target. Useful cloning UI comparisons; do not require proprietary desktop automation for core workflows.
- **Functional overlap:** ALT.
- **Primary source:** https://www.snapgene.com/features
- **Source PDF page:** 21.

C239

SynBioHub

- **Domain rank:** 40.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 5/10; M (practical maturity) = 7/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** JavaScript / Java.
- **Function:** Store, browse and share synthetic-biology designs.
- **Interface / surface:** Web application.
- **Why this rank:** External design repository and exchange surface. Overlaps canonical geneOS/BioRepo storage; add for partner exchange rather than a parallel internal record store.
- **Functional overlap:** ALT.
- **Primary source:** https://synbiohub.org/
- **Source PDF page:** 21.

C022

Benchling

- **Domain rank:** 41.
- **Type:** TOOL.
- **Action:** EXT.
- **Scores:** F (project fit) = 5/10; M (practical maturity) = 9.5/10; V (added integration value) = 1.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Proprietary platform.
- **Function:** Connected sequence-design, cloning and molecular-biology workflows.
- **Interface / surface:** Web application.
- **Why this rank:** Commercial workflow reference or partner import target. Overlaps geneOS, ELN and inventory; no internal-platform dependency assumed.
- **Functional overlap:** ALT.
- **Primary source:** https://www.benchling.com/molecular-biology
- **Source PDF page:** 21.

C038

Cello

- **Domain rank:** 42.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 4.5/10; M (practical maturity) = 7/10; V (added integration value) = 1.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Version-dependent.
- **Function:** Convert supported logic specifications into genetic-circuit designs.
- **Interface / surface:** Engine; separate web UI.
- **Why this rank:** Genetic logic design depends on supported parts, host context and characterization. Defer unless circuit synthesis is an explicit product task.
- **Functional overlap:** COMP.
- **Primary source:** https://github.com/CIDARLAB/Cello-v2
- **Source PDF page:** 21.

C186

pydna

- **Domain rank:** 13.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 8/10; V (added integration value) = 1/10.
- **Revised/new scores:** No.

- **Language / implementation:** Python.
- **Function:** Represent double-stranded DNA and simulate cloning and assembly.
- **Interface / surface:** Python API.
- **Why this rank:** Cloning simulation is valuable, but OpenCloning already uses pydna. Expose additional library functions only for a missing typed operation.
- **Functional overlap:** DEP.
- **Related components / overlap detail:** C150 inherits cloning logic; not a second app.
- **Primary source:** https://pydna.readthedocs.io/
- **Source PDF page:** 17.

fermOS, feedstock and purification

One connected process and calculation owner. The existing full plant includes recovery; specialist engines replace explicit scopes. SFF is a data projection.

C230

Shared streams & units

- **Domain rank:** 2.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = NR; V (added integration value) = 8.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Keep composition, units, conditions, equipment and process revisions consistent.
- **Interface / surface:** Your fermOS interface.
- **Why this rank:** The existing flowsheet needs a durable profile/scenario mapping, including wet/dry basis and chemical IDs. One connected plant graph serves fermOS and pureOS; SFF C270 is an exchange projection, not another authority.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 22.

C106

Kinetics & validation

- **Domain rank:** 1.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = NR; V (added integration value) = 9.5/10.

- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Attach strain-specific kinetics, scale-up assumptions, uncertainty and evidence.
- **Interface / surface:** BioRepo / Runbooks links.
- **Why this rank:** Top scientific gap: fitted, applicable response relationships rather than fixed transferred conversions. Software execution alone cannot optimize pretreatment conditions; retain calibration data, uncertainty, operating domain and independent checks.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 22.

C026

BioSTEAM

- **Domain rank:** 5.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = 8/10; V (added integration value) = 3/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python.
- **Function:** Bioprocess unit operations, connected flowsheets, utilities, costing and uncertainty.
- **Interface / surface:** Python / notebooks / diagrams.
- **Why this rank:** Highest engine fit for the chosen initial path, which is already working. V is further model qualification, not a new engine installation. New feedstock parameterization belongs to C064/C230; preserve full-plant recovery and utilities.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C059: engine alternative or scoped replacement.
- **Primary source:** https://biosteam.readthedocs.io/
- **Source PDF page:** 22.

C064

Engine & AI adapters

- **Domain rank:** 6.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 8/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.

- **Function:** Translate typed, validated requests into engine inputs and retain outputs.
- **Interface / surface:** Postdoc / engine APIs.
- **Why this rank:** The worker and tool registry exist. High V is the missing profile -> apply -> simulate -> read-back -> compare adapter with stable receipts; MCP is an optional calling interface, not the model semantics.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 22.

C030

CADET + CADET-Process

- **Domain rank:** 14.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 8/10; V (added integration value) = 6/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / C++.
- **Function:** Chromatography transport, binding and process optimization.
- **Interface / surface:** Python + native solver.
- **Why this rank:** Chromatography-detail adapter when separation physics and calibration data justify it. Accepted outputs replace the scoped generic operation, never compound its recovery twice.
- **Functional overlap:** SCOPE.
- **Related components / overlap detail:** C026: scoped chromatography replacement.
- **Primary source:** https://cadet-process.readthedocs.io/en/v0.12.0/user_guide/overview.html
- **Source PDF page:** 23.

C059

DWSIM

- **Domain rank:** 11.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 7.5/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** .NET 10 / C# / VB.
- **Function:** General process flowsheets, material/energy balances, equipment and dynamics.
- **Interface / surface:** Desktop + headless / MCP.

- **Why this rank:** Serious primary-engine alternative or scoped property/separation comparator, including the reviewed headless/MCP route. Added V is conditional because BioSTEAM C026 already owns the first full plant; match boundaries before replacing a calculation.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C026: replace an owned calculation explicitly.
- **Primary source:** https://github.com/DanWBR/dwsim10
- **Source PDF page:** 23.

C159

Optuna

- **Domain rank:** 16.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 8.5/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Search simulation parameters and operating choices with black-box objectives.
- **Interface / surface:** Python API; separate web UI.
- **Why this rank:** Black-box simulation search and tuning. Overlaps Ax in candidate selection; retain objective, constraints, budget, seed and result provenance.
- **Functional overlap:** ALT.
- **Primary source:** https://optuna.org/
- **Source PDF page:** 24.

C076

FMI / FMU

- **Domain rank:** 18.
- **Type:** STD.
- **Action:** GROW.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 9/10; V (added integration value) = 4/10.
- **Revised/new scores:** No.
- **Language / implementation:** Interface standard.
- **Function:** Exchange dynamic models and support model exchange/co-simulation.
- **Interface / surface:** Standard; compatible tools.
- **Why this rank:** Model exchange/co-simulation contract, conditional on both endpoints. Test units, states, time stepping, events and supported FMI versions.
- **Functional overlap:** COMP.

- **Primary source:** https://fmi-standard.org/
- **Source PDF page:** 24.

C261

WaterTAP

- **Domain rank:** 23.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 7.5/10; V (added integration value) = 4/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / IDAES.
- **Function:** Water treatment and resource-recovery process modeling.
- **Interface / surface:** Python API / notebooks.
- **Why this rank:** Water-treatment specialist. A zero-order ultrafiltration unit is not automatically a calibrated protein-TFF retention/fouling model.
- **Functional overlap:** SCOPE.
- **Related components / overlap detail:** C095 framework / C026 scoped water work.
- **Primary source:** https://watertap.readthedocs.io/en/stable/
- **Source PDF page:** 25.

C036

CasADi

- **Domain rank:** 19.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 9/10; V (added integration value) = 3.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** C++ / bindings.
- **Function:** Automatic differentiation and numerical tools for dynamic optimization.
- **Interface / surface:** C++ / Python / MATLAB APIs.
- **Why this rank:** Differentiation and dynamic optimization. Overlaps some Pyomo/SciPy tasks; add for a specific continuous optimal-control formulation.
- **Functional overlap:** COMP.
- **Primary source:** https://web.casadi.org/
- **Source PDF page:** 24.

C050

dfba

- **Domain rank:** 20.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 6.5/10; V (added integration value) = 3.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** C++ / Python.
- **Function:** Couple metabolic optimization with extracellular dynamic equations.
- **Interface / surface:** Python API.
- **Why this rank:** Couples metabolic optimization to extracellular dynamics. Compare with COBRApy's illustrative dFBA and a calibrated reactor model; handle infeasibility and alternate optima.
- **Functional overlap:** COMP.
- **Primary source:** https://dynamic-fba.readthedocs.io/
- **Source PDF page:** 24.

C095

IDAES

- **Domain rank:** 21.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8/10; V (added integration value) = 3/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Equation-oriented process modeling and optimization built on Pyomo.
- **Interface / surface:** Python / notebooks.
- **Why this rank:** Equation-oriented modeling built on Pyomo. Add when simultaneous equations and custom optimization justify a separate modeling path.
- **Functional overlap:** ALT.
- **Primary source:** https://idaes.org/overview/
- **Source PDF page:** 24.

C023

Bioindustrial-Park / biorefineries

- **Domain rank:** 4.
- **Type:** TOOL.

- **Action:** BUILD.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = 7/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python.
- **Function:** Reusable BioSTEAM biorefinery models, including ethanol and feedstock preparation.
- **Interface / surface:** Python / notebooks.
- **Why this rank:** The corn-stover reference is directly relevant to phase one, raising fit substantially. Existing recipes reduce model-building work, but new profiles need a separate parameterized adapter and explicit transfer/applicability checks.
- **Functional overlap:** DEP.
- **Primary source:** https://biosteam.readthedocs.io/en/latest/API/biorefineries/index.html
- **Source PDF page:** 22.

C152

OpenFOAM

- **Domain rank:** 24.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 9/10; V (added integration value) = 3/10.
- **Revised/new scores:** No.
- **Language / implementation:** C++.
- **Function:** CFD for mixing, flow, heat transfer and multiphase scale-up studies.
- **Interface / surface:** Command line / post-processing.
- **Why this rank:** CFD for a specific scale-up question and validated subdomain. Boundary conditions, mesh and closure models require expertise; exchange scoped results with the plant model.
- **Functional overlap:** SCOPE.
- **Primary source:** https://openfoam.org/
- **Source PDF page:** 25.

C077

FMPy

- **Domain rank:** 22.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 8/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.

- **Language / implementation:** Python / C.
- **Function:** Inspect and simulate supported FMUs from Python or a graphical interface.
- **Interface / surface:** Python / CLI / desktop GUI.
- **Why this rank:** Execute or inspect FMUs in the selected co-simulation path. No separate model truth; preserve FMU hash and solver configuration.
- **Functional overlap:** DEP.
- **Primary source:** https://fmpy.readthedocs.io/
- **Source PDF page:** 25.

C054

do-mpc

- **Domain rank:** 25.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 8/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / CasADi.
- **Function:** Nonlinear model-predictive control and moving-horizon state estimation.
- **Interface / surface:** Python API / plots.
- **Why this rank:** Model predictive control and estimation, using CasADi. Evaluate offline designs first; live control requires a separately qualified equipment boundary.
- **Functional overlap:** ALT.
- **Primary source:** https://www.do-mpc.com/
- **Source PDF page:** 25.

C153

OpenModelica

- **Domain rank:** 26.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Modelica / C++.
- **Function:** Build dynamic equipment and process models using Modelica.
- **Interface / surface:** Desktop GUI + scripting.
- **Why this rank:** Dynamic physical-system modeling and FMU export. Strong for a scoped

equipment model; avoid importing an additional full plant authority.

- **Functional overlap:** ALT.
- **Primary source:** https://openmodelica.org/
- **Source PDF page:** 25.

C193

python-control

- **Domain rank:** 27.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Analyze feedback systems, responses, stability and controller designs.
- **Interface / surface:** Python API / plots.
- **Why this rank:** Feedback-system analysis for defined models. Does not supply biological kinetics or validate a live controller automatically.
- **Functional overlap:** COMP.
- **Primary source:** https://python-control.readthedocs.io/
- **Source PDF page:** 25.

C168

PharmaPy

- **Domain rank:** 33.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 5/10; M (practical maturity) = 6.5/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Pharmaceutical process dynamics and relevant isolation operations.
- **Interface / surface:** Python API / notebooks.
- **Why this rank:** Specialist pharmaceutical process dynamics and isolation. Add only for relevant product forms/operations and a tested state/stream mapping.
- **Functional overlap:** SCOPE.
- **Related components / overlap detail:** C026: specialist isolation/dynamics scope.
- **Primary source:** https://pharmapy.readthedocs.io/

- **Source PDF page:** 26.

C016

Assimulo

- **Domain rank:** 28.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 7/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / Cython.
- **Function:** Python interfaces to ODE/DAE solvers, including SUNDIALS.
- **Interface / surface:** Python API.
- **Why this rank:** ODE/DAE solver interfaces. Overlaps other numerical wrappers; select only when a required solver or event behavior is missing.
- **Functional overlap:** DEP.
- **Primary source:** https://github.com/modelon-community/Assimulo
- **Source PDF page:** 25.

C034

CAPE-OPEN

- **Domain rank:** 29.
- **Type:** STD.
- **Action:** GROW.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 8/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** Interface standard.
- **Function:** Exchange unit-operation and thermodynamic components with compatible simulators.
- **Interface / surface:** Standard; simulator interfaces.
- **Why this rank:** Compatible simulator/component interoperability. Useful only where the actual engine endpoints and deployment support the interface.
- **Functional overlap:** COMP.
- **Primary source:** https://www.colan.org/specifications/
- **Source PDF page:** 26.

C081

GEKKO

- **Domain rank:** 30.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 8/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / APMonitor.
- **Function:** Dynamic simulation, parameter estimation and optimization/control.
- **Interface / surface:** Python API; solver backend.
- **Why this rank:** Dynamic modeling, estimation and optimization with solver backend. Overlaps other modeler/control paths; add only a concrete differentiating workflow.
- **Functional overlap:** ALT.
- **Primary source:** https://gekko.readthedocs.io/
- **Source PDF page:** 26.

C164

ParaView

- **Domain rank:** 31.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 9/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** C++ / Python.
- **Function:** Inspect and visualize spatial fields and CFD simulation results.
- **Interface / surface:** Desktop GUI / Python.
- **Why this rank:** Post-process spatial simulation fields. Complement to CFD, not a reactor or process engine; link expert outputs into the shared evidence record.
- **Functional overlap:** VIEW.
- **Primary source:** https://www.paraview.org/about/
- **Source PDF page:** 26.

C238

SuperPro Designer

- **Domain rank:** 32.
- **Type:** TOOL.
- **Action:** EXT.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 9/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.

- **Language / implementation:** Proprietary.
- **Function:** Commercial reference for batch/continuous bioprocesses and scheduling.
- **Interface / surface:** Desktop GUI / COM API.
- **Why this rank:** Bioprocess and batch-costing comparator. Export/import assumptions explicitly; avoid a second authoritative throughput and cash-flow model.
- **Functional overlap:** ALT.
- **Primary source:** https://www.intelligen.com/products/superpro-overview/
- **Source PDF page:** 26.

C017

asyncua

- **Domain rank:** 34.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 4.5/10; M (practical maturity) = 8/10; V (added integration value) = 1.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Optional OPC UA client/server connectivity for equipment and process data.
- **Interface / surface:** Python API / CLI tools.
- **Why this rank:** Python OPC UA implementation. Reuse the selected equipment gateway; OPC UA connectivity alone does not define process meaning or authorize actuation.
- **Functional overlap:** DEP.
- **Primary source:** https://github.com/FreeOpcUa/opcua-asyncio
- **Source PDF page:** 26.

C237

SUNDIALS

- **Domain rank:** 15.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 9/10; V (added integration value) = 1/10.
- **Revised/new scores:** No.
- **Language / implementation:** C / Fortran / C++.
- **Function:** Time integration and nonlinear solvers for stiff ODE/DAE systems.
- **Interface / surface:** Native libraries / bindings.
- **Why this rank:** Native time integration and nonlinear solvers used by higher layers. Direct integration requires a defined numerical need, not an extra campaign store.

- **Functional overlap:** DEP.
- **Primary source:** https://computing.llnl.gov/projects/sundials
- **Source PDF page:** 24.

C042

CoolProp

- **Domain rank:** 17.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9/10; V (added integration value) = 1/10.
- **Revised/new scores:** No.
- **Language / implementation:** C++ / bindings.
- **Function:** Fluid properties for utilities, gases, water/steam and supported mixtures.
- **Interface / surface:** Library / language wrappers.
- **Why this rank:** Useful for supported utility fluids and mixtures, often inherited. Add direct operations only for requirements not served by the selected property stack.
- **Functional overlap:** DEP.
- **Primary source:** https://coolprop.org/
- **Source PDF page:** 24.

C014

Aspen Plus

- **Domain rank:** 35.
- **Type:** TOOL.
- **Action:** EXT.
- **Scores:** F (project fit) = 4.5/10; M (practical maturity) = 9.5/10; V (added integration value) = 1/10.
- **Revised/new scores:** No.
- **Language / implementation:** Proprietary.
- **Function:** Commercial reference for integrated chemical and bioprocess simulation.
- **Interface / surface:** Desktop GUI / automation.
- **Why this rank:** Commercial comparison or partner model source. Do not make the open reference workflow depend on unavailable proprietary internals.
- **Functional overlap:** ALT.
- **Primary source:** https://www.aspentech.com/en/products/engineering/aspen-plus
- **Source PDF page:** 26.

C247

ThermoSTEAM

- **Domain rank:** 8.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 8/10; V (added integration value) = 0/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** BioSTEAM stream definitions, reactions, property packages and phase equilibria.
- **Interface / surface:** Python API.
- **Why this rank:** Inherited BioSTEAM stream/property layer. Version and test through the engine; no independent top-level adapter by default. Keep the inherited dependency; V=0 is not removal priority.
- **Functional overlap:** DEP.
- **Primary source:** https://github.com/BioSTEAMDevelopmentGroup/thermosteam
- **Source PDF page:** 23.

C039

chemicals

- **Domain rank:** 12.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8/10; V (added integration value) = 0/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Chemical property correlations, constants and supporting data.
- **Interface / surface:** Python API.
- **Why this rank:** Property correlations and data beneath thermo. Keep provenance and limits; extra catalog presence does not imply another service. Keep the inherited dependency; V=0 is not removal priority.
- **Functional overlap:** DEP.
- **Primary source:** https://github.com/CalebBell/chemicals
- **Source PDF page:** 23.

C246

thermo

- **Domain rank:** 13.
- **Type:** TOOL.

- **Action:** DEP.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8/10; V (added integration value) = 0/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Chemical and mixture property models, flash calculations and phase equilibria.
- **Interface / surface:** Python API.
- **Why this rank:** Inherited physical-property and flash calculations where used. One selected property method per scoped computation, not competing unlabelled values. Keep the inherited dependency; V=0 is not removal priority.
- **Functional overlap:** DEP.
- **Primary source:** https://github.com/CalebBell/thermo
- **Source PDF page:** 23.

C270

Standard Flowsheet Format / SFF toolkit

- **Domain rank:** 3.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = 6/10; V (added integration value) = 8.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** JSON / Python.
- **Function:** Represent process units, streams, chemicals and utilities; export BioSTEAM and validate exchange files.
- **Interface / surface:** JSON schema / Python toolkit.
- **Why this rank:** High V for a reusable process exchange layer with an MIT toolkit. Test topology, chemical mappings and loss on the current model; schema conformance, completeness and reproduced numerical behavior remain separate results.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C230: exchange projection of canonical process data.
- **Primary source:** https://github.com/sustainability-software-lab/pisces-standard-flowsheet-format/tree/e236e3250414f45cf9fd7b8b854f1afc67fc2cad
- **Source PDF page:** 22.

C271

Project PISCES hosted platform

- **Domain rank:** 7.
- **Type:** PLATFORM.

- **Action:** EXT.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 3.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Hosted platform / separate service terms.
- **Function:** Host, explore and develop process/flowsheet data with literature-to-process workflow ambitions.
- **Interface / surface:** Hosted web application.
- **Why this rank:** Very close functional overlap makes comparison/collaboration valuable. Direct code reuse is constrained by separately governed application access; the public MIT SFF toolkit C270 is not a license for all hosted code, data or backends.
- **Functional overlap:** PART.
- **Related components / overlap detail:** C270 is public toolkit; hosted system is distinct.
- **Primary source:** https://projectpisces.org/
- **Source PDF page:** 22.

C272

NSKinetics

- **Domain rank:** 9.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 5.5/10; V (added integration value) = 6.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python.
- **Function:** Connect kinetic biological models to process simulation and downstream TEA through a BioSTEAM bridge.
- **Interface / surface:** Python / SBML-Antimony / documentation.
- **Why this rank:** Potentially avoids custom kinetics-to-plant coupling. Qualify the exact released bridge and a suitable model; it does not infer validated hydrolysis, inhibition or fermentation response for an unfamiliar feedstock.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C114 executes kinetics; this bridges to the plant.
- **Primary source:** https://nskinetics.readthedocs.io/en/latest/tutorial/06_process_tea_bridge.html
- **Source PDF page:** 23.

C273

AutoSynthesis - process design

- **Domain rank:** 10.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 3.5/10; V (added integration value) = 5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python.
- **Function:** Research automated process synthesis/design and agentic interfaces to engineering workflows.
- **Interface / surface:** Public source package / research.
- **Why this rank:** A relevant MIT-licensed process-design lead that may reduce custom synthesis work. Sparse public guidance and unqualified end-to-end behavior limit maturity; request a rerunnable prompt-to-process example before adoption.
- **Functional overlap:** PART.
- **Related components / overlap detail:** C064/C220: process-synthesis/agent overlap.
- **Primary source:** https://github.com/sarangbhagwat/autosynthesis
- **Source PDF page:** 23.

Proforma

Reuse TEA, then add source-linked market assumptions and one reconciled scenario. A live price feed is an observation source, not an economic model.

C176

Price & quote ledger

- **Domain rank:** 1.
- **Type:** CAP.
- **Action:** GROW.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = NR; V (added integration value) = 8/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Version observations, supplier quotes and offtake evidence with terms, freshness and provenance.
- **Interface / surface:** Proforma data review / API.
- **Why this rank:** High V for source-linked prices and quotes with grade, geography, volume, dates and terms. Current TEA parameters are not a live market ledger; refresh creates new observations without changing past scenarios.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 27.

C021

Basis & landed cost

- **Domain rank:** 2.
- **Type:** CAP.
- **Action:** GROW.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 8.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Project code / scientific service.
- **Function:** Match grade, moisture, location, volume and delivery terms; derive traceable delivered costs.
- **Interface / surface:** Shared units / material records.
- **Why this rank:** Normalize specification, moisture, active content and shipment terms through explicit derivations. Separate recoverable taxes, losses, inventories and final delivered costs.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 27.

C061

Economic validation

- **Domain rank:** 3.
- **Type:** CAP.
- **Action:** GROW.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 8.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Project code / scientific service.
- **Function:** Reconcile units and accounts, benchmark cost models and separate forecasts from observations.
- **Interface / surface:** Review / audit / change log.
- **Why this rank:** Reconcile costs and accounts, timing, bases, taxes and double counting. Compare hand-checkable cases and appropriate quotes before presenting economic conclusions.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 27.

C170

Plant cash-flow model

- **Domain rank:** 4.
- **Type:** CAP.
- **Action:** GROW.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 5.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Translate integrated process outputs into capex, opex, taxes, working capital and financing.
- **Interface / surface:** Proforma model / cash-flow UI.
- **Why this rank:** The engine and Proforma already provide TEA machinery. Added work is one reconciled cost/cash-flow basis, not a second authoritative NPV; reuse engine costs and expose financing, volume and exclusions.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C026 / C139: reconcile one financial result.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 27.

C121

Market & offtake model

- **Domain rank:** 5.
- **Type:** CAP.
- **Action:** GROW.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = NR; V (added integration value) = 8/10.
- **Revised/new scores:** No.
- **Language / implementation:** Project code / scientific service.
- **Function:** Constrain sales by qualification, demand, customer volume and contracts; handle coproduct limits.
- **Interface / surface:** Market assumptions / scenarios.
- **Why this rank:** Constrain sellable output by qualification, demand, contracts, ramp-up and coproduct absorption. Nameplate production is not automatically realized revenue.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 27.

C218

Scenario & forecast engine

- **Domain rank:** 6.

- **Type:** CAP.
- **Action:** GROW.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = NR; V (added integration value) = 5.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Preserve correlated market/process paths, assumption versions, data vintages and random seeds.
- **Interface / surface:** Scenario comparison / API.
- **Why this rank:** Scenario/sampling machinery exists. Add forecasting only where historical data, decision timing and information availability support it; separate assumptions, probabilistic forecasts and sensitivity branches.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 27.

C173

Plotly

- **Domain rank:** 7.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 9/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python / JavaScript.
- **Function:** Interactive price history, cost waterfalls, scenario distributions and break-even charts.
- **Interface / surface:** Browser charts / Python.
- **Why this rank:** Existing Recharts C275 covers the initial chart path. Plotly adds value for a specific missing scientific chart/export. Both render canonical artifacts; neither owns calculations or result storage.
- **Functional overlap:** VIEW.
- **Primary source:** https://plotly.com/python/
- **Source PDF page:** 27.

C078

FRED / ALFRED

- **Domain rank:** 8.
- **Type:** DATA.
- **Action:** GROW.

- **Scores:** F (project fit) = 8/10; M (practical maturity) = 9.5/10; V (added integration value) = 6.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Public aggregator; terms vary.
- **Function:** Rates, inflation and macro series; vintages support historical as-of-date analysis.
- **Interface / surface:** REST / web; series-specific.
- **Why this rank:** Macro/rate context with historical vintages. Preserve original source and series-specific rights; an aggregator does not make every series freely redistributable.
- **Functional overlap:** COMP.
- **Primary source:** https://fred.stlouisfed.org/docs/api/fred/
- **Source PDF page:** 27.

C264

World Bank Pink Sheet

- **Domain rank:** 9.
- **Type:** DATA.
- **Action:** GROW.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 9/10; V (added integration value) = 6.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Public commodity dataset.
- **Function:** Long historical commodity benchmarks and scenario context; monthly, not intraday.
- **Interface / surface:** Downloadable tables / web.
- **Why this rank:** Broad historical monthly commodity benchmarks. Useful scenario context; no assumption of grade-specific delivered spot pricing.
- **Functional overlap:** COMP.
- **Primary source:** https://www.worldbank.org/en/research/commodity-markets
- **Source PDF page:** 28.

C234

statsmodels

- **Domain rank:** 10.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 9/10; V (added integration value) = 5.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / Cython.
- **Function:** Econometric regression, state-space and multivariate time-series forecasting.
- **Interface / surface:** Python / notebooks / plots.

- **Why this rank:** Interpretable regression and time-series baselines. Test rolling forecasts with information available then; require future assumptions for exogenous variables.
- **Functional overlap:** COMP.
- **Primary source:** https://www.statsmodels.org/stable/tsa.html
- **Source PDF page:** 28.

C049

Decision optimizer

- **Domain rank:** 11.
- **Type:** CAP.
- **Action:** GROW.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = NR; V (added integration value) = 5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Project code / scientific service.
- **Function:** Compare feasible designs and staged decisions under capacity, supply, demand and downside limits.
- **Interface / surface:** Design comparison / API.
- **Why this rank:** Optimize only after feasible model, commercial bounds and uncertainty are established. Report decision scope, solver gap/status and sensitivity to assumptions.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 28.

C094

ICIS

- **Domain rank:** 12.
- **Type:** DATA.
- **Action:** ALT.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 9/10; V (added integration value) = 5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Commercial data feed.
- **Function:** Licensed chemical price assessments and related data where product/region coverage fits.
- **Interface / surface:** API / portal; series-specific.
- **Why this rank:** Licensed assessments where product, region, frequency and rights fit. Buy relevant coverage only after identifying a missing decision-critical price series.
- **Functional overlap:** ALT.

- **Primary source:** https://developer.icis.com/
- **Source PDF page:** 28.

C254

USDA MyMarketNews

- **Domain rank:** 13.
- **Type:** DATA.
- **Action:** GROW.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9/10; V (added integration value) = 5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Public data; API key.
- **Function:** Published agricultural commodity and coproduct reports for feedstock benchmarks.
- **Interface / surface:** REST / web; report-specific.
- **Why this rank:** Agricultural feedstock/coproduct benchmarks where product and geography fit. Report cadence and grade basis matter more than a generic live-feed label.
- **Functional overlap:** COMP.
- **Primary source:** https://mymarketnews.ams.usda.gov/mymarketnews-api
- **Source PDF page:** 28.

C139

numpy-financial

- **Domain rank:** 15.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 8.5/10; V (added integration value) = 1.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python / Cython.
- **Function:** Periodic NPV, IRR and payment arithmetic; dated cash flows need additional logic.
- **Interface / surface:** Python API.
- **Why this rank:** Small financial helpers can support transparent checks. Added V is limited because the selected TEA already calculates plant finances; these functions do not supply process costing, market evidence or a validated cash-flow model.
- **Functional overlap:** COMP.
- **Primary source:** https://numpy.org/numpy-financial/latest/
- **Source PDF page:** 28.

C189

PyMC

- **Domain rank:** 14.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / PyTensor.
- **Function:** Bayesian parameter uncertainty and hierarchical models; sparse data still require defensible priors.
- **Interface / surface:** Python / diagnostic plots.
- **Why this rank:** Bayesian parameter/hierarchical uncertainty. Priors, identifiability and posterior checks matter; sparse evidence does not become precise through sampling.
- **Functional overlap:** COMP.
- **Primary source:** https://www.pymc.io/
- **Source PDF page:** 28.

C062

EIA Open Data

- **Domain rank:** 16.
- **Type:** DATA.
- **Action:** GROW.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 9.5/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Public data service.
- **Function:** US energy prices, industrial utility context and fuel scenarios; select region and sector.
- **Interface / surface:** REST / web; daily to annual.
- **Why this rank:** US energy benchmark/context adapter. Site tariff, demand charges, service reliability and actual contract terms must be separate scenario inputs.
- **Functional overlap:** COMP.
- **Primary source:** https://www.eia.gov/opendata/
- **Source PDF page:** 29.

C154

openpyxl

- **Domain rank:** 17.
- **Type:** TOOL.

- **Action:** GROW.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 9/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Export reviewable Excel assumptions and results; not a spreadsheet calculation engine.
- **Interface / surface:** Python API / XLSX output.
- **Why this rank:** Reviewable Excel exports and imports. It does not calculate formulas; deliver evaluated results and controlled assumptions with explicit workbook provenance.
- **Functional overlap:** VIEW.
- **Primary source:** https://openpyxl.readthedocs.io/
- **Source PDF page:** 29.

C251

UN Comtrade

- **Domain rank:** 18.
- **Type:** DATA.
- **Action:** GROW.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 9/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Trade statistics service.
- **Function:** Trade value and quantity for market screening; unit values are not grade-specific prices.
- **Interface / surface:** API / web; reporting lag.
- **Why this rank:** Trade screening, partner volumes and unit-value clues. Classification, valuation and reporting lag prevent treating unit values as qualified product prices.
- **Functional overlap:** COMP.
- **Primary source:** https://comtradeplus.un.org/
- **Source PDF page:** 29.

C060

ECB reference FX

- **Domain rank:** 19.
- **Type:** DATA.
- **Action:** GROW.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 9.5/10; V (added integration value) = 4/10.
- **Revised/new scores:** No.
- **Language / implementation:** Public reference data.
- **Function:** Convert currency scenarios with reference exchange rates; not executable FX quotes.

- **Interface / surface:** SDMX API / web; working days.
- **Why this rank:** Reference-rate scenarios with supported currency coverage. Rates are informational, not executable conversion or financing terms.
- **Functional overlap:** COMP.
- **Primary source:** https://data.ecb.europa.eu/help/api/data
- **Source PDF page:** 29.

C011

Argus Media

- **Domain rank:** 20.
- **Type:** DATA.
- **Action:** ALT.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 9/10; V (added integration value) = 3.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Commercial data feed.
- **Function:** Licensed commodity assessments for chemicals, energy and relevant feedstock markets.
- **Interface / surface:** API / feeds / portal.
- **Why this rank:** Alternative/complementary licensed commodity coverage. Compare the actual assessed series and reuse entitlements rather than subscribing to overlapping feeds by brand.
- **Functional overlap:** ALT.
- **Primary source:** https://www.argusmedia.com/en/about-argus/training-and-support
- **Source PDF page:** 29.

C195

QSDsan

- **Domain rank:** 21.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 8/10; V (added integration value) = 3.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** BioSTEAM-based TEA/LCA for treatment and resource recovery; useful for waste-side economics.
- **Interface / surface:** Python / notebooks.
- **Why this rank:** BioSTEAM-based treatment/resource recovery analysis. Reuse process interfaces; add when wastewater or recovery economics materially affects design.

- **Functional overlap:** DEP.
- **Related components / overlap detail:** C026 / specialist environmental assessment.
- **Primary source:** https://qsdsan.com/en/latest/
- **Source PDF page:** 29.

C027

BLS Public Data

- **Domain rank:** 22.
- **Type:** DATA.
- **Action:** ALT.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 9.5/10; V (added integration value) = 3/10.
- **Revised/new scores:** No.
- **Language / implementation:** Public statistics service.
- **Function:** Labor and producer-price indexes for location/context and appropriate escalation.
- **Interface / surface:** REST / web; periodic releases.
- **Why this rank:** US labor and producer-price context. Apply an appropriate series and location; index escalation is not a local equipment or wage quote.
- **Functional overlap:** ALT.
- **Primary source:** https://www.bls.gov/developers/
- **Source PDF page:** 29.

C069

Eurostat

- **Domain rank:** 23.
- **Type:** DATA.
- **Action:** ALT.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 9.5/10; V (added integration value) = 3/10.
- **Revised/new scores:** No.
- **Language / implementation:** Public statistics service.
- **Function:** European producer prices, labor and industry statistics as regional alternatives.
- **Interface / surface:** REST / web; periodic releases.
- **Why this rank:** European industry/labor context. Regional alternative, not another mandatory global economics dependency.
- **Functional overlap:** ALT.
- **Primary source:** https://ec.europa.eu/eurostat/web/user-guides/data-browser/api-data-access/api-getting-started
- **Source PDF page:** 30.

C134

mpi-sppy

- **Domain rank:** 24.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 4.5/10; M (practical maturity) = 8/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Scenario-based multistage stochastic programming, including risk-averse formulations.
- **Interface / surface:** Python / MPI / solvers.
- **Why this rank:** Scenario-based multistage stochastic optimization on Pyomo. Requires justified probabilities, non-anticipative decisions and solver/compute planning.
- **Functional overlap:** DEP.
- **Primary source:** https://mpi-sppy.readthedocs.io/
- **Source PDF page:** 30.

C010

arch

- **Domain rank:** 25.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 4.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / Cython.
- **Function:** Volatility and bootstrap tools when price histories justify time-varying risk models.
- **Interface / surface:** Python / notebooks / plots.
- **Why this rank:** Conditional volatility and bootstrap methods. Use only if adequate price history and a demonstrated risk-model need justify extra complexity.
- **Functional overlap:** COMP.
- **Primary source:** https://arch.readthedocs.io/en/stable/
- **Source PDF page:** 30.

C079

Freightos Baltic Index

- **Domain rank:** 26.
- **Type:** DATA.

- **Action:** GROW.
- **Scores:** F (project fit) = 4.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** Commercial freight index.
- **Function:** Container-shipping benchmarks; supplement with lane, shipment and inland freight quotes.
- **Interface / surface:** Web / licensed data access.
- **Why this rank:** Container lane benchmark. Landed cost still needs shipment specification, inland legs, handling, insurance and quotes for the actual route.
- **Functional overlap:** COMP.
- **Primary source:** https://www.freightos.com/enterprise/terminal/freightos-baltic-index-global-container-pricing-index/
- **Source PDF page:** 30.

C015

Aspen Process Economic Analyzer

- **Domain rank:** 27.
- **Type:** TOOL.
- **Action:** EXT.
- **Scores:** F (project fit) = 4.5/10; M (practical maturity) = 9/10; V (added integration value) = 1.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Proprietary.
- **Function:** Commercial reference for simulation-linked equipment, capital and operating-cost studies.
- **Interface / surface:** Desktop engineering GUI.
- **Why this rank:** Commercial capital/cost-study reference. Benchmark appropriate model scopes; avoid duplicate plant finance ownership.
- **Functional overlap:** ALT.
- **Primary source:** https://www.aspentech.com/en/products/engineering/aspen-process-economic-analyzer
- **Source PDF page:** 30.

C084

gridstatus

- **Domain rank:** 28.
- **Type:** TOOL.
- **Action:** GROW.

- **Scores:** F (project fit) = 3.5/10; M (practical maturity) = 8/10; V (added integration value) = 1.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / public sources.
- **Function:** Retrieve supported wholesale power-market data; apply the actual site tariff separately.
- **Interface / surface:** Python API; source-dependent.
- **Why this rank:** Supported wholesale electricity-market data. Useful only for a matching market/tariff question; not a default proxy for an arbitrary plant's power bill.
- **Functional overlap:** COMP.
- **Primary source:** https://github.com/gridstatus/gridstatus
- **Source PDF page:** 30.

C197

QuantLib

- **Domain rank:** 29.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 3.5/10; M (practical maturity) = 9.5/10; V (added integration value) = 1.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** C++ / bindings.
- **Function:** Optional interest-rate curves and financial instruments for complex financing/hedging studies.
- **Interface / surface:** C++ / language bindings.
- **Why this rank:** Rates, instruments and complex financing/hedging tools. Usually unnecessary for initial plant TEA; add only with a concrete financial instrument model.
- **Functional overlap:** COMP.
- **Primary source:** https://www.quantlib.org/
- **Source PDF page:** 30.

Runbooks

Method -> prospective plan -> actual execution -> reviewed reconciliation. Start computationally; laboratory formats, ELNs and hardware are conditional later integrations.

C120

Locked study plan

- **Domain rank:** 1.
- **Type:** CAP.
- **Action:** BUILD.

- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = NR; V (added integration value) = 8.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Freeze exact materials, samples, conditions, predictions, acceptance criteria and approvals.
- **Interface / surface:** Plan review / signature view.
- **Why this rank:** High V for locked profile/model/assumption revisions in a computational study. Prospective bench approval and amendments extend the same pattern later; a plan, a run and a reviewed conclusion are distinct records.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 31.

C221

Scientific validation gates

- **Domain rank:** 2.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = NR; V (added integration value) = 8.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Check units, bases, controls, randomization, ranges, capacities and required observations.
- **Interface / surface:** Review UI / validation API.
- **Why this rank:** Extend existing structural/tool guards with scientific checks: basis, applicability, input influence, balances and declared acceptance criteria. A valid schema or successful solver exit alone is insufficient.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 31.

C073

Execution event ledger

- **Domain rank:** 5.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 4.5/10.

- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Capture timestamps, operators, instruments, actual values, artifacts and deviations as events.
- **Interface / surface:** Bench UI / equipment adapters.
- **Why this rank:** Tool/audit receipts already exist. Extend them into a coherent study history with status, failures, deviations and stable operation IDs; keep scientific artifacts distinct from transient operational traces.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 31.

C181

Protocol compiler

- **Domain rank:** 4.
- **Type:** CAP.
- **Action:** GROW.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 6/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Compile cited scientific intent into a typed plan graph and multiple human/machine renderings.
- **Interface / surface:** Runbook editor / compiler API.
- **Why this rank:** Generate typed prospective procedures from reviewed methods and selected inputs. Reuse LabOP C110 or an execution format where appropriate; compilation is separate from scientific approval and hardware qualification.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 31.

C206

Reconciliation engine

- **Domain rank:** 3.
- **Type:** CAP.
- **Action:** GROW.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 7.5/10.
- **Revised/new scores:** Yes.

- **Language / implementation:** Project code / scientific service.
- **Function:** Compare predicted, planned and actual results; classify deviations and feed reviewed evidence back.
- **Interface / surface:** Comparison UI / BioRepo link.
- **Why this rank:** Compare locked predictions with actual outcomes and retain discrepancies and reviewer decisions. Initial computational comparisons can establish the pattern; bench observations never silently overwrite literature claims or predictions.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 31.

C033

Capability adapters

- **Domain rank:** 6.
- **Type:** CAP.
- **Action:** GROW.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = NR; V (added integration value) = 5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Map validated abstract operations to people, ELNs, compute workflows or supported equipment.
- **Interface / surface:** Adapter registry / test harness.
- **Why this rank:** The existing tool registry supplies part of the command boundary. Add explicit capability, input/output and receipt mappings when introducing a new model or instrument; hardware execution is a later scoped need.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 31.

C063

eLabFTW

- **Domain rank:** 7.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** PHP / TypeScript / JavaScript.

- **Function:** Open ELN option for experiment templates, bench records, revisions, audit trails and API integration.
- **Interface / surface:** Web UI / REST API.
- **Why this rank:** Established ELN candidate with revisions and traceability for a concrete bench workflow. The initial computational study does not require a new ELN; compare with SciNote C222/openBIS C149 and integrate one chosen path.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C222 / C149: choose one ELN path.
- **Primary source:** https://doc.elabftw.net/
- **Source PDF page:** 31.

C198

Quarto

- **Domain rank:** 9.
- **Type:** TOOL.
- **Action:** GROW.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 9/10; V (added integration value) = 5.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** TypeScript / Pandoc ecosystem.
- **Function:** Render readable, cited protocols and reports from text, metadata and executable calculations.
- **Interface / surface:** Text editor / IDE / rendered docs.
- **Why this rank:** Reusable rendering for scientific reports and releases. Read canonical run/evidence snapshots; choose one rendering path for a given output rather than rebuilding quantities inside a report or duplicating Jupyter Book C103.
- **Functional overlap:** VIEW.
- **Related components / overlap detail:** C103: select one rendering route per output.
- **Primary source:** https://quarto.org/docs/guide/
- **Source PDF page:** 32.

C213

RO-Crate

- **Domain rank:** 8.
- **Type:** STD.
- **Action:** LATER.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 7.5/10; V (added integration value) = 6/10.
- **Revised/new scores:** No.

- **Language / implementation:** Open spec / Python.
- **Function:** Package plans, executions, inputs, outputs, code and machine-readable metadata.
- **Interface / surface:** JSON-LD package / Python API.
- **Why this rank:** Package workflow inputs, activities, outputs and provenance for export. Map canonical IDs and exact bytes; packaging does not establish scientific validation.
- **Functional overlap:** COMP.
- **Primary source:** https://www.researchobject.org/workflow-run-crate/
- **Source PDF page:** 32.

C187

PyDOE

- **Domain rank:** 10.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 7.5/10; V (added integration value) = 6/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Generate classical factorial, response-surface, mixture and related experimental designs.
- **Interface / surface:** Python API.
- **Why this rank:** Conventional experimental designs. Add randomization, blocking, replication and analysis plans explicitly; design generation alone does not justify experimental conclusions.
- **Functional overlap:** COMP.
- **Primary source:** https://pydoe.github.io/pydoe/
- **Source PDF page:** 32.

C160

OR-Tools

- **Domain rank:** 11.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9/10; V (added integration value) = 5/10.
- **Revised/new scores:** No.
- **Language / implementation:** C++ / bindings.
- **Function:** Allocate samples, plates, staff, instruments and time under discrete constraints.
- **Interface / surface:** Python / native APIs.
- **Why this rank:** Discrete resource scheduling and routing. Complement to numerical process

models; ownership of equipment calendar and run authorization remains in domain services.

- **Functional overlap:** COMP.
- **Primary source:** https://developers.google.com/optimization
- **Source PDF page:** 32.

C019

Ax

- **Domain rank:** 12.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 8.5/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Manage adaptive experiments and generate sequential trials using suitable strategies.
- **Interface / surface:** Python service API.
- **Why this rank:** Adaptive experiment management using models such as BoTorch. Compare with Optuna; keep candidate proposals separate from locked, authorized study runs.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C029 dependency; C159 optimization overlap.
- **Primary source:** https://github.com/facebook/Ax
- **Source PDF page:** 32.

C110

LabOP

- **Domain rank:** 13.
- **Type:** STD.
- **Action:** LATER.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 5.5/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / RDF ontology.
- **Function:** Open linked-data representation for lab protocols and execution; promising interchange layer.
- **Interface / surface:** Python / RDF / exporters.
- **Why this rank:** Protocol semantics, data/control flow and exporters. Evaluate a narrow supported profile before inventing a complete custom representation or assuming turnkey execution.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C018 / XDL: overlapping representation roles.

- **Primary source:** https://bioprotocols.github.io/labop/
- **Source PDF page:** 32.

C232

Snakemake

- **Domain rank:** 14.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Reproducible file-oriented scientific computation with software and provenance metadata.
- **Interface / surface:** Python-based DSL / CLI.
- **Why this rank:** Reproducible computational subworkflow option. The selected application/campaign owner may submit one pipeline job and retain its receipt; avoid duplicating retries and approval ownership between workflow layers.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C137 / C046: computational subworkflow choices.
- **Primary source:** https://snakemake.readthedocs.io/
- **Source PDF page:** 33.

C126

MLflow

- **Domain rank:** 15.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 9/10; V (added integration value) = 4/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python ecosystem.
- **Function:** Track parameters, code versions, metrics and artifacts for simulation or analysis runs.
- **Interface / surface:** Python / REST / web UI.
- **Why this rank:** Track computational model/experiment artifacts and evaluations. Overlaps parts of observability and run tracking; link records rather than using it as the ELN or scientific ledger.
- **Functional overlap:** COMP.
- **Primary source:** https://mlflow.org/docs/latest/ml/tracking/
- **Source PDF page:** 33.

C188

PyLabRobot

- **Domain rank:** 16.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8/10; V (added integration value) = 4/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Hardware-agnostic Python control and resource models for supported laboratory devices.
- **Interface / surface:** Python SDK.
- **Why this rank:** Hardware-agnostic API for supported devices. Qualify each backend, labware, firmware and operation; simulated deck state does not validate physical liquid handling.
- **Functional overlap:** ALT.
- **Primary source:** https://docs.pylabrobot.org/
- **Source PDF page:** 33.

C046

CWL + cwltool

- **Domain rank:** 17.
- **Type:** STD.
- **Action:** ALT.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 3.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Open standard / Python runner.
- **Function:** Portable specification and reference execution for command-line computational workflows.
- **Interface / surface:** YAML / JSON / CLI.
- **Why this rank:** Portable computational workflow description plus runner. Competes with other subworkflow paths; use when partner portability merits the mapping burden.
- **Functional overlap:** ALT.
- **Primary source:** https://www.commonwl.org/features/
- **Source PDF page:** 33.

C137

Nextflow

- **Domain rank:** 18.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 9/10; V (added integration value) = 3.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Groovy / Java.
- **Function:** Scalable dataflow pipelines, especially useful for genomics and computational analysis.
- **Interface / surface:** Groovy DSL / CLI / reports.
- **Why this rank:** Strong computational pipeline ecosystem. Prefer existing qualified workflows over rewriting; do not make it another prospective wet-lab study-plan owner.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C232 / C046: workflow alternatives.
- **Primary source:** https://www.nextflow.io/docs/latest/
- **Source PDF page:** 33.

C058

DVC

- **Domain rank:** 22.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 8.5/10; V (added integration value) = 3.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Version data-dependent computational experiments and compare parameters and metrics.
- **Interface / surface:** CLI / Git / IDE extension.
- **Why this rank:** Version large datasets/model artifacts with Git-linked metadata. Add where useful; avoid competing blob identity, retention and entitlement controls.
- **Functional overlap:** COMP.
- **Primary source:** https://doc.dvc.org/user-guide/experiment-management
- **Source PDF page:** 34.

C158

Opentrons Protocol API

- **Domain rank:** 23.
- **Type:** TOOL.
- **Action:** ALT.

- **Scores:** F (project fit) = 6/10; M (practical maturity) = 8.5/10; V (added integration value) = 3.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / hardware runtime.
- **Function:** Write and analyze automated biology protocols for supported Opentrons robots.
- **Interface / surface:** Python API / robot app.
- **Why this rank:** Vendor-specific supported robot execution path. Overlaps PyLabRobot for some devices; use one qualified command route per operation and record calibration/consumables.
- **Functional overlap:** ALT.
- **Primary source:** https://docs.opentrons.com/python-api/
- **Source PDF page:** 34.

C103

Jupyter Book / MyST

- **Domain rank:** 24.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 8.5/10; V (added integration value) = 3/10.
- **Revised/new scores:** No.
- **Language / implementation:** JavaScript / Python ecosystem.
- **Function:** Combine narrative, equations and executable computation in published technical material.
- **Interface / surface:** Markdown / notebooks / web.
- **Why this rank:** Alternative executable-document and teaching publication stack. Select one primary rendering pipeline unless distinct outputs justify both.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C198: executable-document/teaching alternative.
- **Primary source:** https://jupyterbook.org/stable/
- **Source PDF page:** 34.

C182

protocols.io

- **Domain rank:** 25.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 8.5/10; V (added integration value) = 3/10.
- **Revised/new scores:** No.
- **Language / implementation:** Commercial platform.

- **Function:** Collaboratively develop, version and publish stepwise protocols; optional external repository.
- **Interface / surface:** Hosted web UI / API.
- **Why this rank:** External protocol collaboration/publication option. Reconcile versions and permissions; do not create a competing internal execution ledger.
- **Functional overlap:** ALT.
- **Primary source:** https://www.protocols.io/features
- **Source PDF page:** 34.

C007

AnIML

- **Domain rank:** 28.
- **Type:** STD.
- **Action:** LATER.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 7.5/10; V (added integration value) = 3/10.
- **Revised/new scores:** No.
- **Language / implementation:** Analytical-data standard.
- **Function:** Open ASTM XML representation for analytical instrument data and associated context.
- **Interface / surface:** XML files / compatible tools.
- **Why this rank:** Analytical-data exchange for compatible techniques. Preserve original instrument files and technique metadata; format support varies by exporter and consumer.
- **Functional overlap:** COMP.
- **Primary source:** https://www.animl.org/
- **Source PDF page:** 35.

C108

LabCompiler

- **Domain rank:** 19.
- **Type:** RESEARCH.
- **Action:** LATER.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 3.5/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Emerging AI system.
- **Function:** Research framework separating protocol repair, device-agnostic graphs and hardware assignment.
- **Interface / surface:** Research prototype / paper.

- **Why this rank:** October 2026 preprint offers protocol repair and device-agnostic compilation ideas. Evaluate methods and released implementation separately; reported results are not openFerment validation.
- **Functional overlap:** COMP.
- **Primary source:** https://arxiv.org/html/2610.19550
- **Source PDF page:** 33.

C222

SciNote

- **Domain rank:** 20.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Ruby / JavaScript ecosystem.
- **Function:** ELN option with protocol tasks, inventory links, review/signature features and integrations.
- **Interface / surface:** Web UI / REST API.
- **Why this rank:** ELN alternative with overlapping notebook/inventory/workflow scope. Compare chosen edition, API and deployment rather than implementing two notebooks.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C063 / C149: notebook alternatives.
- **Primary source:** https://www.scinote.net/product/
- **Source PDF page:** 34.

C149

openBIS

- **Domain rank:** 26.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 9/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Java / web ecosystem.
- **Function:** FAIR data platform combining lab notebook, inventory and research-data management.
- **Interface / surface:** Web UI / Python API.
- **Why this rank:** Broad research-data, inventory and ELN platform. Significant overlap with BioRepo;

consider institutional integration before adding another internal platform.

- **Functional overlap:** ALT.
- **Related components / overlap detail:** C063 / C222: broader data-platform overlap.
- **Primary source:** https://openbis.ch/
- **Source PDF page:** 35.

C141

OPC UA

- **Domain rank:** 29.
- **Type:** STD.
- **Action:** LATER.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 9/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Interface standard.
- **Function:** Industrial interoperability option for equipment data and control where supported.
- **Interface / surface:** Client/server / SDKs.
- **Why this rank:** Equipment data/control interchange. Separate observation and actuation permissions, timestamps, quality flags and the locally enforced operational boundary.
- **Functional overlap:** COMP.
- **Primary source:** https://opcfoundation.org/about/opc-technologies/opc-ua/
- **Source PDF page:** 35.

C231

SiLA 2

- **Domain rank:** 30.
- **Type:** STD.
- **Action:** ALT.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 8/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Interface standard.
- **Function:** Service-oriented standard for discovering and communicating with lab devices/services.
- **Interface / surface:** gRPC / HTTP2 services.
- **Why this rank:** Lab service interoperability where instruments expose compatible features. Connectivity and device discovery do not establish scientific protocol equivalence.
- **Functional overlap:** ALT.
- **Primary source:** https://sila-standard.com/standards/
- **Source PDF page:** 35.

C183

ProtoPilot

- **Domain rank:** 27.
- **Type:** RESEARCH.
- **Action:** LATER.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 4/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** Emerging agentic system.
- **Function:** Research direction for validated protocol/code generation with execution feedback.
- **Interface / surface:** Research prototype / paper.
- **Why this rank:** Research on layered protocol-to-code validation and feedback. Hardware/task demonstrations do not establish general bioprocess-runbook reliability.
- **Functional overlap:** COMP.
- **Primary source:** https://arxiv.org/html/2606.31763v1
- **Source PDF page:** 35.

C018

Autoprotocol

- **Domain rank:** 31.
- **Type:** STD.
- **Action:** ALT.
- **Scores:** F (project fit) = 5/10; M (practical maturity) = 6.5/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** Protocol standard / Python.
- **Function:** Express life-science operations in a machine-readable form mappable to automation.
- **Interface / surface:** JSON specification / Python.
- **Why this rank:** Machine-readable protocol/export path for compatible destinations. LabOP and internal plans overlap; mapping must preserve operations, quantities and unsupported steps.
- **Functional overlap:** ALT.
- **Primary source:** https://autoprotocol.org/specification/
- **Source PDF page:** 35.

C029

BoTorch

- **Domain rank:** 21.
- **Type:** TOOL.

- **Action:** DEP.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 1.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / PyTorch.
- **Function:** Research-level Bayesian optimization building blocks beneath adaptive-design systems.
- **Interface / surface:** Python API.
- **Why this rank:** Bayesian optimization methods used by Ax. Add direct hooks only when the selected campaign API cannot express the required acquisition/model logic.
- **Functional overlap:** DEP.
- **Related components / overlap detail:** C019 can supply higher-level experiment management.
- **Primary source:** https://botorch.org/
- **Source PDF page:** 34.

Postdoc and agent operations

The current bounded loop is the baseline. Alternative runtimes, protocols, model serving, tracing and scientific review solve different roles.

C005

Agent & tool registry

- **Domain rank:** 3.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 2.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Version manifests, schemas, capabilities, model policies, scopes and eval suites.
- **Interface / surface:** Admin UI / registry API.
- **Why this rank:** A bounded agent loop and tool whitelist already exist. Add the narrow study tools and their schemas, budgets and guards there; V is extension and evaluation, not building a new agent platform.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 36.

C070

Evaluation & decision ledger

- **Domain rank:** 1.

- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 6.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Record evidence, assumptions, checks, approvals, versions and replayable outcomes.
- **Interface / surface:** Review UI / append-only API.
- **Why this rank:** Retain evidence selections, accepted/rejected outcomes, failures and evaluation labels. Existing tool tests help, but do not establish end-to-end engineering quality; benchmark accepted studies and human verification effort.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 36.

C130

Module adapter SDK

- **Domain rank:** 2.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 5.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Wrap geneOS, fermOS, pureOS, Proforma, BioRepo and Runbooks behind typed calls.
- **Interface / surface:** Python SDK / MCP / OpenAPI.
- **Why this rank:** Build on the shared registry, service APIs and write boundary. Added V is consistent adapters and receipts across domain tools; SDK convenience must not create another source-of-truth or execution owner.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 36.

C185

PydanticAI

- **Domain rank:** 12.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 7.5/10; V (added integration value) = 2.5/10.

- **Revised/new scores:** Yes.
- **Language / implementation:** Python.
- **Function:** Create typed model agents with validated dependencies, tools and structured outputs.
- **Interface / surface:** Python API / MCP support.
- **Why this rank:** Typed-agent framework candidate, not the current baseline. Compare against the implemented Postdoc loop before migration; its structured output/MCP features do not remove scientific applicability or durable-operation requirements.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C113 / C145 / current Postdoc loop: alternatives.
- **Primary source:** https://pydantic.dev/docs/ai/overview/
- **Source PDF page:** 37.

C147

OpenAPI

- **Domain rank:** 4.
- **Type:** STD.
- **Action:** KEEP.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 10/10; V (added integration value) = 1/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Open standard.
- **Function:** Describe stable HTTP services and generate clients, validators and documentation.
- **Interface / surface:** YAML / JSON specification.
- **Why this rank:** The FastAPI service already exposes an HTTP contract. Keep documentation aligned with typed commands; OpenAPI describes operations but does not supply an agent loop or domain adapter semantics.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C123: HTTP contract and agent protocol complement.
- **Primary source:** https://spec.openapis.org/oas/latest.html
- **Source PDF page:** 36.

C123

MCP

- **Domain rank:** 7.
- **Type:** STD.
- **Action:** KEEP.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 8/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** Yes.

- **Language / implementation:** Open protocol.
- **Function:** Expose tools, resources and prompts to model clients through a standard protocol.
- **Interface / surface:** JSON-RPC; local or HTTP transport.
- **Why this rank:** The agent extra already declares an MCP SDK and the project describes a tool-registry server. Added V is qualified client/adapter work, including Sci RAG Kit C269, rather than a universal connector or new scientific semantics.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C147 protocol interface; C270 data representation.
- **Primary source:** https://modelcontextprotocol.io/specification/2026-07-28
- **Source PDF page:** 36.

C220

Scientific supervisor

- **Domain rank:** 6.
- **Type:** CAP.
- **Action:** GROW.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = NR; V (added integration value) = 6.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Plan a task graph, select specialists, enforce budgets and integrate results.
- **Interface / surface:** Workspace UI / orchestration API.
- **Why this rank:** Supervision means applicability, evidence and budget decisions with accountable review. Start with deterministic gates and the current loop; add a separate reviewer agent only after demonstrated error reduction.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 36.

C112

Langfuse

- **Domain rank:** 10.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** TypeScript / Python SDK.
- **Function:** Trace prompts, model calls, tools, costs, feedback and evaluation datasets.

- **Interface / surface:** SDKs / web UI / API.
- **Why this rank:** Candidate backend for model/tool traces, cost and evaluations. Pair one selected backend with OpenTelemetry C157; distinguish MIT core from enterprise controls and compare against current logging before adding operations.
- **Functional overlap:** COMP.
- **Related components / overlap detail:** C012 alternative backend; C157 instrumentation.
- **Primary source:** https://langfuse.com/docs
- **Source PDF page:** 37.

C013

Artifact compiler

- **Domain rank:** 9.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = NR; V (added integration value) = 4.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Produce cited, versioned decisions, models, plans and reports from structured results.
- **Interface / surface:** Document / object / API outputs.
- **Why this rank:** Artifacts already carry inputs, hashes and replay information. Extend compilation into a readable study package and concise explanation; generated prose remains a view over records, not a new scientific authority.
- **Functional overlap:** VIEW.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 37.

C179

promptfoo

- **Domain rank:** 8.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8/10; V (added integration value) = 6.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** TypeScript.
- **Function:** Run repeatable prompt, model, red-team and behavioral regression tests.
- **Interface / surface:** CLI / YAML / web UI.
- **Why this rank:** Versioned regression cases and adversarial tool/permission tests. Prefer

deterministic domain checks where possible; model grading remains noisy evidence.

- **Functional overlap:** COMP.
- **Primary source:** https://www.promptfoo.dev/docs/
- **Source PDF page:** 37.

C116

LiteLLM

- **Domain rank:** 13.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 5.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Normalize many model APIs, routing, fallbacks, budgets and usage accounting.
- **Interface / surface:** Proxy / Python SDK.
- **Why this rank:** Shared multi-provider gateway, quotas and accounting if needed. Domain tool authorization and scientific acceptance remain outside model routing.
- **Functional overlap:** COMP.
- **Primary source:** https://docs.litellm.ai/
- **Source PDF page:** 37.

C200

Ragas

- **Domain rank:** 18.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8/10; V (added integration value) = 4/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python.
- **Function:** Evaluate retrieval and generated answers with configurable metrics and test sets.
- **Interface / surface:** Python API.
- **Why this rank:** Retrieval/answer evaluation metrics. Complement a curated scientific test set; relevance or faithfulness does not establish factual biological validity.
- **Functional overlap:** COMP.
- **Primary source:** https://docs.ragas.io/
- **Source PDF page:** 38.

C117

llama.cpp

- **Domain rank:** 19.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 9/10; V (added integration value) = 3.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** C / C++.
- **Function:** Run quantized local models efficiently across commodity and edge hardware.
- **Interface / surface:** CLI / server / native library.
- **Why this rank:** Local/edge inference path for supported hardware and models. Compare actual quantization quality and latency; use the same governed model-access contract.
- **Functional overlap:** ALT.
- **Primary source:** https://github.com/ggml-org/llama.cpp
- **Source PDF page:** 38.

C004

AG-UI

- **Domain rank:** 25.
- **Type:** STD.
- **Action:** EVAL.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 7/10; V (added integration value) = 2/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Open protocol.
- **Function:** Stream messages, tool events, state and approvals between an agent backend and UI.
- **Interface / surface:** Event stream / client SDKs.
- **Why this rank:** Agent-to-interface event contract candidate. Useful only where it simplifies actual status/tool rendering; existing server job receipts still own execution, and an animated event is not evidence that a simulation ran.
- **Functional overlap:** COMP.
- **Primary source:** https://docs.ag-ui.com/introduction
- **Source PDF page:** 39.

C113

LangGraph

- **Domain rank:** 11.

- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8.5/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python / TypeScript.
- **Function:** Build stateful agent graphs with checkpoints, interrupts, subgraphs and replay.
- **Interface / surface:** Python / TypeScript APIs.
- **Why this rank:** Agent/state-machine alternative with persistence and interrupts. Compare replay, budgets and tool correctness against the current Postdoc loop and other runtimes; migration is conditional, not a second simultaneous agent owner.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C185 / C145 / current Postdoc loop: alternatives.
- **Primary source:** https://docs.langchain.com/oss/python/langgraph/overview
- **Source PDF page:** 37.

C259

vLLM

- **Domain rank:** 20.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 9/10; V (added integration value) = 3/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / CUDA / C++.
- **Function:** Serve supported open-weight language models with high-throughput inference.
- **Interface / surface:** OpenAI-compatible HTTP API.
- **Why this rank:** Local model-serving candidate when workload, hardware and economics justify hosting. Benchmark supported models/tool calling; not a second agent framework.
- **Functional overlap:** ALT.
- **Primary source:** https://docs.vllm.ai/
- **Source PDF page:** 38.

C125

Microsoft Agent Framework

- **Domain rank:** 14.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 7.5/10; V (added integration value) = 2/10.

- **Revised/new scores:** Yes.
- **Language / implementation:** .NET / Python.
- **Function:** Compose agents, middleware and workflows across enterprise model stacks.
- **Interface / surface:** .NET / Python SDKs.
- **Why this rank:** Multi-provider agent framework alternative. Assess concrete runtime, tool, state and deployment behavior against the working loop; framework availability alone provides little new scientific capability.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C185 / C145 / current loop: alternatives.
- **Primary source:** https://learn.microsoft.com/en-us/agent-framework/overview/agent-framework-overview
- **Source PDF page:** 37.

C145

OpenAI Agents SDK

- **Domain rank:** 15.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 8/10; V (added integration value) = 2/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python / TypeScript.
- **Function:** Build tool-using agents, handoffs, guardrails, sessions and traces.
- **Interface / surface:** Python / TypeScript SDKs.
- **Why this rank:** Agent-loop/tool framework alternative. Compare providers, guard behavior, receipts and evaluation cases with the current implementation; adding it does not itself justify replacing the working application boundary.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C185 / C125 / current loop: alternatives.
- **Primary source:** https://openai.github.io/openai-agents-python/
- **Source PDF page:** 38.

C228

SGLang

- **Domain rank:** 21.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8.5/10; V (added integration value) =

2.5/10.

- **Revised/new scores:** No.
- **Language / implementation:** Python / CUDA.
- **Function:** Serve and program model inference with optimized structured and agentic workloads.
- **Interface / surface:** Server / Python API.
- **Why this rank:** Alternative model-serving runtime. Compare against vLLM on the pinned model, hardware, correctness and load rather than vendor speed claims.
- **Functional overlap:** ALT.
- **Primary source:** https://docs.sglang.ai/
- **Source PDF page:** 38.

C140

Ollama

- **Domain rank:** 24.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 8.5/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Go / native backends.
- **Function:** Package and run local models behind a simple developer-facing service.
- **Interface / surface:** CLI / HTTP API / desktop.
- **Why this rank:** Convenient local model management and serving. Often overlaps lower-level inference tools; use as a development route before adding redundant production endpoints.
- **Functional overlap:** DEP.
- **Primary source:** https://docs.ollama.com/
- **Source PDF page:** 39.

C009

Arazzo

- **Domain rank:** 26.
- **Type:** STD.
- **Action:** EVAL.
- **Scores:** F (project fit) = 5/10; M (practical maturity) = 6.5/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Open standard.
- **Function:** Describe goal-oriented sequences and dependencies across OpenAPI operations.
- **Interface / surface:** YAML / JSON specification.

- **Why this rank:** Describes API workflow sequences. Useful for interoperable examples/tests; does not replace durable execution or scientific runbook semantics.
- **Functional overlap:** COMP.
- **Primary source:** https://spec.openapis.org/arazzo/latest.html
- **Source PDF page:** 39.

C012

Arize Phoenix

- **Domain rank:** 16.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 8/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / TypeScript.
- **Function:** Inspect traces, retrieval and evaluations with an open observability stack.
- **Interface / surface:** Python / web UI / OTLP.
- **Why this rank:** Observability/evaluation alternative under ELv2 repository terms. Normal internal use and restricted hosted-service cases differ; assess the actual deployment.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C112 alternative backend; verify edition/license.
- **Primary source:** https://arize.com/docs/phoenix
- **Source PDF page:** 38.

C249

Transformers

- **Domain rank:** 17.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9.5/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / PyTorch.
- **Function:** Load, train and run a broad range of model architectures and tokenizers.
- **Interface / surface:** Python API.
- **Why this rank:** Model-loading/inference/training ecosystem. Usually inherited beneath selected models; direct use is a separate deliberate workload, not another service by default.
- **Functional overlap:** DEP.
- **Primary source:** https://huggingface.co/docs/transformers/

- **Source PDF page:** 38.

C090

Haystack Agents

- **Domain rank:** 22.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8/10; V (added integration value) = 1.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python.
- **Function:** Combine deterministic pipelines, retrieval, tools and specialist agents.
- **Interface / surface:** Python API / pipelines.
- **Why this rank:** Agent layer tied to the Haystack ecosystem. It overlaps the current loop and alternative runtimes; adopt only if an evaluated Haystack assembly offers a concrete advantage without duplicate state and write paths.
- **Functional overlap:** ALT.
- **Primary source:** https://docs.haystack.deepset.ai/docs/agents
- **Source PDF page:** 39.

C119

LlamaIndex Workflows

- **Domain rank:** 23.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 7.5/10; V (added integration value) = 1.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python / TypeScript.
- **Function:** Coordinate event-driven agents and retrieval-centric workflows.
- **Interface / surface:** Python / TypeScript APIs.
- **Why this rank:** Workflow/agent orchestration in the LlamaIndex ecosystem. Connector and retrieval reuse does not require adopting its whole execution owner; compare one scoped workflow against the existing path.
- **Functional overlap:** ALT.
- **Primary source:** https://developers.llamaindex.ai/python/workflows/
- **Source PDF page:** 39.

C001

A2A

- **Domain rank:** 27.
- **Type:** STD.
- **Action:** EVAL.
- **Scores:** F (project fit) = 4/10; M (practical maturity) = 6.5/10; V (added integration value) = 1/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Open protocol.
- **Function:** Let independently operated agent services exchange tasks, messages and artifacts.
- **Interface / surface:** HTTP / JSON-RPC / gRPC.
- **Why this rank:** Cross-agent interoperability protocol. Added V is low before there is a real independently operated agent boundary; ordinary internal services and MCP tools do not require an A2A layer.
- **Functional overlap:** COMP.
- **Primary source:** https://a2a-protocol.org/latest/
- **Source PDF page:** 39.

C278

Anthropic Python SDK

- **Domain rank:** 5.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 9/10; V (added integration value) = 0/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Python.
- **Function:** Connect the current agent loop to configured model calls and tool-use messages.
- **Interface / surface:** Python API / hosted-model client.
- **Why this rank:** Already declared by the service and used by its documented model-calling path. Retain the client; budgets, scientific guards, tool authority and provenance belong to the application, not the provider SDK.
- **Functional overlap:** DEP.
- **Primary source:** https://github.com/anthropics/anthropic-sdk-python
- **Source PDF page:** 36.

Interface and scientific workbench

Extend the existing UI. Shared quantity, evidence, revision and job views matter more than replacing the shell; visualizations read canonical records.

C196

Quantity and evidence inspector

- **Domain rank:** 1.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = NR; V (added integration value) = 8/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Inspect quantity meaning, source anchors, conditions and derivations.
- **Interface / surface:** Browser + API.
- **Why this rank:** High V for the first demonstration: inspect units, basis, source anchors, applied assumptions and actual artifacts from a comparison. Reuse existing evidence components rather than implementing a second quantity store.
- **Functional overlap:** VIEW.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 40.

C210

Revision and impact review

- **Domain rank:** 2.
- **Type:** CAP.
- **Action:** GROW.
- **Scores:** F (project fit) = 9.5/10; M (practical maturity) = NR; V (added integration value) = 7.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Review semantic changes, stale descendants and approval scope.
- **Interface / surface:** Browser + API.
- **Why this rank:** Expose changed inputs/evidence and affected revisions while retaining historical run bases. Current audit records are a starting point; a visual diff does not itself determine scientific impact or review authority.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 40.

C202

React

- **Domain rank:** 4.
- **Type:** TOOL.

- **Action:** KEEP.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 10/10; V (added integration value) = 0/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** JavaScript / TypeScript.
- **Function:** Compose the shared shell, module pages, inspectors and reusable scientific UI components.
- **Interface / surface:** Browser UI library.
- **Why this rank:** The application already uses React. Keep the compatible frontend and extend the study workflow; high practical standing does not justify a framework rewrite or a new scientific state owner.
- **Functional overlap:** COMP.
- **Primary source:** https://react.dev/learn
- **Source PDF page:** 40.

C219

Scientific editor bridge

- **Domain rank:** 3.
- **Type:** CAP.
- **Action:** GROW.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 6/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Bind graphical edits to validated scientific commands.
- **Interface / surface:** Browser + API.
- **Why this rank:** Connect specialist editors and inspectors to canonical domain records. Start with the needed profile/process views; a molecular/sequence editor is later work and must not establish another independent project store.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 40.

C250

TypeScript

- **Domain rank:** 5.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = 10/10; V (added integration value) = 0/10.

- **Revised/new scores:** Yes.
- **Language / implementation:** TypeScript / Go toolchain.
- **Function:** Type frontend models, API clients, component contracts and module boundaries.
- **Interface / surface:** Language / compiler / IDE.
- **Why this rank:** The client is already TypeScript. Retain static client contracts and align with server schemas; type checking is not numerical or biological validation.
- **Functional overlap:** COMP.
- **Primary source:** https://www.typescriptlang.org/docs/
- **Source PDF page:** 40.

C100

Job and approval inbox

- **Domain rank:** 7.
- **Type:** CAP.
- **Action:** BUILD.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = NR; V (added integration value) = 5.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Present durable jobs, budgets, approvals and recoverable failures.
- **Interface / surface:** Browser + API.
- **Why this rank:** Reuse actual worker/job events for queued, running, failed and completed states. Add approval scopes as studies grow; an inbox or loading animation must read authoritative events rather than invent activity.
- **Functional overlap:** VIEW.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 40.

C172

Playwright

- **Domain rank:** 10.
- **Type:** TOOL.
- **Action:** BUILD.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 9.5/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** TypeScript / clients.
- **Function:** Verify complete user journeys, browser behavior, access boundaries and revision

workflows.

- **Interface / surface:** Browser test runner / traces.
- **Why this rank:** Playwright is already declared for browser tooling. Extend coverage with meaningful profile/study/status/evidence cases; this is test coverage work, not introducing a new browser framework or certifying scientific model validity.
- **Functional overlap:** COMP.
- **Primary source:** https://playwright.dev/docs/intro
- **Source PDF page:** 41.

C178

Project and module shell

- **Domain rank:** 11.
- **Type:** CAP.
- **Action:** KEEP.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = NR; V (added integration value) = 0.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** Project code / scientific service.
- **Function:** Shared project, object and revision navigation.
- **Interface / surface:** Browser + API.
- **Why this rank:** Home and module navigation already exist. Preserve them for the first phase; little additional shell work is needed compared with the profile-to-run path. A workspace label does not require a separate deployment.
- **Functional overlap:** VIEW.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 41.

C203

React Flow

- **Domain rank:** 8.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** TypeScript.
- **Function:** Build interactive process and workflow graphs with custom scientific nodes and ports.
- **Interface / surface:** React graph editor / canvas.
- **Why this rank:** Interactive process/runbook graph candidate. Added V is scientific

editing/inspection when required; layout is separate from model semantics, and the existing numerical worker still calculates the plant.

- **Functional overlap:** VIEW.
- **Related components / overlap detail:** C047: graph editing versus network visualization.
- **Primary source:** https://reactflow.dev/learn
- **Source PDF page:** 41.

C241

TanStack Query

- **Domain rank:** 9.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 9/10; V (added integration value) = 5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** TypeScript.
- **Function:** Fetch and cache revision-scoped server data; coordinate mutations and invalidation.
- **Interface / surface:** React hooks / async-state APIs.
- **Why this rank:** Server-state caching/query layer candidate over existing APIs. It can reduce UI state bugs but must respect revision keys and authoritative server responses; it is not another persistent scientific database.
- **Functional overlap:** VIEW.
- **Primary source:** https://tanstack.com/query/latest/docs/framework/react/overview
- **Source PDF page:** 41.

C242

TanStack Table

- **Domain rank:** 6.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 9/10; V (added integration value) = 6/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** TypeScript.
- **Function:** Build evidence tables, stream tables, registers and comparison views with custom cells.
- **Interface / surface:** Headless table APIs.
- **Why this rank:** Useful headless table foundation for scenario and evidence comparison. Compare against existing table components; it complements a canonical result API rather than calculating or storing independent results.

- **Functional overlap:** VIEW.
- **Related components / overlap detail:** C003: overlapping table/grid surfaces.
- **Primary source:** https://tanstack.com/table/latest/docs/introduction
- **Source PDF page:** 40.

C020

axe-core

- **Domain rank:** 13.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 9/10; V (added integration value) = 6.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** JavaScript.
- **Function:** Detect many accessibility defects; supplement with manual keyboard and assistive-tech checks.
- **Interface / surface:** Automated accessibility engine.
- **Why this rank:** Automated accessibility checks. Supplement with keyboard, assistive-tech and scientific-table/graph review; automated coverage is partial.
- **Functional overlap:** COMP.
- **Primary source:** https://github.com/dequelabs/axe-core
- **Source PDF page:** 41.

C204

React Hook Form

- **Domain rank:** 16.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 9/10; V (added integration value) = 5.5/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** TypeScript.
- **Function:** Manage nested scientific forms, field arrays, draft values and validation feedback.
- **Interface / surface:** React hooks / form APIs.
- **Why this rank:** Typed-form state/validation candidate for profile entry. Reuse if it reduces form complexity; server scientific validation remains authoritative and must agree with any client constraints.
- **Functional overlap:** COMP.
- **Primary source:** https://react-hook-form.com/get-started

- **Source PDF page:** 42.

C205

React Router

- **Domain rank:** 17.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 9/10; V (added integration value) = 0/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** TypeScript.
- **Function:** Keep module, project, object and revision context in navigable, shareable URLs.
- **Interface / surface:** Browser routing / framework API.
- **Why this rank:** React Router is already declared by the frontend. Reuse compatible routing and record deep links; new study screens do not justify a second router or rebuilding the module shell.
- **Functional overlap:** COMP.
- **Primary source:** https://reactrouter.com/start/modes
- **Source PDF page:** 42.

C248

Tiptap

- **Domain rank:** 14.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8.5/10; V (added integration value) = 6.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** TypeScript / ProseMirror.
- **Function:** Author structured narratives, annotations and reports; custom nodes can reference scientific objects.
- **Interface / surface:** Headless rich-text editor.
- **Why this rank:** Structured narrative drafts with references to canonical scientific objects. Separate editor content, version compatibility and reviewed scientific revisions.
- **Functional overlap:** VIEW.
- **Related components / overlap detail:** C132: rich scientific document versus code editor.
- **Primary source:** https://tiptap.dev/docs/editor/getting-started/overview
- **Source PDF page:** 41.

C199

Radix UI

- **Domain rank:** 15.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8.5/10; V (added integration value) = 6/10.
- **Revised/new scores:** No.
- **Language / implementation:** TypeScript / React.
- **Function:** Provide accessible dialogs, menus, tabs and controls beneath openFerment styling.
- **Interface / surface:** Unstyled React primitives.
- **Why this rank:** Accessible interaction primitives. Integrate labels, focus order and keyboard behavior; primitive selection does not certify the complete scientific workbench.
- **Functional overlap:** COMP.
- **Primary source:** https://www.radix-ui.com/primitives/docs/overview/introduction
- **Source PDF page:** 42.

C258

Vite

- **Domain rank:** 18.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 9/10; V (added integration value) = 0/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** TypeScript / JavaScript.
- **Function:** Develop and bundle a modular browser application; load heavy editors only when needed.
- **Interface / surface:** Developer CLI / build tooling.
- **Why this rank:** The existing client build uses Vite. Keep its compatible path; it is build tooling, not a scientific or workflow service requiring another integration project.
- **Functional overlap:** DEP.
- **Primary source:** https://vite.dev/guide/
- **Source PDF page:** 42.

C093

i18next

- **Domain rank:** 22.
- **Type:** TOOL.
- **Action:** EVAL.

- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9/10; V (added integration value) = 5.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** JavaScript.
- **Function:** Localize interface text and regional presentation while preserving canonical scientific values.
- **Interface / surface:** JavaScript translation framework.
- **Why this rank:** Localization and regional presentation. Preserve canonical units, precision, currency basis and timestamps; translation must not rewrite scientific meaning.
- **Functional overlap:** COMP.
- **Primary source:** https://www.i18next.com/overview/getting-started
- **Source PDF page:** 43.

C253

Uppy

- **Domain rank:** 23.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 8.5/10; V (added integration value) = 5.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** TypeScript / JavaScript.
- **Function:** Handle file selection, progress, resumable upload options and recoverable intake transfers.
- **Interface / surface:** Browser uploader / plugins.
- **Why this rank:** Browser upload and recovery UI. Server owns size/type checks, content hashes, access, original storage and source-ingestion state.
- **Functional overlap:** COMP.
- **Primary source:** https://uppy.io/docs/
- **Source PDF page:** 43.

C266

Zod

- **Domain rank:** 20.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 9/10; V (added integration value) = 5/10.
- **Revised/new scores:** No.
- **Language / implementation:** TypeScript.

- **Function:** Validate browser inputs and typed API payloads against frontend schema contracts.
- **Interface / surface:** TypeScript schema API.
- **Why this rank:** Frontend runtime validation where useful. Share supported schema contracts; some Zod types/transforms cannot round-trip through JSON Schema.
- **Functional overlap:** COMP.
- **Primary source:** https://zod.dev/json-schema
- **Source PDF page:** 42.

C235

Storybook

- **Domain rank:** 24.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9/10; V (added integration value) = 5/10.
- **Revised/new scores:** No.
- **Language / implementation:** TypeScript / JavaScript.
- **Function:** Develop and review shared UI components across loading, error and permission states.
- **Interface / surface:** Component workbench / test tooling.
- **Why this rank:** Review reusable components in loading, error, empty, permission and review states. Add as shared UI infrastructure rather than per-module installations.
- **Functional overlap:** COMP.
- **Primary source:** https://storybook.js.org/docs
- **Source PDF page:** 43.

C102

jsondiffpatch

- **Domain rank:** 26.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8/10; V (added integration value) = 4/10.
- **Revised/new scores:** No.
- **Language / implementation:** TypeScript.
- **Function:** Compute structural JSON differences beneath human and Postdoc change-review screens.
- **Interface / surface:** JavaScript / TypeScript library.
- **Why this rank:** Structural change display underneath domain-aware review. JSON path differences are not proof of scientific equivalence or a sufficient impact analysis.

- **Functional overlap:** VIEW.
- **Primary source:** https://github.com/benjamine/jsondiffpatch
- **Source PDF page:** 43.

C265

Yjs

- **Domain rank:** 27.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 4/10.
- **Revised/new scores:** No.
- **Language / implementation:** JavaScript.
- **Function:** Synchronize collaborative document drafts through conflict-free shared data types.
- **Interface / surface:** CRDT library / providers.
- **Why this rank:** Collaborative draft synchronization. A converged CRDT document can still be scientifically inconsistent; approvals bind to a reviewed snapshot.
- **Functional overlap:** DEP.
- **Related components / overlap detail:** C092: shared-document synchronization layers.
- **Primary source:** https://docs.yjs.dev/
- **Source PDF page:** 43.

C047

Cytoscape.js

- **Domain rank:** 28.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 9/10; V (added integration value) = 3.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** JavaScript.
- **Function:** Explore pathways, evidence dependencies and other biological or scientific networks.
- **Interface / surface:** Graph visualization / analysis API.
- **Why this rank:** Scientific network exploration. Overlaps graph display, but its network analysis differs from a process editor; share IDs, not a competing process model.
- **Functional overlap:** VIEW.
- **Related components / overlap detail:** C203: visualization and editing differ.
- **Primary source:** https://js.cytoscape.org/
- **Source PDF page:** 44.

C240

Tailwind CSS

- **Domain rank:** 21.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 9/10; V (added integration value) = 0/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** TypeScript / Rust.
- **Function:** Implement consistent spacing, typography, responsive layouts and visual design tokens.
- **Interface / surface:** CSS / build-time utilities.
- **Why this rank:** Tailwind is already in the frontend build dependencies. Preserve compatible styles and accessible layout tokens; no separate CSS-framework integration is needed for the first study.
- **Functional overlap:** ALT.
- **Primary source:** https://tailwindcss.com/docs
- **Source PDF page:** 42.

C104

JupyterLab

- **Domain rank:** 29.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 9.5/10; V (added integration value) = 3.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** TypeScript / Python kernels.
- **Function:** Provide a linked expert notebook environment for exploration and reproducible analysis.
- **Interface / surface:** Separate web application / extensions.
- **Why this rank:** Linked expert analysis environment. Export reproducible artifacts and assumptions; notebook state cannot silently replace approved platform inputs.
- **Functional overlap:** COMP.
- **Primary source:** https://jupyterlab.readthedocs.io/en/stable/
- **Source PDF page:** 44.

C003

AG Grid Community

- **Domain rank:** 25.
- **Type:** TOOL.

- **Action:** ALT.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** TypeScript.
- **Function:** Alternative prebuilt grid for dense editable data; Enterprise features are separate.
- **Interface / surface:** Browser data-grid component.
- **Why this rank:** Prebuilt grid alternative to the headless table path. Verify which editing/grouping features are in Community versus Enterprise before committing.
- **Functional overlap:** ALT.
- **Related components / overlap detail:** C242: grid alternative; edition features differ.
- **Primary source:** https://www.ag-grid.com/react-data-grid/community-vs-enterprise/
- **Source PDF page:** 43.

C092

Hocuspocus

- **Domain rank:** 30.
- **Type:** TOOL.
- **Action:** DEP.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 8/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** TypeScript.
- **Function:** Host Yjs synchronization with authentication hooks and persistence integrations.
- **Interface / surface:** WebSocket service / client provider.
- **Why this rank:** Yjs synchronization/persistence server when collaboration is adopted. Reuse identity and access checks, and avoid treating transport history as scientific history.
- **Functional overlap:** DEP.
- **Primary source:** https://tiptap.dev/docs/hocuspocus/introduction
- **Source PDF page:** 44.

C227

Sentry SDK

- **Domain rank:** 31.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 9/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Multi-language SDKs.

- **Function:** Optional browser error and performance reporting; service and server terms are separate.
- **Interface / surface:** Browser SDK / diagnostic service.
- **Why this rank:** Browser error/performance reporting if operational gaps remain. SDK, server and service terms differ; avoid duplicating sensitive payload capture in multiple backends.
- **Functional overlap:** ALT.
- **Primary source:** https://docs.sentry.io/platforms/javascript/
- **Source PDF page:** 44.

C268

Zustand

- **Domain rank:** 19.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8.5/10; V (added integration value) = 0/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** TypeScript.
- **Function:** Hold small UI-only state such as panel layout, selection and temporary local controls.
- **Interface / surface:** React hooks / store API.
- **Why this rank:** Zustand is already declared for frontend state. Keep UI-only state separate from canonical server records; adding TanStack Query C241 must not create duplicate scientific ownership or replace a working store without need.
- **Functional overlap:** COMP.
- **Primary source:** https://zustand.docs.pmnd.rs/getting-started/introduction
- **Source PDF page:** 42.

C132

Monaco Editor

- **Domain rank:** 32.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 5/10; M (practical maturity) = 9.5/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** TypeScript.
- **Function:** Offer optional expert editing of scripts, configuration and structured model files.
- **Interface / surface:** Embeddable browser code editor.
- **Why this rank:** Expert script/configuration surface. Route execution through the governed job

boundary; an embedded editor does not justify unrestricted server code execution.

- **Functional overlap:** VIEW.
- **Related components / overlap detail:** C248: code editing and prose editing differ.
- **Primary source:** https://microsoft.github.io/monaco-editor/
- **Source PDF page:** 44.

C131

Mol*

- **Domain rank:** 33.
- **Type:** TOOL.
- **Action:** EVAL.
- **Scores:** F (project fit) = 4.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** TypeScript.
- **Function:** Inspect molecular structures and associated annotations when a biology task requires it.
- **Interface / surface:** WebGL viewer / TypeScript API.
- **Why this rank:** On-demand structure viewer for protein tasks. Lazy-load and link exact structure/model revisions; no need in every workbench.
- **Functional overlap:** VIEW.
- **Primary source:** https://molstar.org/
- **Source PDF page:** 44.

C275

Recharts

- **Domain rank:** 12.
- **Type:** TOOL.
- **Action:** KEEP.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 0/10.
- **Revised/new scores:** Yes.
- **Language / implementation:** TypeScript / JavaScript.
- **Function:** Render React charts over study results, scenarios and evidence-backed quantities.
- **Interface / surface:** React / SVG chart components.
- **Why this rank:** Already declared in the frontend. Reuse it for the first comparison/evidence views; Plotly C173 is a conditional extension for a missing capability, not a required replacement or independent calculator.
- **Functional overlap:** VIEW.
- **Primary source:** https://recharts.org/en-US/

- **Source PDF page:** 41.

Guild and Primer

Retain the mission of demonstrated competence, scoped sign-off and teaching from reviewed evidence. Full institutional platforms are later work.

C086

Guild and Primer spine

- **Domain rank:** 1.
- **Type:** CAP.
- **Action:** LATER.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = NR; V (added integration value) = 7/10.
- **Revised/new scores:** No.
- **Language / implementation:** Project code / scientific service.
- **Function:** Connect competence evidence, membership and scoped authority.
- **Interface / surface:** Browser + API.
- **Why this rank:** Separate learning evidence, competence assessment, membership and scoped authority. Record validity periods, conflicts, review responsibilities and revocation behavior.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 45.

C044

CRediT

- **Domain rank:** 2.
- **Type:** STD.
- **Action:** LATER.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 9/10; V (added integration value) = 6/10.
- **Revised/new scores:** No.
- **Language / implementation:** Open standard.
- **Function:** Describe research contributions such as data curation, software, methodology and validation.
- **Interface / surface:** Contributor-role taxonomy.
- **Why this rank:** Contribution-role taxonomy. Records who contributed what; does not by itself determine authorship, inventorship, competence or permission.
- **Functional overlap:** COMP.
- **Primary source:** https://www.niso.org/standards-committees/credit

- **Source PDF page:** 45.

C087

H5P

- **Domain rank:** 3.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** JavaScript / host integration.
- **Function:** Create interactive lessons, practice questions and branching educational activities.
- **Interface / surface:** Content packages / LMS integration.
- **Why this rank:** Interactive educational content in a supported host. Separate learning attempts, assessment evidence and competence decisions; package/accessibility support varies.
- **Functional overlap:** COMP.
- **Primary source:** https://h5p.org/documentation
- **Source PDF page:** 45.

C133

Moodle

- **Domain rank:** 4.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 9.5/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** PHP / JavaScript.
- **Function:** Optional LMS for larger curricula, assessments, learning progress and course administration.
- **Interface / surface:** Web application / APIs / integrations.
- **Why this rank:** Full LMS only when course operations justify one. Connect via supported APIs/LTI; keep scoped signing authority in Guild governance records.
- **Functional overlap:** ALT.
- **Primary source:** https://docs.moodle.org/en/External_tool
- **Source PDF page:** 45.

C142

Open Badges

- **Domain rank:** 5.
- **Type:** STD.
- **Action:** LATER.
- **Scores:** F (project fit) = 5/10; M (practical maturity) = 8/10; V (added integration value) = 2.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Open standard.
- **Function:** Optional portable representation of assessed achievements and their supporting evidence.
- **Interface / surface:** JSON / JSON-LD credentials.
- **Why this rank:** Portable signed achievement assertions when useful. Verify issuer, evidence, status and criteria; receipt of a badge does not automatically grant approval authority.
- **Functional overlap:** VIEW.
- **Primary source:** https://www.1edtech.org/standards/open-badges
- **Source PDF page:** 45.

C051

Discourse

- **Domain rank:** 6.
- **Type:** TOOL.
- **Action:** LATER.
- **Scores:** F (project fit) = 4.5/10; M (practical maturity) = 9/10; V (added integration value) = 2/10.
- **Revised/new scores:** No.
- **Language / implementation:** Ruby / JavaScript.
- **Function:** Optional discussion and working-group forum; object-specific reviews stay linked to artifacts.
- **Interface / surface:** Web forum / API / SSO.
- **Why this rank:** Community discussion when needed. Link artifact-specific review decisions to their canonical objects; forum moderation and scientific approval have different owners.
- **Functional overlap:** COMP.
- **Primary source:** https://docs.discourse.org/
- **Source PDF page:** 45.

Dominion and public release

Later patent research and reviewed disclosure. Documents, jurisdictional claims, clearance judgment, released bytes and public-access evidence are separate records.

C208

Release and public-access ledger

- **Domain rank:** 1.
- **Type:** CAP.
- **Action:** LATER.
- **Scores:** F (project fit) = 9/10; M (practical maturity) = NR; V (added integration value) = 8.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Project code / scientific service.
- **Function:** Reviewed release snapshots, public endpoints and availability evidence.
- **Interface / surface:** Dominion review UI + API.
- **Why this rank:** Freeze exact disclosure bytes, approvals, rights, public URL, access verification and publication evidence. Private project storage and timestamps alone do not establish public dissemination.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 46.

C056

Dominion workbench

- **Domain rank:** 2.
- **Type:** CAP.
- **Action:** LATER.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = NR; V (added integration value) = 7.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Project code / scientific service.
- **Function:** Review patent claims and assemble disclosure releases.
- **Interface / surface:** Browser + API.
- **Why this rank:** Claim-limitation charts, jurisdiction/member-specific status and dated research assessments. Reviewed release workflow is distinct from patentability/FTO judgments.
- **Functional overlap:** COMP.
- **Primary source:** No external source URL displayed for this project capability/entry in the PDF.
- **Source PDF page:** 46.

C065

EPO OPS

- **Domain rank:** 3.
- **Type:** DATA.
- **Action:** LATER.

- **Scores:** F (project fit) = 8/10; M (practical maturity) = 9/10; V (added integration value) = 7/10.
- **Revised/new scores:** No.
- **Language / implementation:** External data service.
- **Function:** Retrieve supported patent bibliographic, family and other records through EPO services.
- **Interface / surface:** Registered XML / HTTP API.
- **Why this rank:** Patent bibliographic, family, text/image and legal-event inputs within coverage and terms. Preserve member-level claims and dated status sources.
- **Functional overlap:** COMP.
- **Primary source:** https://www.epo.org/en/searching-for-patents/data/web-services/ops
- **Source PDF page:** 46.

C255

USPTO Open Data Portal

- **Domain rank:** 4.
- **Type:** DATA.
- **Action:** LATER.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 6/10.
- **Revised/new scores:** No.
- **Language / implementation:** External data service.
- **Function:** Retrieve supported US patent data and file-wrapper material through registered services.
- **Interface / surface:** Web portal / APIs / bulk data.
- **Why this rank:** US patent/file-wrapper data through current registered services. Qualify actual endpoints, authentication and limits; US coverage is not a worldwide search.
- **Functional overlap:** COMP.
- **Primary source:** https://data.uspto.gov/apis/getting-started
- **Source PDF page:** 46.

C211

RFC 3161 timestamping

- **Domain rank:** 5.
- **Type:** STD.
- **Action:** LATER.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9/10; V (added integration value) = 5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Open standard.
- **Function:** Obtain third-party time-stamp tokens over release digests; preserve verification

material.

- **Interface / surface:** Time-stamp protocol / TSA service.
- **Why this rank:** Third-party timestamp over release digest with verification material. Establishes a time assertion about data, not public accessibility or disclosure sufficiency.
- **Functional overlap:** COMP.
- **Primary source:** https://datatracker.ietf.org/doc/html/rfc3161
- **Source PDF page:** 46.

C263

WIPO PATENTSCOPE

- **Domain rank:** 6.
- **Type:** DATA.
- **Action:** LATER.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9/10; V (added integration value) = 4.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** External information service.
- **Function:** Supplement international patent searches and inspect relevant documents and classifications.
- **Interface / surface:** External search web UI.
- **Why this rank:** International search surface plus conditional paid PCT webservices. Distinguish accessible search from licensed programmatic coverage; no assumed unrestricted global API.
- **Functional overlap:** COMP.
- **Primary source:** https://www.wipo.int/en/web/patentscope/data/index
- **Source PDF page:** 46.

C098

InvenioRDM

- **Domain rank:** 7.
- **Type:** TOOL.
- **Action:** ALT.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 3.5/10.
- **Revised/new scores:** No.
- **Language / implementation:** Python / JavaScript.
- **Function:** Optional repository platform for managed research releases, metadata, versions and access.
- **Interface / surface:** Web application / REST API.

- **Why this rank:** Repository/release platform option. External deposition can be simpler than operating a new repository; either route needs access and release verification.
- **Functional overlap:** ALT.
- **Primary source:** https://inveniordm.docs.cern.ch/
- **Source PDF page:** 46.

Broader landscape: 72 records

Described scope from the 25 October review; vendor/research claims were not reproduced here. Linked components use the catalog scores; groups with several components refer to those entries.

These records overlap the component catalog and include papers, institutions and commercial platforms. They are not 72 extra dependencies. NR maturity/value for a research team or unqualified access route is intentional.

E01

Project PISCES / Standard Flowsheet Format

- **Access / role:** Platform / toolkit.
- **Scores:** Linked to the multiple component dossiers below; the PDF displays "linked," not a synthetic group score.
- **Linked components:** C270, C271.
- **Function and reason for position:** Literature/process-data and flowsheet workflow. Separate hosted-platform access from the open SFF toolkit; two different reuse decisions.
- **Primary sources:**
  - https://projectpisces.org/
  - https://github.com/sustainability-software-lab/pisces-standard-flowsheet-format
  - https://energyanalysis.lbl.gov/tools
- **Source PDF page:** 47.

E02

AutoSynthesis

- **Access / role:** Public code.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 3.5/10; V (added integration value) = 5/10.
- **Linked components:** C273.
- **Function and reason for position:** Automated process synthesis and agentic interfaces. Relevant code reuse lead; a complete rerunnable prompt-to-process example remains the adoption gate.
- **Primary sources:**
  - https://eta.lbl.gov/people/sarang-bhagwat
  - https://github.com/sarangbhagwat/autosynthesis

- https://pypi.org/project/autosynthesis/

- **Source PDF page:** 47.

E03

JFAB and the JBEI software ecosystem

- **Access / role:** Ecosystem.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = NR; V (added integration value) = NR.
- **Function and reason for position:** JBEI/JFAB connects predictive models, automation, data and design-build-test-learn. High collaboration fit; this is an institutional tool ecosystem with mixed access terms.
- **Primary sources:**
  - https://www.jbei.org/jfab/
  - https://www.jbei.org/jfab-accelerating-discovery-in-biomanufacturing/
  - https://github.com/JBEI/ART
  - https://github.com/JBEI/media_compiler
- **Source PDF page:** 47.

E04

untangle.bio

- **Access / role:** Hosted / compare.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = NR; V (added integration value) = 3.5/10.
- **Function and reason for position:** Broth definition, downstream routes, balances and economics. Very close recovery/TEA overlap; test a shared case and verify model/data access before integration.
- **Primary sources:**
  - https://untangle.bio/
  - https://untangle.bio/docs.html
  - https://untangle.bio/about
- **Source PDF page:** 47.

E05

New Wave Biotech / Bioprocess Foresight

- **Access / role:** Commercial.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = NR; V (added integration value) = 3/10.
- **Function and reason for position:** Hybrid process models and downstream/full-process decision support. Strong comparator; vendor demonstrations do not establish open reusable code or independent predictive accuracy.

- **Primary sources:**
  - https://www.newwavebiotech.com/
  - https://www.newwavebiotech.com/about
- **Source PDF page:** 47.

E06

SDAI Labs / BioFlow Studio / BioSTEAM MCP

- **Access / role:** Mixed access.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 4.5/10; V (added integration value) = 4.5/10.
- **Function and reason for position:** BioSTEAM/DWSIM-oriented process and AI/MCP interfaces. Public adapter evidence is useful, but broader studio and multi-engine claims need their own runnable qualification.
- **Primary sources:**
  - https://sdai-labs.com/platforms/bioflow-studio
  - https://github.com/SDAI-institute/Biosteam-mcp
  - https://github.com/SDAI-institute
- **Source PDF page:** 47.

E07

Evolrix AI / Kairos

- **Access / role:** Vendor / research.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = NR; V (added integration value) = 3/10.
- **Function and reason for position:** Expression-construct design, checks and evidence/claim provenance. Close geneOS/Postdoc overlap; it does not establish qualified fermentation-to-economics behavior.
- **Primary sources:**
  - https://evolrix.bio/product/
  - https://evolrix.bio/
  - https://evolrix.bio/research/platform-whitepaper/
- **Source PDF page:** 47.

E08

BioMedAna

- **Access / role:** Vendor platform.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = NR; V (added integration value) = 3.5/10.
- **Function and reason for position:** Evidence/data hub, specialist models, study planning and

human review. Broad conceptual overlap; an inspectable integrated deployment and independent evaluation were not established.

- **Primary sources:**
  - https://www.biomedana.ai/
  - https://www.biomedana.ai/twin
- **Source PDF page:** 47.

E09

CADET ecosystem / CADET-Hub

- **Access / role:** Public tools.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 8/10; V (added integration value) = 6/10.
- **Linked components:** C030.
- **Function and reason for position:** Specialist reactor and downstream process-chain models. High value for qualified chromatography scope; a connected specialist chain is not a whole literature-to-design institution.
- **Primary sources:**
  - https://github.com/cadet
  - https://orbit.dtu.dk/en/publications/cadet-hub-online-platform-for-biomanufacturing-modeling-and-stabi/
  - https://www.fz-juelich.de/en/ibg/ibg-1/researchgroups/systems-biotechnology/modeling-and-simulation
- **Source PDF page:** 47.

E10

TeselaGen / Tesela AI / CoScientist

- **Access / role:** Commercial.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = NR; V (added integration value) = 3/10.
- **Function and reason for position:** DBTL platform with sequence design, lab work, data and AI assistance. Substantial precedent for shared records; whole-plant process/TEA scope needs separate demonstration.
- **Primary sources:**
  - https://teselagen.com/platform
  - https://teselagen.com/tesela-ai
  - https://www.teselagen.com/leadership
- **Source PDF page:** 47.

E11

Benchling Bioprocess

- **Access / role:** Commercial.
- **Scores:** F (project fit) = 5/10; M (practical maturity) = 9.5/10; V (added integration value) = 1.5/10.
- **Linked components:** C022.
- **Function and reason for position:** Bioprocess recipes, experiments, execution and contextual data. A useful production workflow benchmark; it is not an open general gene-to-plant solver.
- **Primary sources:**
  - https://help.benchling.com/hc/en-us/articles/39279130261133-How-to-Use-Benchling-Bioprocess
  - https://www.benchling.com/blog/announcing-benchling-bioprocess-for-process-development
- **Source PDF page:** 47.

E12

Ark Biotech / Ark open-source schemas

- **Access / role:** Mixed access.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = NR; V (added integration value) = 3.5/10.
- **Function and reason for position:** Bioreactor modelling and optimization plus public schemas. Reuse schemas only after semantic checks; open schemas do not establish open predictive-engine code.
- **Primary sources:**
  - https://www.ark-biotech.com/
  - https://www.ark-biotech.com/open-source
  - https://pypi.org/project/ark-opensource/0.1.0/
- **Source PDF page:** 47.

E13

KBase Narrative

- **Access / role:** Public platform.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8.5/10; V (added integration value) = 4/10.
- **Function and reason for position:** Typed research objects, executable narrative analyses and retained results. Strong data/workflow precedent; plant design, economics and review scope require additional domain work.
- **Primary sources:**
  - https://www.kbase.us/develop/
  - https://docs.kbase.us/getting-started/narrative/explore-data
  - https://docs.kbase.us/getting-started/narrative/share
- **Source PDF page:** 48.

E14

Biomni / Phylo Biomni Lab

- **Access / role:** Mixed access.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 6.5/10; V (added integration value) = 3.5/10.
- **Function and reason for position:** General biological agents with retrieval, tools and execution. Useful Postdoc benchmark; biomedical breadth does not establish whole-plant bioprocess modelling.
- **Primary sources:**
  - https://github.com/snap-stanford/Biomni
  - https://phylo.bio/
  - https://www.biorxiv.org/content/10.1101/2025.05.30.656746v1
- **Source PDF page:** 48.

E15

BioSTEAM / Thermosteam / Bioindustrial-Park

- **Access / role:** Public code.
- **Scores:** Linked to the multiple component dossiers below; the PDF displays "linked," not a synthetic group score.
- **Linked components:** C026, C247, C023.
- **Function and reason for position:** Flowsheets, thermodynamics, utilities, costing and biorefinery recipes. The selected phase-one numerical route; use linked component scores for different layers.
- **Primary sources:**
  - https://biosteam.readthedocs.io/en/latest/
  - https://github.com/BioSTEAMDevelopmentGroup/biosteam
  - https://cde.nus.edu.sg/chbe/staff/cortes-pena-yoel-rene/
- **Source PDF page:** 48.

E16

QSDsan / EXPOsan / DMsan

- **Access / role:** Public code.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 8/10; V (added integration value) = 3.5/10.
- **Linked components:** C195.
- **Function and reason for position:** Sanitation/resource-recovery TEA/LCA and decision workflows. Strong uncertainty/study precedent; biological kinetics and product-specific unit models still need qualification.
- **Primary sources:**
  - https://yalinli.group/

- https://github.com/QSD-Group/QSDsan
- https://github.com/QSD-Group/EXPOsan
- https://github.com/QSD-Group/DMsan

- **Source PDF page:** 48.

E17

NSKinetics

- **Access / role:** Public code.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 5.5/10; V (added integration value) = 6.5/10.
- **Linked components:** C272.
- **Function and reason for position:** Connect supplied biological kinetics to BioSTEAM and TEA. Valuable coupling candidate; distinguish source-development features from a qualified released bridge.
- **Primary sources:**
  - https://nskinetics.readthedocs.io/en/latest/
  - https://github.com/sarangbhagwat/nskinetics
- **Source PDF page:** 48.

E18

DWSIM and its AI interfaces

- **Access / role:** Public code.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 7.5/10; V (added integration value) = 4.5/10.
- **Linked components:** C059.
- **Function and reason for position:** General process simulation with automation and AI interfaces. Serious engine alternative; correct operation of a flowsheet does not validate transferred biological assumptions.
- **Primary sources:**
  - https://github.com/DanWBR/dwsim
  - https://dwsim.org/help/main/02-contact-information.html
  - https://dwsim.org/help/main/15-ai-powered-flowsheet-analysis.html
- **Source PDF page:** 48.

E19

DTU LLM-AVEVA / MCP framework

- **Access / role:** Research.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = NR; V (added integration value) = 4.5/10.

- **Function and reason for position:** LLM/MCP interaction with an AVEVA simulator. Especially valuable engineering-evaluation precedent: successful tools and numbers can coexist with flawed interpretation.
- **Primary sources:**
  - https://orbit.dtu.dk/en/publications/large-language-model-agent-for-user-friendly-chemical-process-sim/
  - https://arxiv.org/html/2601.11650v2
- **Source PDF page:** 48.

E20

BASF Chemasim agentic process design

- **Access / role:** Research / industry.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = NR; V (added integration value) = 3.5/10.
- **Function and reason for position:** Separate conceptual process engineering from simulator implementation. Relevant planning architecture; published case studies do not establish an open general synthesis suite.
- **Primary sources:**
  - https://arxiv.org/html/2603.12813
  - https://aiche.onlinelibrary.wiley.com/doi/pdf/10.1002/aic.70488
- **Source PDF page:** 48.

E21

From Text to Simulation

- **Access / role:** Research.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = NR; V (added integration value) = 4/10.
- **Function and reason for position:** Text-to-process interpretation, topology and simulator configuration. Good execution benchmark; benchmark success is not arbitrary feedstock-model validity.
- **Primary sources:**
  - https://arxiv.org/html/2601.06776v1
  - https://ojs.aaai.org/index.php/AAAI/article/view/40215
- **Source PDF page:** 48.

E22

Sketch2Simulation

- **Access / role:** Research.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = NR; V (added integration value) = 3/10.

- **Function and reason for position:** Convert process diagrams into executable simulator flowsheets. Useful diagram-ingestion lead; biological parameter evidence and cross-simulator fidelity remain separate.
- **Primary sources:**
  - https://arxiv.org/abs/2603.24629
  - https://aiche.confex.com/aiche/2026/prelim.cgi/Paper/731164
- **Source PDF page:** 48.

E23

Process Architect

- **Access / role:** Publication lead.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = NR; V (added integration value) = 2.5/10.
- **Function and reason for position:** AI-assisted conceptual process design with early TEA/LCA screening. Relevant direction, but full implementation and maturity were not established by the recovered record.
- **Primary sources:**
  - https://www.sciencedirect.com/science/article/abs/pii/S0098135426003418
  - https://doi.org/10.1016/j.compchemeng.2026.109889
- **Source PDF page:** 48.

E24

BioProcessNexus

- **Access / role:** Public tools.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 6.5/10; V (added integration value) = 5/10.
- **Function and reason for position:** Share Monte Carlo and surrogate process TEA/LCA models. Reuse a model and its training domain; publication does not independently validate all assumptions.
- **Primary sources:**
  - https://bioprocessnexus.readthedocs.io/en/latest/
  - https://forschung.boku.ac.at/en/publications/192281
- **Source PDF page:** 48.

E25

IDAES / WaterTAP

- **Access / role:** Public code.
- **Scores:** Linked to the multiple component dossiers below; the PDF displays “linked,” not a synthetic group score.

- **Linked components:** C095, C261.
- **Function and reason for position:** Equation-oriented optimization and water-treatment property/unit models. Useful scoped infrastructure; energy/water applicability differs from biological production coverage.
- **Primary sources:**
  - https://idaes.org/
  - https://github.com/watertap-org/watertap
  - https://watertap.readthedocs.io/en/latest/index.html
- **Source PDF page:** 49.

E26

PharmaPy

- **Access / role:** Public code.
- **Scores:** F (project fit) = 5/10; M (practical maturity) = 6.5/10; V (added integration value) = 2.5/10.
- **Linked components:** C168.
- **Function and reason for position:** Dynamic pharmaceutical unit operations and flowsheet optimization. A specialist isolation-model candidate, not a fermentation evidence or general biological reasoning engine.
- **Primary sources:**
  - https://github.com/CryPTSys/PharmaPy
  - https://pmc.ncbi.nlm.nih.gov/articles/PMC10793241/
  - https://pharmapy.readthedocs.io/
- **Source PDF page:** 49.

E27

SimulKnock / SimulKnockReactor

- **Access / role:** Research / code.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 6.5/10; V (added integration value) = 4.5/10.
- **Function and reason for position:** Joint microbial-intervention and fermentation/reactor optimization. Valuable geneOS-fermOS coupling; mathematical tractability and biological assumptions bound applicability.
- **Primary sources:**
  - https://arxiv.org/abs/2401.16188
  - https://juser.fz-juelich.de/record/1052756/files/1-s2.0-S0098135425003916-main.pdf?version=1
  - https://gitlab.git.nrw/rwth-avt-svt/public/simulknock/-/tree/main/02_SimulKnock/SimulKnock?ref_type=heads

- **Source PDF page:** 49.

E28

SuperPro Designer / SchedulePro

- **Access / role:** Commercial.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = 9/10; V (added integration value) = 2/10.
- **Linked components:** C238.
- **Function and reason for position:** Established integrated batch/continuous process, scheduling and economics. Strong practical benchmark; proprietary access limits direct open-platform integration value.
- **Primary sources:**
  - https://www.intelligen.com/products/superpro-designer/
  - https://www.intelligen.com/products/superpro-product-features/
- **Source PDF page:** 49.

E29

BioSolve Process

- **Access / role:** Commercial.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8/10; V (added integration value) = 2.5/10.
- **Function and reason for position:** Biopharmaceutical process costing, resource and sustainability assessment. Useful Proforma comparator; public descriptions do not supply an open scientific-design backend.
- **Primary sources:**
  - https://biopharmservices.com/biosolve-software/biosolve-process/
  - https://biopharmservices.com/biosolve-software/biosolve-process/use-cases-for-biosolve-process/
- **Source PDF page:** 49.

E30

Siemens gPROMS bioprocess digital twins

- **Access / role:** Commercial.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 9/10; V (added integration value) = 2.5/10.
- **Function and reason for position:** Mechanistic process digital twins, scale-up and optimization. High specialist standing; model qualification, licenses and domain-specific deployment still govern use.
- **Primary sources:**
  - https://www.siemens.com/en-us/products/gproms/

- https://resources.sw.siemens.com/en-US/bioprocess-digital-twins
- https://blog.siemens.com/2024/12/optimizing-the-development-of-biopharmaceuticals-with-bioprocess-digital-twins/

- **Source PDF page:** 49.

E31

DataHowLab

- **Access / role:** Commercial.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = NR; V (added integration value) = 3/10.
- **Function and reason for position:** Hybrid mechanistic/ML bioprocess models and experimental design. Valuable learning-from-data precedent; sparse-data prediction is not universal first-principles plant design.
- **Primary sources:**
  - https://datahow.ch/products/datahowlab/
  - https://datahow.ch/about/core-technologies/
  - https://datahow.ch/about/datahow/
- **Source PDF page:** 49.

E32

Insilico Biotechnology digital twins

- **Access / role:** Commercial.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = NR; V (added integration value) = 3/10.
- **Function and reason for position:** Genome-based and process/ML models for manufacturing. Close cell-to-process ambition; public pages do not establish an open arbitrary-host platform.
- **Primary sources:**
  - https://www.insilico-biotechnology.eu/en/products/insilico-software
- **Source PDF page:** 49.

E33

Culture Biosciences / Console

- **Access / role:** Service / platform.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = NR; V (added integration value) = 3/10.
- **Function and reason for position:** Bioreactors, experiment execution, contextual data and models. Relevant experimental feedback benchmark; hardware access and service engagement are central.
- **Primary sources:**
  - https://www.culturebiosciences.com/console

- https://www.culturebiosciences.com/services/process-model-development

- **Source PDF page:** 49.

E34

Invert

- **Access / role:** Commercial.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = NR; V (added integration value) = 3/10.
- **Function and reason for position:** Equipment connectivity, live data, reporting and predictive workflows. Strong ingestion/execution infrastructure; discovery-to-plant design coverage is a separate question.
- **Primary sources:**
  - https://invertbio.com/
- **Source PDF page:** 49.

E35

BioRaptor

- **Access / role:** Commercial.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = NR; V (added integration value) = 3/10.
- **Function and reason for position:** Contextualize online/offline data and compare bioprocess runs. Useful BioRepo/interpretation precedent; cross-run analysis is narrower than novel process synthesis.
- **Primary sources:**
  - https://www.bioraptor.ai/
  - https://www.bioraptor.ai/how-it-works
- **Source PDF page:** 49.

E36

Synthace

- **Access / role:** Commercial.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = NR; V (added integration value) = 3/10.
- **Function and reason for position:** Design, simulate and execute lab experiments with contextual records. Strong Runbooks precedent; experiment simulation does not imply whole-plant physics or economics.
- **Primary sources:**
  - https://www.synthace.com/how-we-enable-powerful-experiments
  - https://www.synthace.com/about/partners/tecan
  - https://www.synthace.com/miniaturized-purification

- **Source PDF page:** 49.

E37

Dynochem Biologics

- **Access / role:** Commercial.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 8.5/10; V (added integration value) = 3/10.
- **Function and reason for position:** Upstream/downstream modelling and scale-up libraries. Useful specialist comparison when a relevant model is licensed; not an open evidence-to-design community.
- **Primary sources:**
  - https://www.mt.com/gb/en/home/products/L1_AutochemProducts/scale-up-systems/dynochem-biologics.html
- **Source PDF page:** 50.

E38

BIOVIA / 3DEXPERIENCE bioprocess development

- **Access / role:** Enterprise suite.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = NR; V (added integration value) = 2.5/10.
- **Function and reason for position:** Connect process development, recipes, models and transfer data. Important shared-data precedent; suite-level continuity does not establish every model or interface in one installation.
- **Primary sources:**
  - https://www.3ds.com/assets/invest/2026-03/biovia-bioprocessing-and-manufacturing-intelligence.pdf
  - https://blog.3ds.com/brands/biovia/bioprocessing-4-0-with-biovia/
- **Source PDF page:** 50.

E39

Atinary SDLabs

- **Access / role:** Commercial.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = NR; V (added integration value) = 4/10.
- **Function and reason for position:** Experimental optimization with optional hardware integration. Useful active-learning benchmark; optimizing experiments is not a validated general process-plant model.
- **Primary sources:**
  - https://atinary.com/applications/ai-biotech-rd-platform/
- **Source PDF page:** 50.

E40

Reflux

- **Access / role:** Vendor / early.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = NR; V (added integration value) = 3.5/10.
- **Function and reason for position:** Agents inspect/edit/run simulations and review artifacts. Close tool-operation overlap; advertised simulator adapters need runtime and source-access qualification.
- **Primary sources:**
  - https://www.reflux.sh/
- **Source PDF page:** 50.

E41

pyFOOMB

- **Access / role:** Public code.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 7.5/10; V (added integration value) = 5/10.
- **Function and reason for position:** Kinetic bioprocess fitting and analysis. Useful model-qualification component; a fitted kinetic model is distinct from a complete plant, evidence repository or research agent.
- **Primary sources:**
  - https://pmc.ncbi.nlm.nih.gov/articles/PMC7923582/
  - https://github.com/MicroPhen/pyFOOMB
- **Source PDF page:** 50.

E42

COBRApy / cameo

- **Access / role:** Public code.
- **Scores:** Linked to the multiple component dossiers below; the PDF displays “linked,” not a synthetic group score.
- **Linked components:** C041, C032.
- **Function and reason for position:** Constraint-based metabolic modelling and intervention analysis. A geneOS foundation; flux feasibility does not itself predict measured kinetics or equipment economics.
- **Primary sources:**
  - https://www.biosustain.dtu.dk/technologies/scientific-tools
  - https://opencobra.github.io/cobrapy/packages/
  - https://github.com/biosustain/cameo

- **Source PDF page:** 50.

E43

EPFL pyTFA / ETFL family

- **Access / role:** Public code.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 7/10; V (added integration value) = 3.5/10.
- **Linked components:** C192.
- **Function and reason for position:** Thermodynamic and expression constraints on metabolic models. Specialist geneOS refinement; data availability and modelling assumptions limit applicability.
- **Primary sources:**
  - https://people.epfl.ch/vassily.hatzimanikatis/?lang=en
  - https://github.com/EPFL-LCSB/pytfa
  - https://gitlab.com/EPFL-LCSB/etfl
- **Source PDF page:** 50.

E44

RAVEN / GECKO ecosystem

- **Access / role:** Public code.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 7/10; V (added integration value) = 5/10.
- **Linked components:** C080.
- **Function and reason for position:** Metabolic reconstruction and enzyme-constrained modelling. Complementary host-model capabilities; preserve model/version and scientific assumption boundaries.
- **Primary sources:**
  - https://sysbiochalmers.github.io/RAVEN/
  - https://sysbiochalmers.github.io/GECKO/
  - https://github.com/sysbiochalmers
- **Source PDF page:** 50.

E45

Asimov Kernel

- **Access / role:** Commercial.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = NR; V (added integration value) = 2.5/10.
- **Function and reason for position:** Construct design, hybrid models and expression-system libraries. Relevant geneOS comparison; expression/cell-line workflows do not supply generic whole-process economics.

- **Primary sources:**
  - https://www.asimov.com/kernel
  - https://www.asimov.com/approach
- **Source PDF page:** 50.

E46

Cradle

- **Access / role:** Commercial.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = NR; V (added integration value) = 2.5/10.
- **Function and reason for position:** AI-assisted protein design with assay feedback. Relevant future molecular design; improving a protein is narrower than selecting and pricing a production process.
- **Primary sources:**
  - https://www.cradle.bio/platform
- **Source PDF page:** 50.

E47

ToolUniverse

- **Access / role:** Public code.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 6.5/10; V (added integration value) = 5/10.
- **Function and reason for position:** Scientific tool catalogue, APIs and datasets for agent calling. Useful Postdoc infrastructure; a large registry does not qualify coupled assumptions or scientific results.
- **Primary sources:**
  - https://zitniklab.hms.harvard.edu/ToolUniverse/
  - https://aiscientist.tools/
- **Source PDF page:** 50.

E48

BioDiscoveryAgent

- **Access / role:** Research / code.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 5.5/10; V (added integration value) = 3.5/10.
- **Function and reason for position:** Experiment-design agents combining literature, code and biological screening. Good closed-loop benchmark; genetic perturbation tasks differ from industrial plant design.
- **Primary sources:**
  - https://proceedings.iclr.cc/paper_files/paper/2025/hash/4252dc945318330290000f85dc5fac792-

Abstract-Conference.html
    - https://github.com/snap-stanford/BioDiscoveryAgent
- **Source PDF page:** 50.

E49

Coscientist

- **Access / role:** Research / code.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 5.5/10; V (added integration value) = 3/10.
- **Function and reason for position:** Chemical documentation retrieval and experimental planning/execution. Important agent precedent; narrow demonstrations do not establish general fermentation-scale capability.
- **Primary sources:**
    - https://www.nature.com/articles/s41586-023-06792-0
    - https://github.com/gomesgroup/coscientist
- **Source PDF page:** 51.

E50

PaperQA

- **Access / role:** Public code.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = 7.5/10; V (added integration value) = 4.5/10.
- **Function and reason for position:** Evidence retrieval and cited scientific answers. Strong comparison with current retrieval/Sci RAG Kit; citations alone do not preserve every numerical context or verify downstream model use.
- **Primary sources:**
    - https://github.com/Future-House/paper-qa
    - https://github.com/Future-House/paper-qa/releases
- **Source PDF page:** 51.

E51

ChemCrow

- **Access / role:** Research / code.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 5.5/10; V (added integration value) = 2.5/10.
- **Function and reason for position:** Tool-augmented chemistry reasoning with databases and code. Relevant specialist-agent precedent; not a whole bioprocess evidence, plant and finance stack.
- **Primary sources:**
    - https://github.com/ur-whitelab/chemcrow-public

- **Source PDF page:** 51.

E52

Galaxy

- **Access / role:** Public platform.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 9/10; V (added integration value) = 3/10.
- **Function and reason for position:** Reproducible scientific tools, workflows, histories and sharing. Strong computational precedent; existing workflow coverage limits migration value, and domain semantics still need adapters.
- **Primary sources:**
  - https://galaxyproject.org/
  - https://iwc.galaxyproject.org/about/
  - https://pmc.ncbi.nlm.nih.gov/articles/PMC13355071/
- **Source PDF page:** 51.

E53

BioSimulations / BioSimulators

- **Access / role:** Public platform.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 8/10; V (added integration value) = 4/10.
- **Function and reason for position:** Standardized biological model execution, registries and results. Useful reproducibility layer; supported formats do not imply universal process-engine interoperability.
- **Primary sources:**
  - https://biosim.biosimulations.org/
  - https://github.com/biosimulations/biosimulations
  - https://docs.biosimulations.org/about/team/
- **Source PDF page:** 51.

E54

BioModels

- **Access / role:** Model repository.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 8.5/10; V (added integration value) = 5/10.
- **Linked components:** C024.
- **Function and reason for position:** Curated published models with identifiers and releases. Valuable reusable-model source; curated metadata does not validate every model for a new process task.
- **Primary sources:**

- https://www.ebi.ac.uk/training/online/courses/biomodels-quick-tour/what-is-biomodels/
- https://journals.plos.org/ploscompbiol/article?id=10.1371%2Fjournal.pcbi.1013239

- **Source PDF page:** 51.

E55

FAIRDOM-SEEK

- **Access / role:** Public platform.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 8.5/10; V (added integration value) = 5/10.
- **Function and reason for position:** Link investigations, studies, assays, data and models. Strong BioRepo/FAIR architecture precedent; it does not automatically infer causal process behavior.
- **Primary sources:**
  - https://fair-dom.org/fairdomseek
- **Source PDF page:** 51.

E56

openBIS

- **Access / role:** Public platform.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 9/10; V (added integration value) = 2.5/10.
- **Linked components:** C149.
- **Function and reason for position:** ELN/research data and durable provenance. Useful existing records platform; metadata configuration and notebook scope differ from an engineering process simulator.
- **Primary sources:**
  - https://csb.ethz.ch/tools/software/openbis-lims-eln.html
- **Source PDF page:** 51.

E57

EnzymeML / PyEnzyme

- **Access / role:** Open standard/code.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 7.5/10; V (added integration value) = 5/10.
- **Function and reason for position:** Exchange enzyme reaction/kinetic data with conditions and software interfaces. Useful quantity/context bridge; whole-plant, campaign and finance records require additional contracts.
- **Primary sources:**
  - https://www.ibc.uni-stuttgart.de/en/tbc/research/bioinformatics/EnzymeML-a-data-exchange-format-for-biocatalysis-and-enzymology/
  - https://github.com/EnzymeML/PyEnzyme

- **Source PDF page:** 51.

E58

WorkflowHub / RO-Crate

- **Access / role:** Registry / standard.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = 7.5/10; V (added integration value) = 6/10.
- **Linked components:** C213.
- **Function and reason for position:** Discover and package workflows with metadata/provenance. Valuable release/replay packaging; identify what ran separately from review of scientific correctness.
- **Primary sources:**
  - https://workflowhub.org/
  - https://pmc.ncbi.nlm.nih.gov/articles/PMC11386446/
- **Source PDF page:** 51.

E59

LabOP

- **Access / role:** Standard / code.
- **Scores:** F (project fit) = 7/10; M (practical maturity) = 5.5/10; V (added integration value) = 4.5/10.
- **Linked components:** C110.
- **Function and reason for position:** Typed machine-readable lab procedure semantics and execution hooks. Useful protocol representation; valid syntax is not an approved, validated experimental method.
- **Primary sources:**
  - https://github.com/Bioprotocols/labop
  - https://bioprotocols.github.io/labop/quickstart/
- **Source PDF page:** 51.

E60

MADSci

- **Access / role:** Public code.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 7/10; V (added integration value) = 5/10.
- **Function and reason for position:** Modular autonomous research/lab orchestration. Useful future execution foundation; orchestrating equipment does not validate a production-plant model.
- **Primary sources:**
  - https://ad-sdl.github.io/MADSci/
  - https://github.com/AD-SDL

- **Source PDF page:** 51.

E61

PyLabRobot

- **Access / role:** Public code.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 8/10; V (added integration value) = 4/10.
- **Linked components:** C188.
- **Function and reason for position:** Hardware-neutral lab automation interfaces. Good protocol/driver separation; hardware support requires actual qualification and is later than the simulation demonstrator.
- **Primary sources:**
  - https://github.com/PyLabRobot/pylabrobot
  - https://docs.pylabrobot.org/stable/
- **Source PDF page:** 52.

E62

ChemOS / ChemOS 2.0

- **Access / role:** Public code.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 7/10; V (added integration value) = 4/10.
- **Function and reason for position:** Coordinate experiments, instruments and selection algorithms. Strong autonomous-lab precedent; it is not a source-grounded fermentation/TEA application by itself.
- **Primary sources:**
  - https://github.com/aspuru-guzik-group/ChemOS
  - https://a3md.utoronto.ca/chemos-an-orchestration-software-to-democratize-autonomous-discovery/
  - https://pubs.rsc.org/en/content/articlehtml/2024/dd/d4dd00115j
- **Source PDF page:** 52.

E63

Chemputer / XDL

- **Access / role:** Open formats/code.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 7.5/10; V (added integration value) = 4.5/10.
- **Function and reason for position:** Represent and compile chemical procedures to modular hardware. Useful reproducible execution pattern; declared hardware/procedure scope does not cover every bioprocess model.
- **Primary sources:**

    - https://github.com/croningp/chemputation
    - https://pubs.acs.org/doi/10.1021/jacsau.1c00303
- **Source PDF page:** 52.

E64

protocols.io

- **Access / role:** Hosted methods.
- **Scores:** F (project fit) = 6/10; M (practical maturity) = 8.5/10; V (added integration value) = 3/10.
- **Linked components:** C182.
- **Function and reason for position:** Publish and collaborate on versioned protocols. Useful source methods and revision history; publication is not scientific validation, endorsement or model execution.
- **Primary sources:**
    - https://www.protocols.io/
    - https://www.protocols.io/help/publishing/publish
    - https://www.nature.com/nprot/protocolsio
- **Source PDF page:** 52.

E65

BayBE

- **Access / role:** Public code.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 7.5/10; V (added integration value) = 4/10.
- **Function and reason for position:** Bayesian experiment planning for mixed, low-data campaigns. Relevant Runbooks optimizer; it needs a defined domain and objective and does not supply biological validity.
- **Primary sources:**
    - https://doi.org/10.1039/D5DD00050E
    - https://www.emdgroup.com/en/news/opensource-ai-experimentation-planner-06-12-2023.html
- **Source PDF page:** 52.

E66

BoFire

- **Access / role:** Public code.
- **Scores:** F (project fit) = 7.5/10; M (practical maturity) = 7.5/10; V (added integration value) = 4.5/10.
- **Function and reason for position:** Bayesian/black-box and mixed-variable experimental design. Useful optimization candidate; select within declared constraints and compare with existing optimizer families.

- **Primary sources:**
  - https://experimental-design.github.io/bofire/
  - https://github.com/experimental-design/bofire
- **Source PDF page:** 52.

E67

Summit

- **Access / role:** Public code.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = 7/10; V (added integration value) = 3/10.
- **Function and reason for position:** Benchmarks and strategies for chemical experiment optimization. Good evaluation lineage; verify current supported strategies before adoption as a process-design component.
- **Primary sources:**
  - https://github.com/sustainable-processes/summit
- **Source PDF page:** 52.

E68

DTU Biosustain Research Software team

- **Access / role:** Research team.
- **Scores:** F (project fit) = 8/10; M (practical maturity) = NR; V (added integration value) = NR.
- **Function and reason for position:** Research software, catalogues, metadata and lab-system integration. Strong collaboration fit; team capabilities are not one reusable openFerment-equivalent product.
- **Primary sources:**
  - https://www.biosustain.dtu.dk/technologies/biofoundry/informatics/research-software
- **Source PDF page:** 52.

E69

JBEI LEAD tools / DSP Designer / BioC2G

- **Access / role:** Mixed access.
- **Scores:** F (project fit) = 8.5/10; M (practical maturity) = NR; V (added integration value) = 4/10.
- **Function and reason for position:** Feedstock/site assessment, TEA/LCA and downstream design tools. High practical comparison value; each tool has its own access, maturity and supported model scope.
- **Primary sources:**
  - https://lead.jbei.org/
  - https://energyanalysis.lbl.gov/tools

    - https://dspdesigner.lbl.gov/
    - https://lead.jbei.org/tea_lca_tool/f2f_landing
- **Source PDF page:** 52.

E70

OxTium / BioFord intelligent fermentation

- **Access / role:** Indexed vendor lead.
- **Scores:** F (project fit) = 5.5/10; M (practical maturity) = NR; V (added integration value) = 1.5/10.
- **Function and reason for position:** AI-assisted fermentation prediction/control and agent-based lab claims. Keep as a watchlist: limited indexed documentation and no inspectable implementation established.
- **Primary sources:**
    - https://www.oxtium.com/en/ai-biomanufacturing
    - https://www.oxtium.com/en/news/oxtium-2026-ai-biopharma-conference
- **Source PDF page:** 52.

E71

SIAT Shenzhen synthetic-biology Design-Learn platform

- **Access / role:** Institutional lead.
- **Scores:** F (project fit) = 6.5/10; M (practical maturity) = NR; V (added integration value) = NR.
- **Function and reason for position:** Synthetic-biology design-learn platform and applications. Worth follow-up; retrieved descriptions do not establish unrestricted access, complete simulation or one integrated product.
- **Primary sources:**
    - https://infrasynbio.siat.ac.cn/platform/our/Design-Learn
- **Source PDF page:** 52.

E72

UCL intelligent automation / agent-based bioprocess design

- **Access / role:** Research lineage.
- **Scores:** F (project fit) = 5/10; M (practical maturity) = NR; V (added integration value) = NR.
- **Function and reason for position:** Earlier agents, automation and multi-agent bioprocess research. Useful precedent; current software artifacts and present capability were not established in the review.
- **Primary sources:**
    - https://pmc.ncbi.nlm.nih.gov/articles/PMC4113973/
    - https://pubmed.ncbi.nlm.nih.gov/20014420/

- **Source PDF page:** 52.

Reuse decisions that change the assembly

Source PDF page 53; INTEGRATION DECISIONS.

The most valuable candidates remove duplicated work while keeping scientific records and calculation ownership intact.

DECISION	FIRST QUALIFICATION	ADOPTION / OWNERSHIP GATE
SFF C270 and PISCES C271	Export the current reference. Inspect topology, chemicals, units, metadata, missing fields and source mappings.	MIT toolkit reuse is separate from hosted PISCES access and restricted source deposits. Conformance, completeness and numerical reproduction are different checks.
Sci RAG Kit C269	Compare current retrieval and the smallest kit configuration on the same real corpus and held-out questions.	Map evidence IDs/rights/review. Its Postgres index can be derived while BioRepo retains authority; a later migration transfers ownership explicitly.
Retrieval complexity	Add semantic, graph, community or hypothetical-answer layers one at a time.	Measure retrieval/citation/context correctness, missing-evidence behavior, latency, indexing cost and spend per accepted study. Synthetic demo numbers never become evidence.
Kinetics and process synthesis	Test a suitable model through NSKinetics; request a rerunnable AutoSynthesis process example.	Coupling does not supply validated parameters. The separate meta-analysis paper is an evidence-method lead, with code access unverified.
Alternative execution owners	Compare existing Postdoc/worker behavior with a proposed runtime or durable platform.	Use the same tools, budgets, replay/failure cases and scientific acceptance criteria. Add agents or migrate only for demonstrated gains.

A fair baseline also includes a strong general agent with the same sources and tools. Measure total model/engine cost and human verification effort per accepted study. Correct tool execution does not guarantee sound engineering interpretation; the DTU simulator-agent work is a relevant failure-pattern precedent.

Note: The inspected Sci RAG Kit source is alpha, BSD-3-Clause, with documented REST/MCP serving. Offline local-hash retrieval is not equivalent semantic retrieval or a fully local graph/answer stack. Embedding-model changes require index migration. SFF's inspected schema is 0.2.4; neither integration was runtime-qualified in this export.

Six-month path and handoff basis

Source PDF page 54; CONTINUE THE PROJECT.

Keep the larger mission while qualifying one tractable scientific domain at a time.

PERIOD	TARGET	EVIDENCE TO PRODUCE
Months 2-3	One supported feedstock family and pretreatment response model.	Applicable characterization/kinetics, retained conditions, held-out checks and declared error/domain limits.
Months 4-5	Resource/cost tradeoffs; another route/family when justified.	Common dry-feed/product/process basis, sensitivity, conditional uncertainty and explicit model replacements.
Month 6	A bounded prototype ready for independent technical review.	Restart/replay, failure behavior, traceable studies and reviewer-assessed applicability.

Next AI: inspect the current repository and available runtime data first. Preserve the reference model and implement one saved profile -> recalculated full plant -> comparison -> evidence path. Resolve quantities from records and explicit assumptions. Settle the supported feedstock/route, available characterization/corpus and builder/reviewer capacity when implementation depends on them. Corpus size follows coverage and held-out performance; no fixed target paper count for the new feedstock path was established.

Source snapshots

- **Application snapshot:** bytesizedyogurt/openFermentUI @ d19224ff0b6eb809b21492d1b7a2e8fd895a0f44; runtime audit 25 Oct; same-commit README/manifests reviewed 28 Oct. Source: https://github.com/bytesizedyogurt/openFermentUI/tree/d19224ff0b6eb809b21492d1b7a2e8fd895a0f44
- **SFF source:** sustainability-software-lab/pisces-standard-flowsheet-format @ e236e3250414f45cf9fd7b8b854f1afc67fc2cad; schema 0.2.4. Source: https://github.com/sustainability-software-lab/pisces-standard-flowsheet-format/tree/e236e3250414f45cf9fd7b8b854f1afc67fc2cad
- **Sci RAG Kit source:** sustainability-software-lab/sci-rag-kit @ 26b3990c6264b68f29e833e22b04eb323e06e11b; source inspected 27 Oct. Source: https://github.com/sustainability-software-lab/sci-rag-kit/tree/26b3990c6264b68f29e833e22b04eb323e06e11b
- **Research basis:** Recovered 25 Oct component atlas and 72-record landscape; 20-page development guide; supplied October brief; visible user instructions.

Note: Coverage: cross-chat retrieval was unavailable on 27-28 October. This synthesis uses the visible conversation, recovered brief/atlas/development guide/landscape and previously available relevant excerpts; unseen later-chat decisions are not certified. The 279 component dossiers and 72 ecosystem records overlap and must not be summed as unique software products. Maturity scores are editorial; no new application/runtime benchmark was run for this export.